The
Ultimate SuperComputer-on-a-Chip
for
Massive Big Data and Highly Iterative Algorithms

Veljko Milutinović

The Flynn's Classification of Computer Systems:

- SISD
- SIMD
- MISD
- MIMD

Early 80s: DARPA's First GaAs Microprocessor (SISD)

MICROPROCESSORS

DARPA EYES 100-MIPS GaAs CHIP FOR STAR WARS

PALO ALTO
For its Star Wars program, the Department of Defense intends to push well beyond the current limits of technology. And along with lasers and particle beams, one piece of hardware it has in mind is a microprocessor chip having as much computing power as 100 of Digital Equipment Corp.'s VAX-11/780 superminicomputers.

One candidate for the role of basic computing engine for the program, officially called the Strategic Defense Initiative [*ElectronicsWeek*, May 13, 1985, p. 28], is a gallium arsenide version of the Mips reduced-instruction-set computer (RISC) developed at Stanford University. Three teams are now working on the processor. And this month, the Defense Advanced Projects Research Agency closed the request-for-proposal (RFP) process for a 1.25-μm silicon version of the chip.

Last October, Darpa awarded three contracts for a 32-bit GaAs microprocessor and a floating-point coprocessor. One went to McDonnell Douglas Corp., another to a team formed by Texas Instruments Inc. and Control Data Corp., and the third to a team from RCA Corp. and Tektronix Inc. The three are now working on processes to get useful yields. After a year, the program will be reduced to one or two teams. Darpa's target is to have a 10,000-gate GaAs chip by the beginning of 1988.

If it is as fast as Darpa expects, the chip will be the basic engine for the Advanced Onboard Signal Processor, one of the baseline machines for the SDI. "We went after RISC because we needed something small enough to put on GaAs," says Sheldon Karp, principal scientist for strategic technology at Darpa. The agency had been working with the Motorola Inc. 68000 microprocessor, but Motorola wouldn't even consider trying to put the complex 68000 onto GaAs, Karp says.

A natural. The Mips chip, which was originally funded by Darpa, was a natural for GaAs. "We have only 10,000 gates to work with," Karp notes. "And the Mips people had taken every possible step to reduce hardware requirements. There are no hardware interlocks, and only 32 instructions."

Even 10,000 gates is big for GaAs; the first phase of the work is intended to make sure that the RISC architecture can be squeezed into that size at respectable yields, Karp says.

Mips was designed by a group under John Hennessey at Stanford. Hennessey, who has worked as a consultant with Darpa on the SDI project, recently took the chip into the private sector by forming Mips Computer Systems of Mountain View, Calif. [*ElectronicsWeek*, April 29, 1985, p. 36]. Computer-aided-design software came from the Mayo Clinic in Rochester, Minn.

The GaAs chip will be clocked at 200 MHz, the silicon at 40 MHz

The silicon Mips chip will come from a two-year effort using the 1.25-μm design rules developed for the Very High Speed Integrated Circuit program. (The Darpa chip was not made part of VHSIC in order to open the RFP to contractors outside that program.)

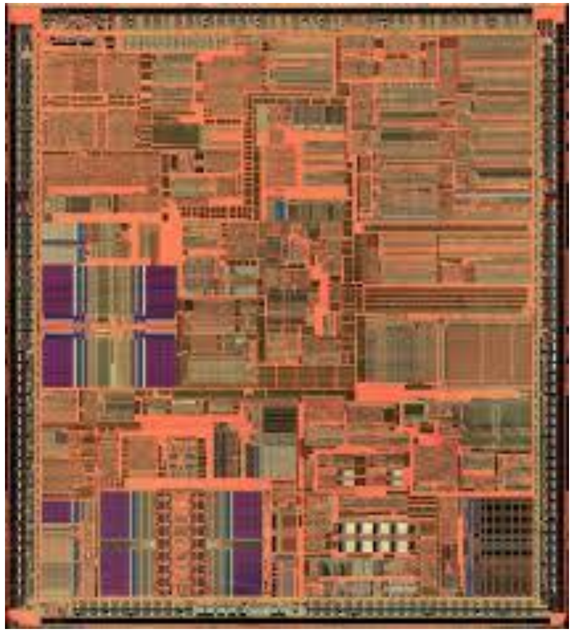
Both the silicon and GaAs microprocessors will be full 32-bit engines sharing 90% of a common instruction core. Pascal and Air Force 1750A compilers will be targeted for the core instruction set, so that all software will be interchangeable.

The GaAs requirement specifies a clock frequency of 200 MHz and a computation rate of 100 million instructions per second. The silicon chip will be clocked at 40 MHz.

Eventually, the silicon chip must be made radiation-hard; the GaAs chip will be intrinsically rad-hard.

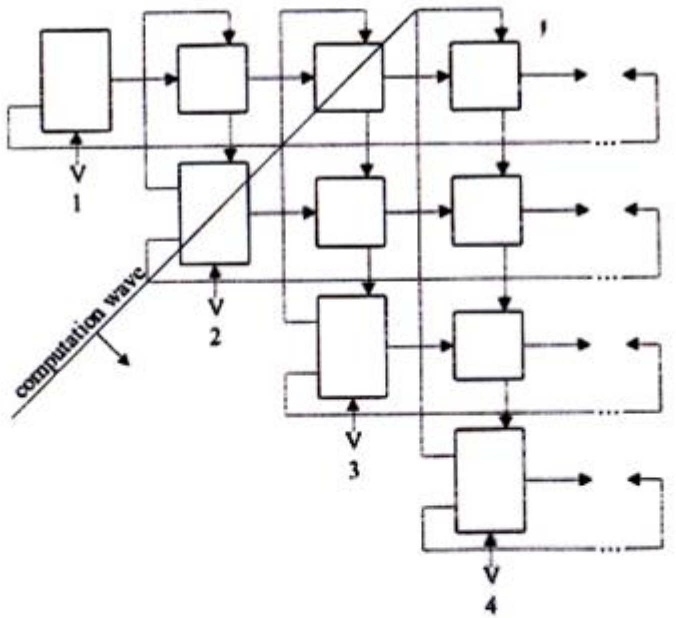
Darpa will not release figures on the size of its RISC effort. The silicon version is being funded through the Air Force's Air Development Center in Rome, N.Y.

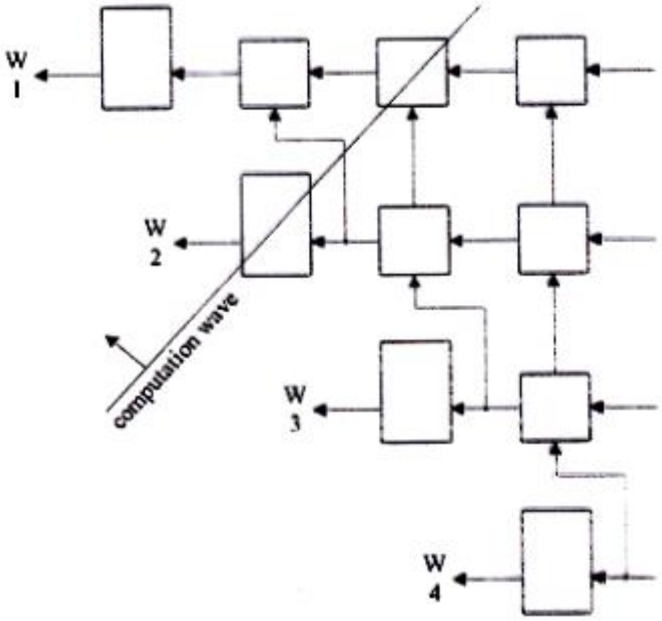
–Clifford Barney



Lesson Learned:
Mere technological change is NOT a solution, due to large OFF-CHIP/ON-CHIP delays, leading fast to asymptotic saturations of the achievable speedup!

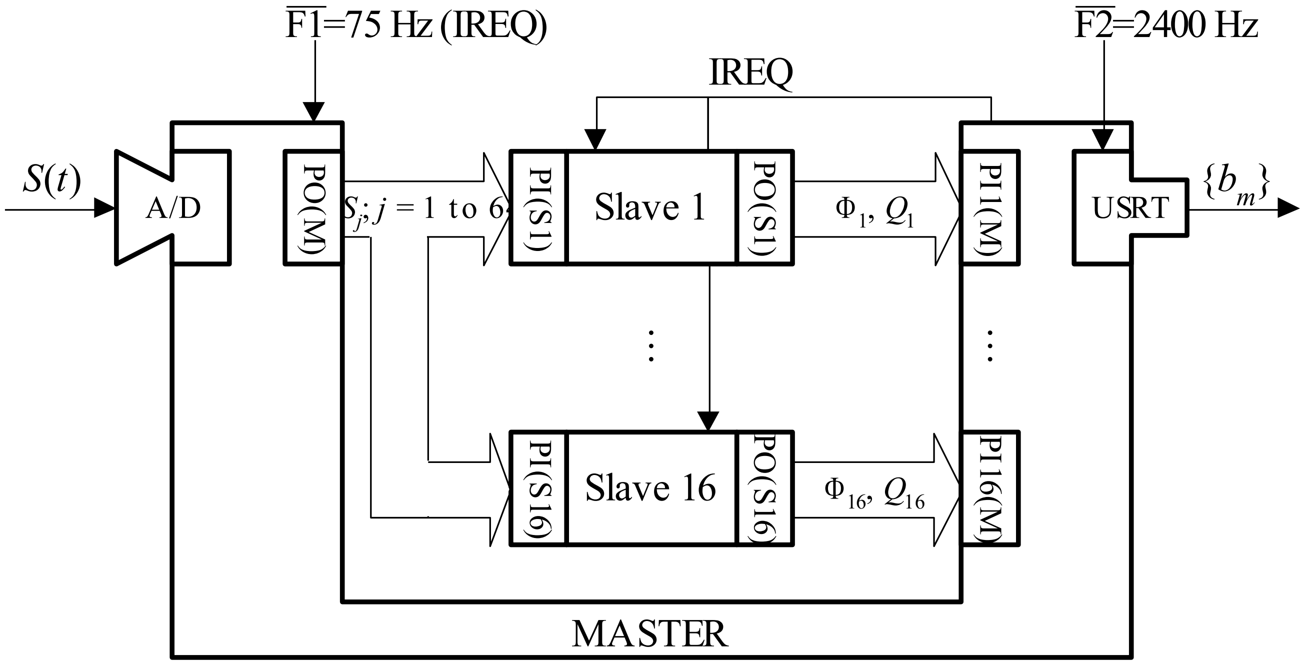
Mid 80s: DARPA's First GaAs Systolic Array 4096 (SIMD)





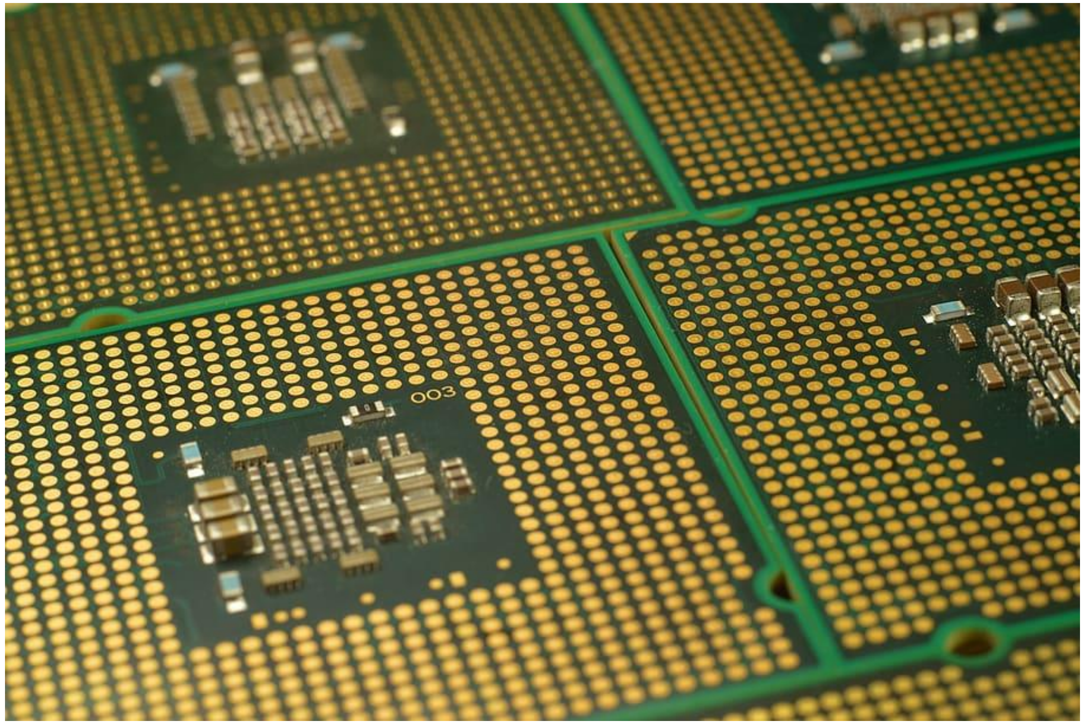
Lesson Learned: Good Mostly for TENSOR CALCULUS!
(now optimal: Google TPU)

Late 80s: The AeroSpace Fourier Transform Engine (MISD)



Lesson Learned: Good Mostly for TRANSFORMATIONAL ALGORITHMS
(now optimal: Intel MultiCore)

End 80s: The NCR Vertical Migration Engine (MIMD)



Lesson Learned: Good Mostly for IMAGE UNDERSTANDING
(now optimal: nVidia ManyCore)

ControlFlow: Inherent Characteristics Represent Limitations

- SISD: IEEE T/Computers

  Helbig, W., & Milutinovic, V. (1989). A dcfl e/d-mesfet gaas experimental risc machine. *IEEE transactions on computers*, *38*(2), 263-274.

- SIMD: IEEE T/Communications

  Milutinovic, V. (1988). A comparison of suboptimal detection algorithms applied to the additive mix of orthogonal sinusoidal signals. *IEEE transactions on communications*, *36*(5), 538-543.

- MISD: IEEE T/ASSP

  Milutinovic, V., Fortes, J., & Jamieson, L. (1986). A multimicroprocessor architecture for real-time computation of a class of DFT algorithms. *IEEE transactions on acoustics, speech, and signal processing*, *34*(5), 1301-1309.

- MIMD: IEEE T/Software

  Milutinovic, V. (1987). A simulation study of the Vertical-Migration microprocessor architecture. *IEEE transactions on software engineering*, (12), 1265-1277.

DataFlow: New Paradigms Badly Needed,
NOT to Substitute,
BUT to Supplement!

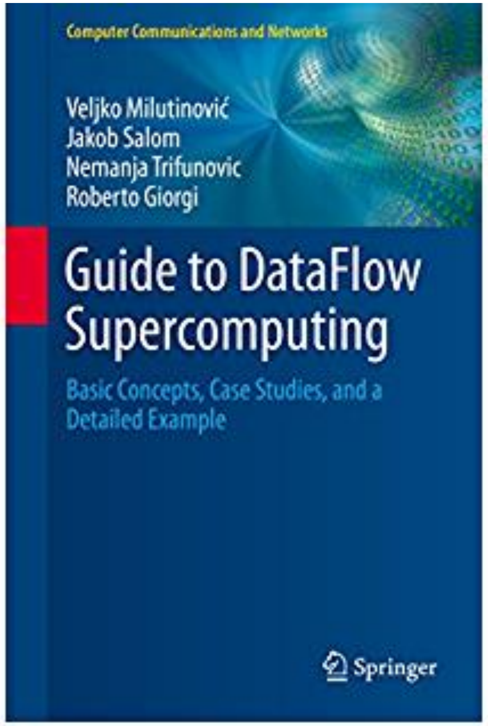
- The First FineGrain DataFlow Idea: Proceedings of the IEEE

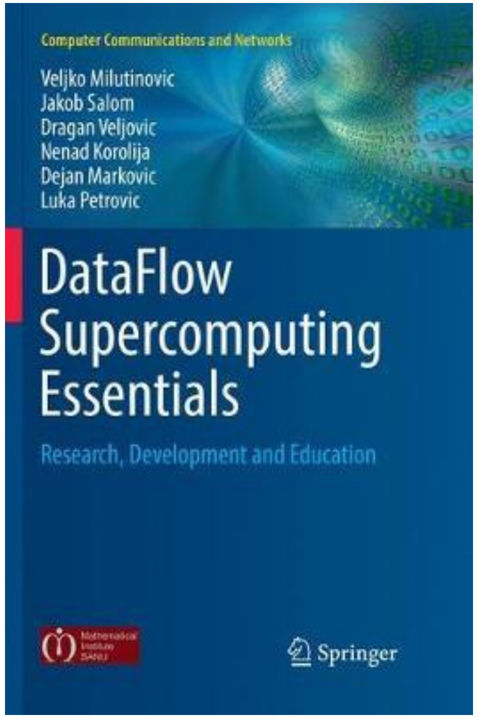
  Milutinovic, V. (1989).
  Mapping of neural networks on the honeycomb architecture.
  Proceedings of the IEEE, *77*(12), 1875-1878.

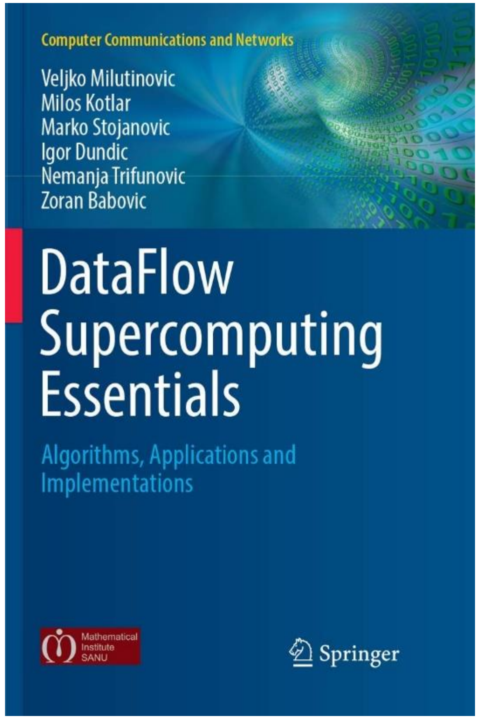
- The First FineGrain DataFlow Implementation: Communications of the ACM

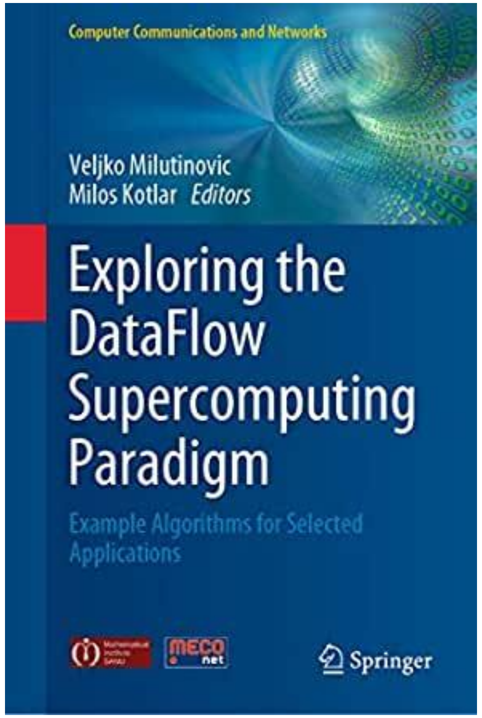
  Flynn, M. J., Mencer, O., Milutinovic, V., Rakocevic, G., Stenstrom, P., Trobec, R., & Valero, M. (2013).
  Moving from petaflops to petadata.
  Communications of the ACM, *56*(5), 39-42.

Conclusion: Synergy of ControlFlow and DataFlow









N.B. Successful marriage is THE solution! 

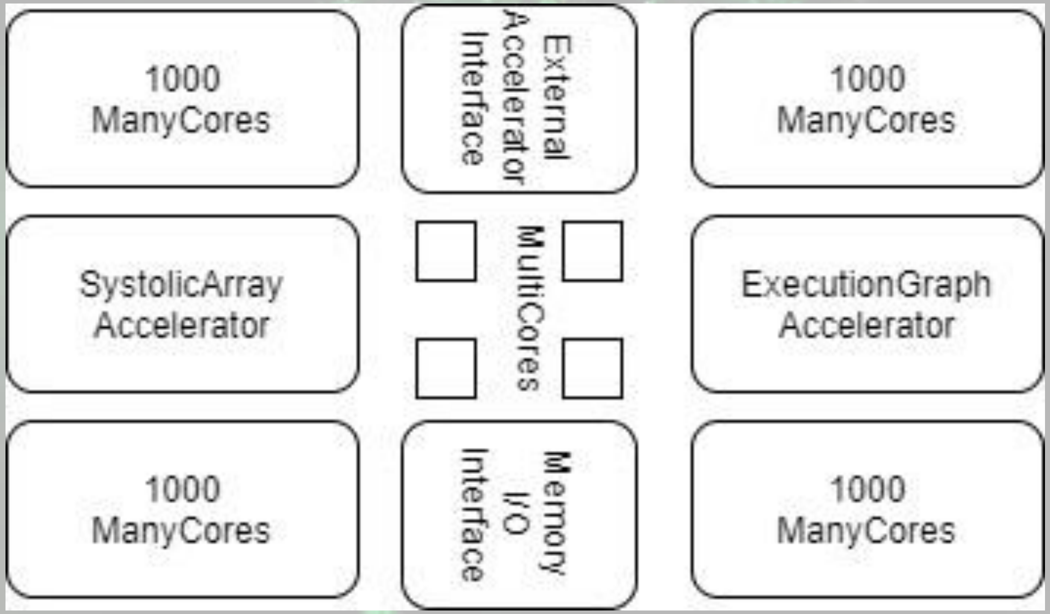
The Optimal Architecture

CF (ControlFlow):
- **N MultiCore CPUs**
- **1000N ManyCore GPUs (10000N, etc…)**

DF (DataFlow):
- **An ASIC SystolicArray (Fixed)**
- **An FPGA ExecutionGraph (Reconfigurable)**

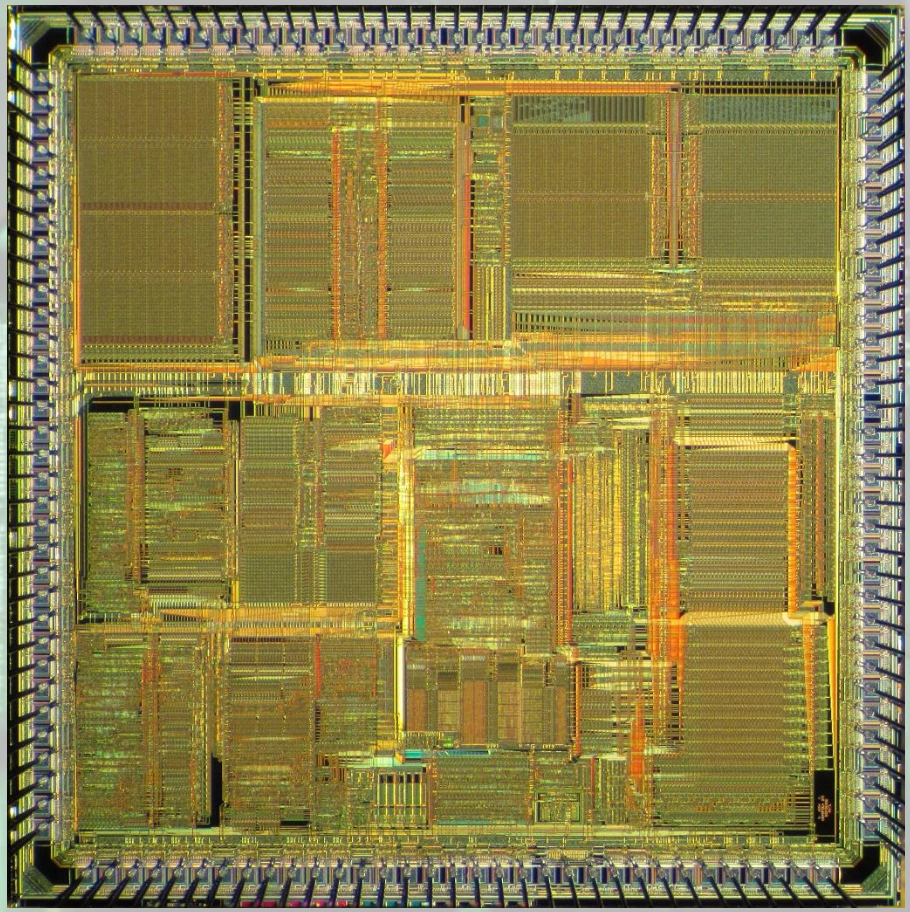
Peripherals to External Accelerators (EnergyFlow):
- **Chemical**
- **Molecular**
- **Opto**
- **Quantum**

+ Memory and I/O + Peripherals to WSN and IoT (DiffusionFlow)



Optimal Distribution of Transistor Budget

- **Ceiling-Dependent:**
  - **VLSI: 100BTr**
  - **WSI: 1TTr**
- **Application-Dependent:**
  - **SW DataFlow**
  - **ML ControlFlow**
- **Strategy-Dependent:**
  - **CF-Oriented**
  - **DF-Oriented**
- **Memory-Dependent!**



Pioneering Efforts:

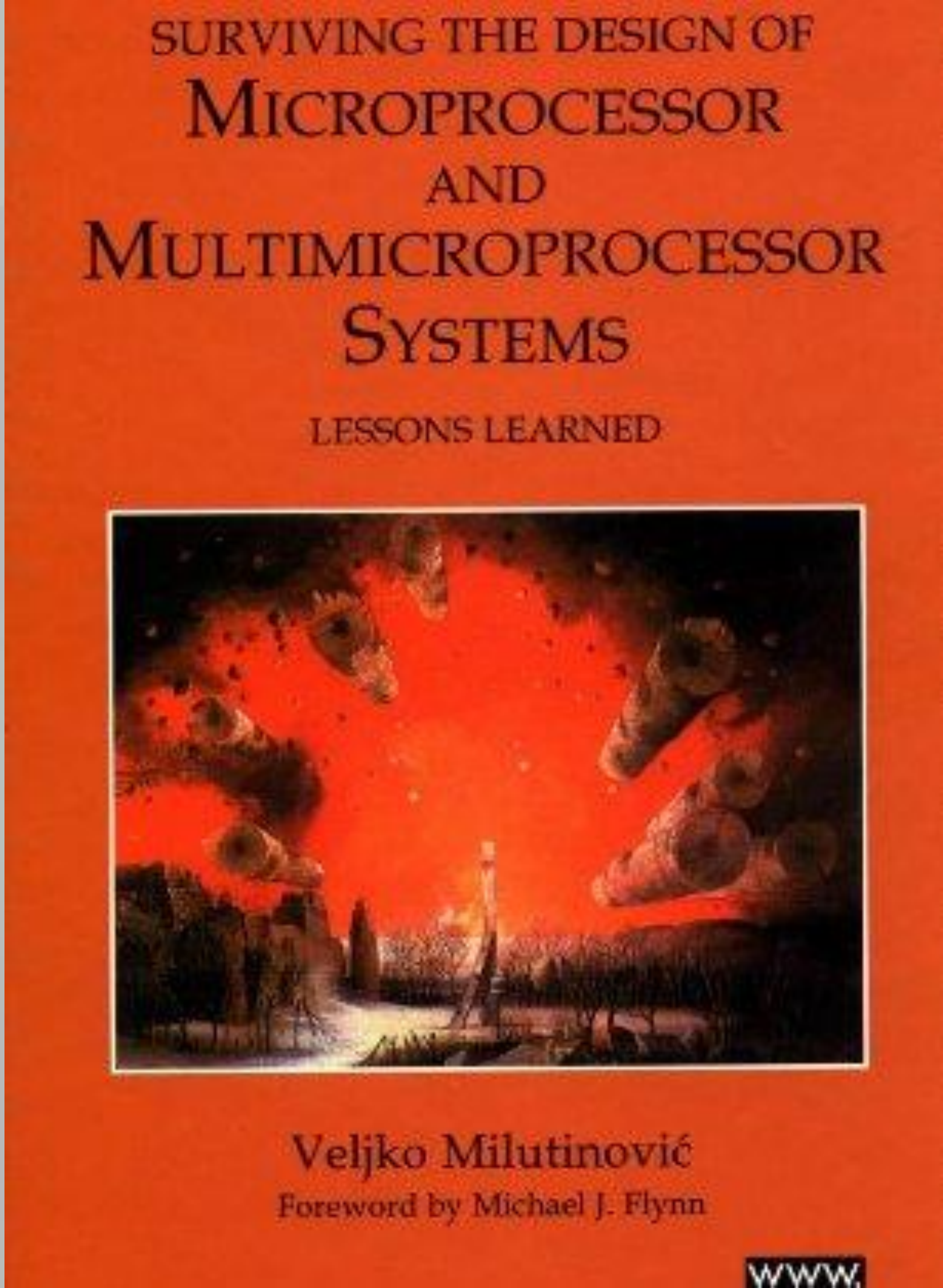
Company	Product name	Country	City	Number of CPU cores	Number of GPU cores	CPU clock rate	Launched
Alibaba	XT910	China	Hangzhou	1/2/4 per cluster	N/A	2.0 – 2.5 GHz	2020
RISC-V	Micro Magic RISC-V Core [39]	US	San Francisco	1	N/A	4.25 – 5.19 GHz	2020
SiFive	FU740 RISC-V SoC [25]	US	San Francisco	4	1	1.4 – 1.5 GHz	2020
AMD	AMD Ryzen™ 5 3400G with Radeon™ RX Vega 11 Graphics [24]	US	Santa Clara	4	11	3.7 – 4.2 GHz	2019
Nvidia	Tegra Xavier [28]	US	Santa Clara	8	512 CUDA	N/A	2019
Esperanto Technologies		US	Mountain View	16 ET-Maxion cores	4096 ET-Minion cores + ET-Graphics [26,27]	2+GHz	2018
Intel	Intel Sandy Bridge [23]	US	Santa Clara	1-4 (4-6 Extreme, 2-8 Xeon)	6	1.60 – 3.60 GHz	2011
Moscow Center of SPARC Technologies (MCST)	Elbrus-2S+ (1891ВМ7Я) [29]	Russia	Moscow	2 Elbrus 2000 cores	4 DSP Elcore-09 cores	300 – 800 MHz	2011

State of the Art:

- **Biren (77BTr@2023) = Chip Integration (China)**
- **Cerebras (1.4TTr@2023) = Wafer Integration (California)**

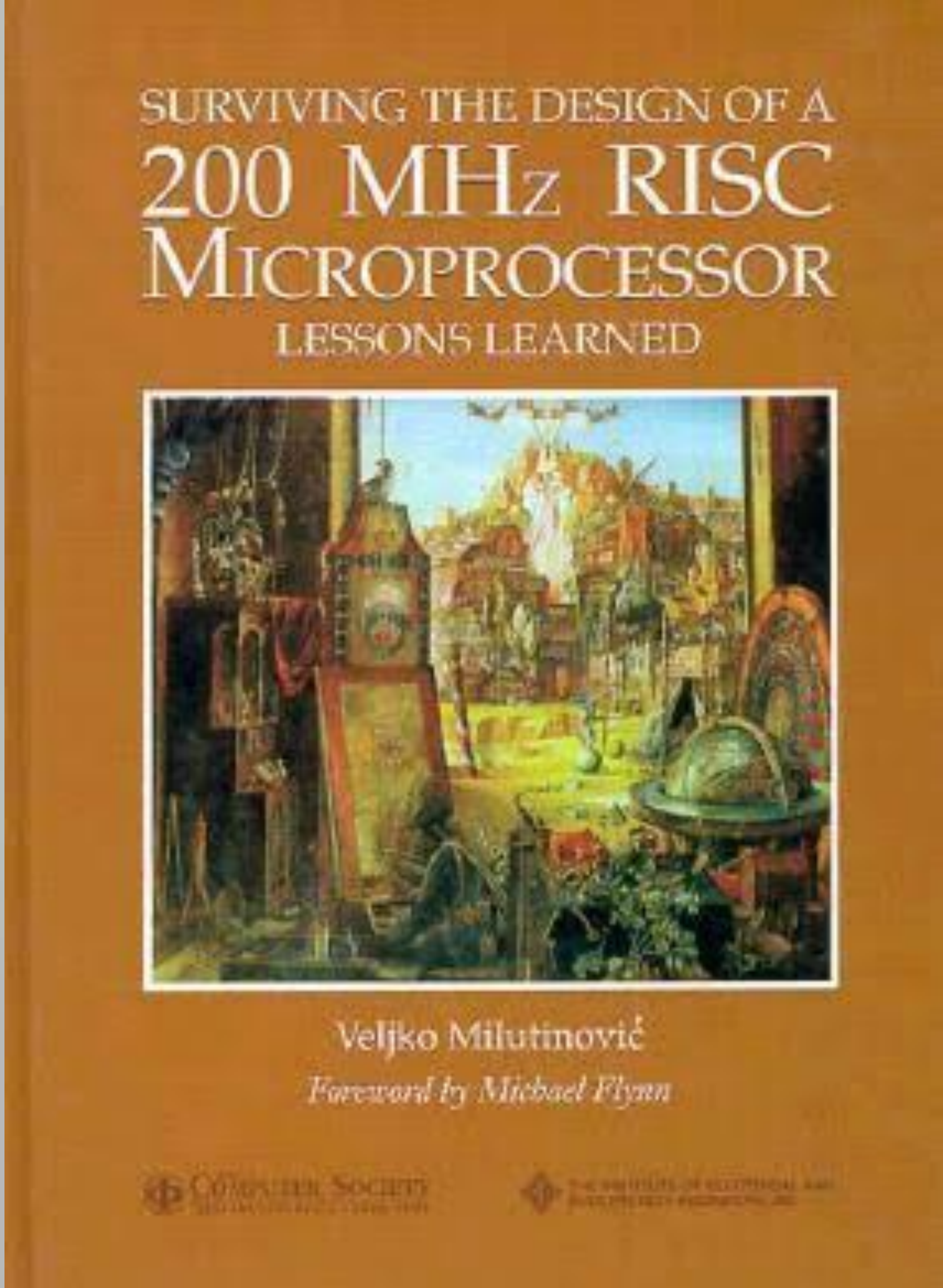
Past Experiences: MultiCore

- **Split Cache**
  - **Milutinovic, V. (1996).**
    The Split Temporal/Spatial Cache:
    Initial Performance Analysis.
    ***SCIzzL-5, March 1996*, 63-69.**



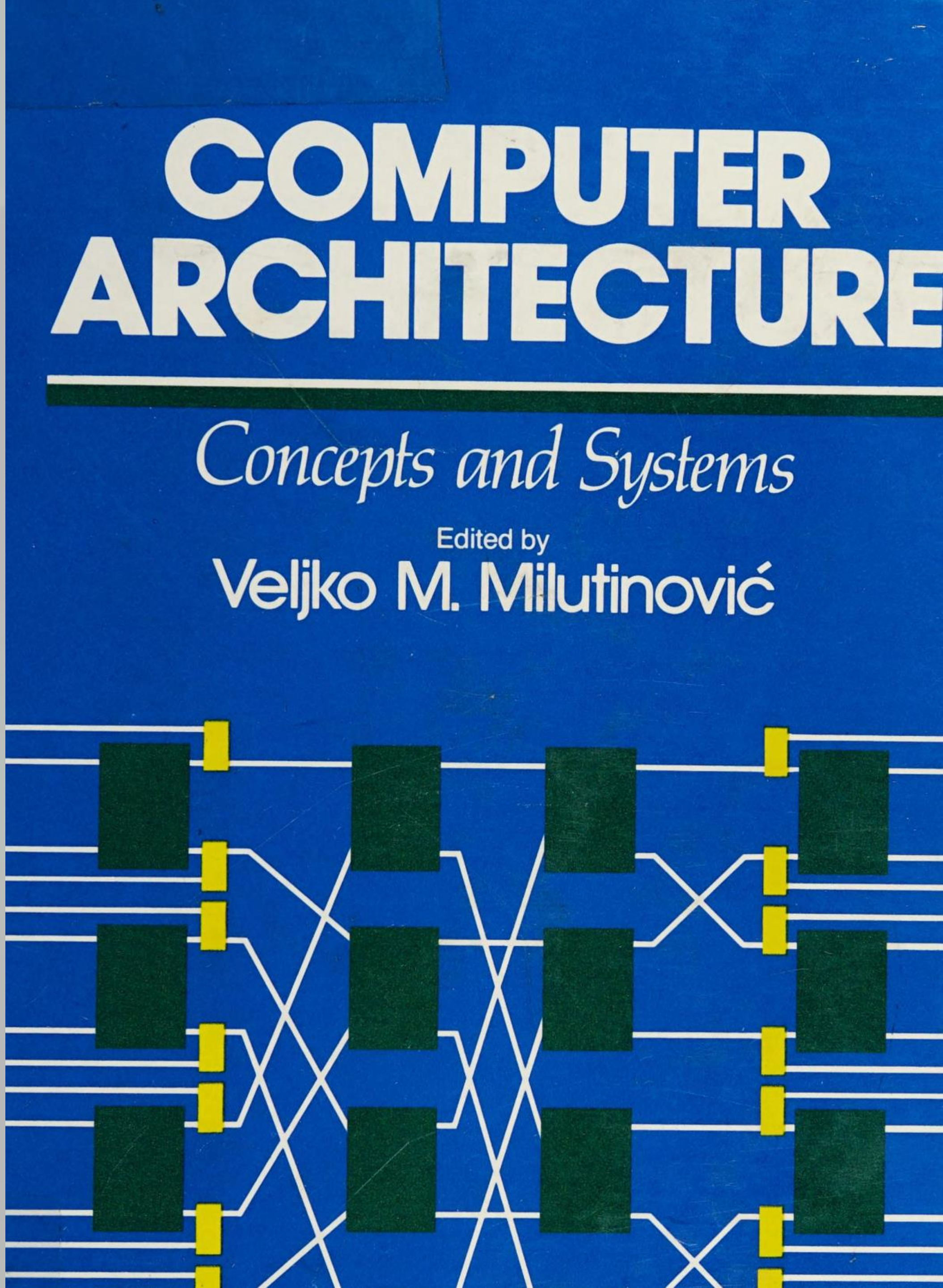
Past Experiences: ManyCore

- **GaAs Microprocessor**
    - **Milutinovic, V., Fura, D., & Helbig, W. (1986).**
      An Introduction To GaAs Microprocessor Architecture for VLSI.
      ***Computer*, (3), 30-42.**



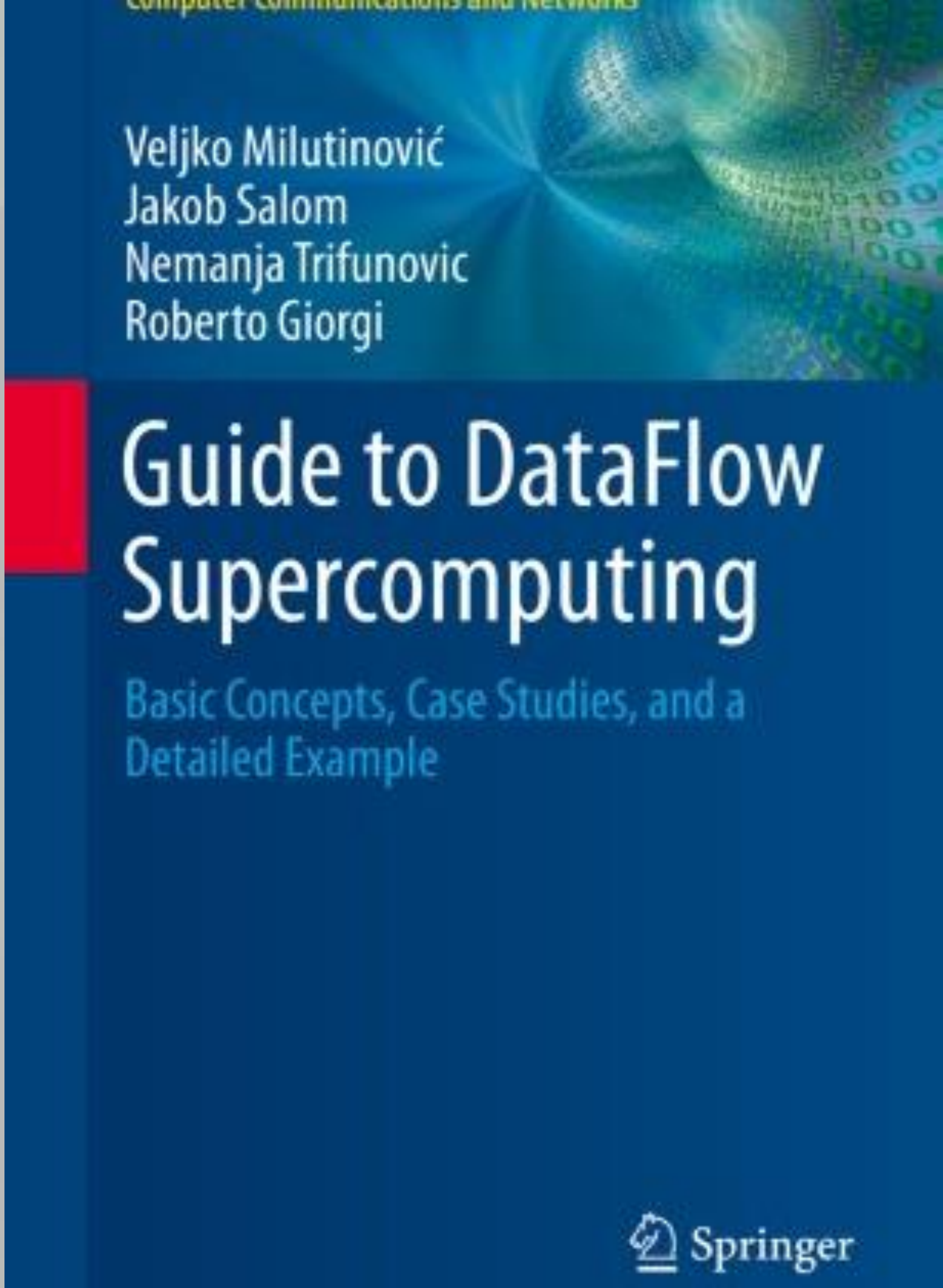
Past Experiences: SystolicArrays

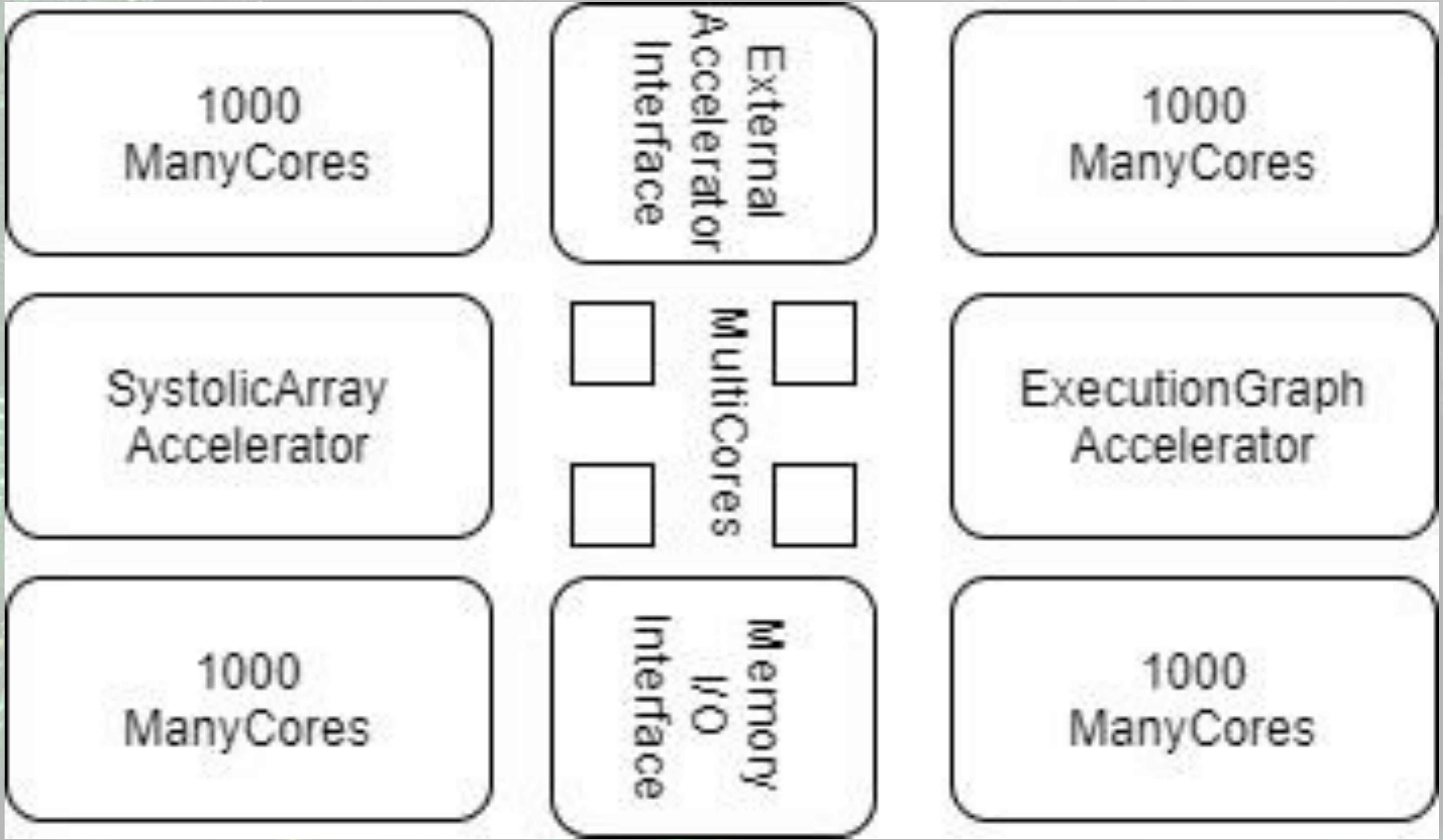
- **GaAs Systolics**
  - **Fortes, J. A., Milutinovic, V., (1986, January). A High-Level Systolic Architecture for GaAs. In Proc. 19th Ann. Hawaii Int'l Conf. System Sciences (pp. 253-258).**



Past Experiences: ExecutionFlow

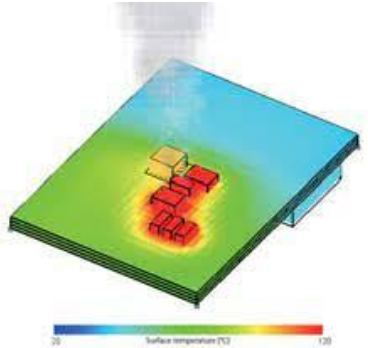
- **DataFlow**
  - **Milutinović, V., Salom, J., Trifunović, N., & Giorgi, R. (2015). Guide to DataFlow Supercomputing. *Cham: Springer Nature*.**

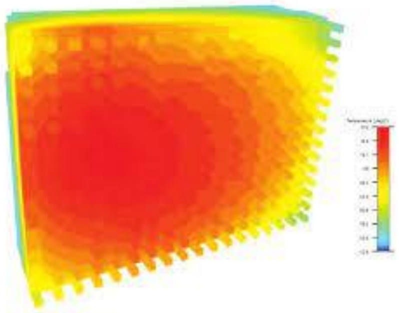



1000 ManyCores
External Accelerator Interface
1000 ManyCores
SystolicArray Accelerator
MultiCores
ExecutionGraph Accelerator
1000 ManyCores
Memory I/O Interface
1000 ManyCores

3D Implementations Offer Effective Cooling

- In a hybrid processor,
  multi-layer semiconductor structure fabrication can spread:
  - MultiCore on the entire chip surface,
  - ManyCore on the entire chip surface, and
  - DataFlow on the entire chip surface.
- Thus, for jobs that use predominantly one paradigm,
  the frequencies of other chiplets could be lowered by:
  - Hardware (automatically) [1], or
  - Software (scheduled) [2].





[1] Lakhani, S., Wang, Y., Milenković, A., & Milutinović, V. (1996). 2D matrix multiplication on a 3D systolic array. Microelectronics journal, 27(1), 11-22.
[2] Milenkovic, A., & Milutinovic, V. (1998). A quantitative analysis of wiring lengths in 2D and 3D VLSI. Microelectronics journal, 29(6), 313-321.

Current Research:

January 31, 2020

The Ultimate DataFlow for Ultimate SuperComputers-on-a-Chips

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Miroslav Kosanic, MIT, Cambridge, MA, USA

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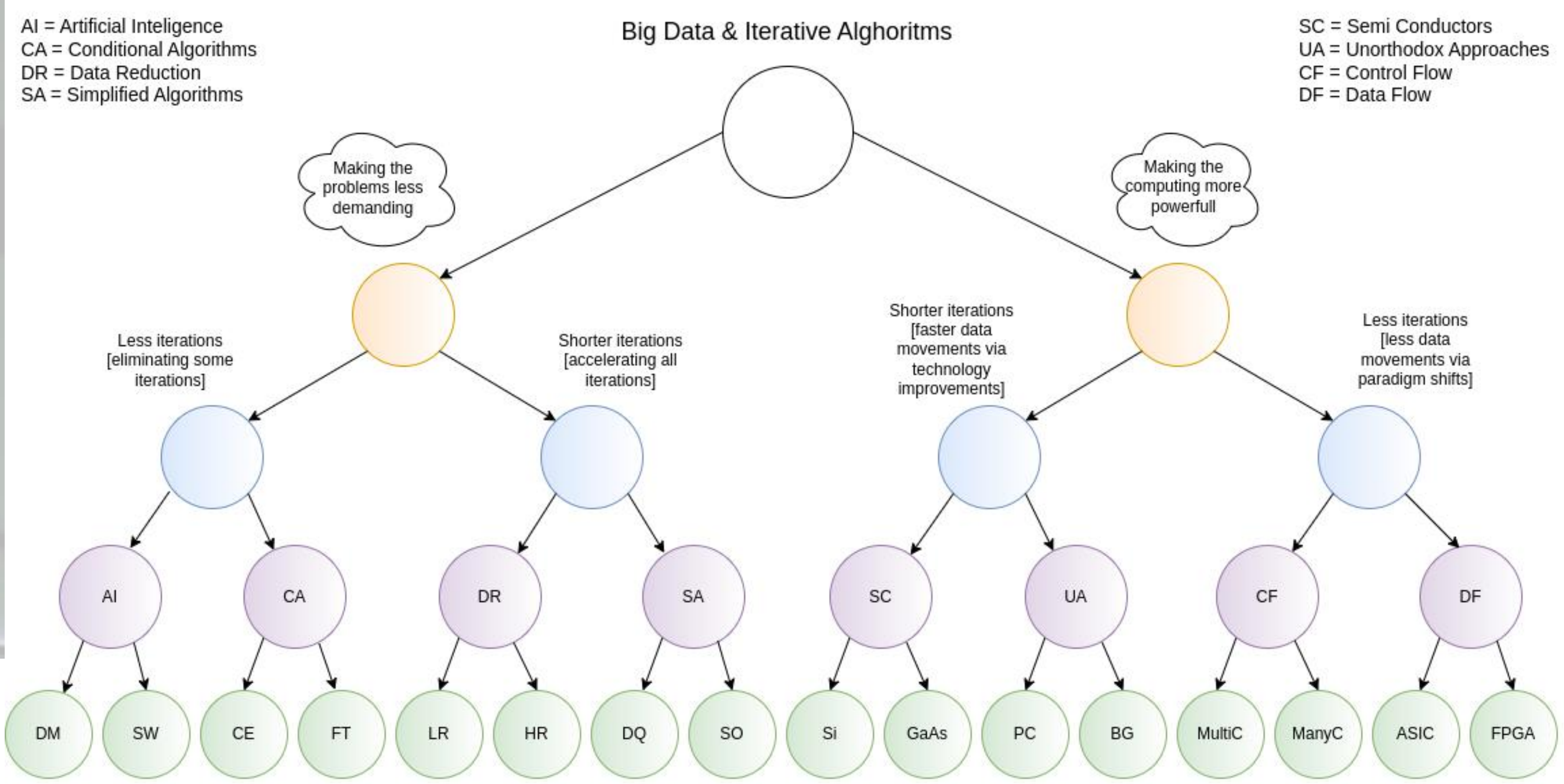
Jelena Skoručak, University of Zurich and ETH, Switzerland

Laura Dipietro, Highland Instruments, Cambridge, MA, USA

Ivan Ratkovic, Esperanto Technologies, Belgrade, Serbia and San Francisco, California, USA

This article starts from the assumption that near future 100BillionTransistor (100BT) SuperComputers-on-a-Chip will include N big multi-core processors, 1000N small many-core processors, an ASIC TPU-like fixed-structure systolic array accelerator for the most frequently used Machine Learning algorithms needed in bandwidth-bound applications and an FPGA flexible-structure re-programmable accelerator for less frequently used Machine Learning



Making the problems less demanding:

DM = Data Mining
SW = Semantic Web
CE = Conditional Execution
FT = Fast Transforms
LR = Low Level Reductions
HR = High Level Reductions
DQ = Data Quantizations (eg. binary arithmetic)
SO = Simplified Operations (eg. no MLTP)

Making the computing more powerfull:

Si = Silicon
GaAs = Gallium Arsenide
PC = Physics/Chemistry
BG = Biology/Genomics
MultiC = Multi Cores
ManyC = Many Cores
ASIC = Aplication Specific Integrated Circuits (eg. Google TPU)
FPGA = Field Programmable Gate Arrays (eg. Maxeler DFE)

Research in computing-intensive simulations for nature-oriented civil-engineering and related scientific fields, using machine learning and big data: an overview of open problems
Zoran Babović, Branislav Bajat, Vladan Đokić, Filip Đorđević, Dražen Drašković, Nenad Filipović, Borko Furht, Nikola Gačić, Igor Ikodinović, Marija Ilić, Ayhan Irfanoglu, Branislav Jelenković, Aleksandar Kartelj, Gerhard Klimeck, Nenad Korolija, Miloš Kotlar, Miloš Kovačević, Vladan Kuzmanović, Marko Marinković, Slobodan Marković, Avi Mendelson, Veljko Milutinović, Aleksandar Nešković, Nataša Nešković, Nenad Mitić, Boško Nikolić, Konstantin Novoselov, Arun Prakash, Ivan Ratković, Zoran Stojadinović, Andrey Ustyuzhanin, Stan Zak
Journal of Big Data volume 10, Article number: 73 (2023)

Handbook of Research on Methodologies and Applications of Supercomputing

Table of Contents #1

Table of Contents #2

Reviews and Testimonials of 8 Nobel Laureates

- "I want to commend Dr. Veljko Milutinovic and Dr. Milos Kotlar for having completed this volume in the timely field of supercomputing in these difficult times, and I want to share their optimism regarding its use as a textbook around the world."
- *– Prof. Kurt Wüthrich, Nobel Laureate, Switzerland*

"Our complex and fast-moving world meets big data issues that call for reliable, efficient analysis and prompt response. A most advanced approach to dealing with these issues is presented in this book where algorithms of Control Flow represent the host architecture of supercomputers and Data Flow represents the acceleration architecture. A comprehensive list of applications is illustrated in the book including natural language processing, medical research, customer-oriented studies, and many more.“

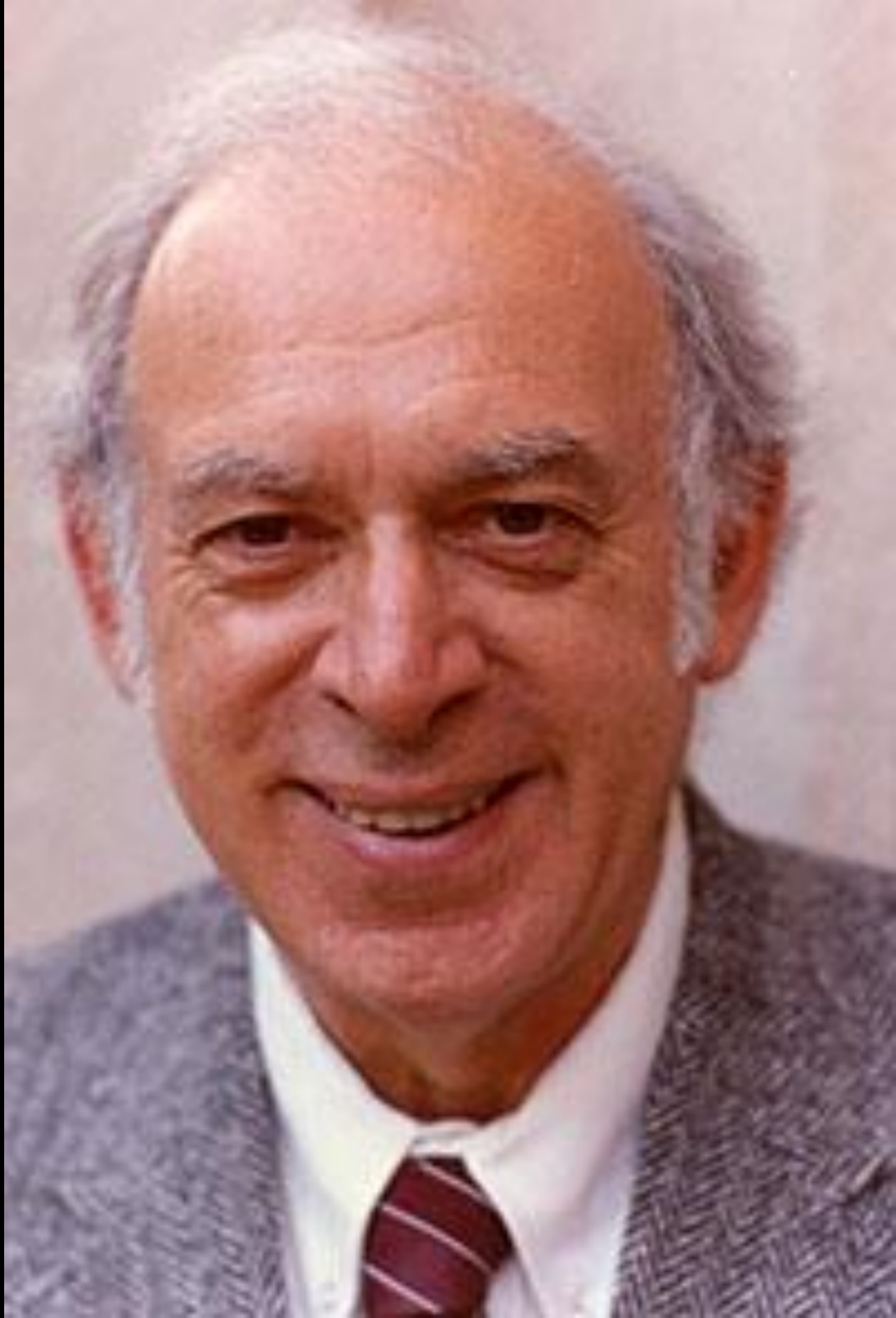
– Prof. Dan Shechtman, Nobel Laureate, Israel

- "Supercomputers have become a ubiquitous instrument in many areas of science and technology. Very hard to imagine modern physics, biology or chemistry without the use of this versatile tool. The breakthroughs in the development of supercomputers expand the range of problems we can tackle. Supercomputers, as well as specialised computers will undoubtedly contribute significantly to the overall landscape of discoveries in many different disciplines in the future."
- *– Prof. Konstantin Novoselov, Nobel Laureate, National University of Singapore, Singapore– Prof. Kurt Wüthrich, Nobel Laureate, Switzerland*

- "It’s clear that computers can do anything, as the pioneers already recognised, and it’s probably only a matter of time before they can perform any of the kinds of tasks we humans take for granted quicker than blinking. I guess this book represents a stage on this exciting and very important journey."
- *– Prof. Tim Hunt, Nobel Laureate, UK*

- "Computers have become essential tools for the pursuit of both experimental and theoretical physics, as well as synthetic chemistry, paleontology, the medical sciences, economics the social sciences and so much more. Science, being the most international of all endeavors, will be well served by this important book, which honors the golden anniversary of the creation of Montinegro’s Academy of Science."
- *– Prof. Sheldon Glashow, Nobel Laureate, Boston University and Harvard University, USA*

- "Humankind’s continuing quests to uncover, understand, and utilize the secrets of nature have been greatly enhanced and will be further extended by the power of supercomputers."
- *– Prof. Jerome I Friedman, Nobel Laureate, USA*



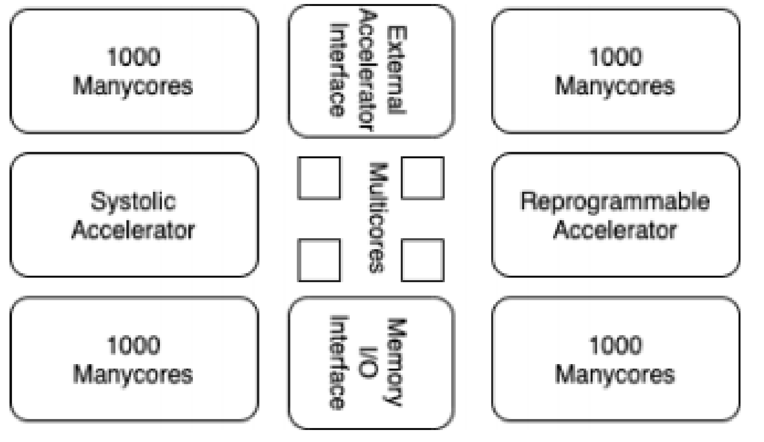
- "I wish to congratulate very warmly the Montenegrin Academy of Arts and Sciences on the occasion of its 50th Anniversary and wish the CANU a highly successful future. Science shapes the Future of Mankind."
- *– Prof. Jean-Marie Lehn, Nobel Laureate, France*

- "Aim high, stay grounded! These four crisp words of wisdom accompany the best wishes for the 50th anniversary of the Montenegrin Academy of Sciences and Arts."
- *– Prof. Stefan Hell, Nobel Laureate, Germany*

Ultimate DataFlow SuperComputing for BigData DeepAnalytics

V. Milutinovic, University of Indiana, Bloomington, USA

Acknowledgements: O. Mencer, Imperial College, London, GBR, **M.J. Flynn,** Stanford University, Palo Alto, USA,
M. Kotlar, School of Electrical Engineering, University of Belgrade, SRB



Chip Hardware Type	**Estimated Transistor Count**
One Manycore with Memory	3.29 million
4000 Manycores with Memory	11 800 million [17]
One Multicore with Memory	1 billion [18]
4 Multicore with Memory	4 billion
One Systolic Array	<1 billion [19]
One Reprogrammable Ultimate Dataflow	<69 billion [20]
Interface to I/O with external Memory	<100 million
Interface to External Accelerators	<100 million
TOTAL	<100 billion

Table 1. Basically, current efforts include about 30 billion transistors on a chip, and this article advocates that, for future 100 billion transistor chips, the most effective resources to include are those based on the dataflow principle. For some important applications, such resources bring significant speedups, that would fully justify the incorporation of additional 70 billion transistors. The speedups could be, in reality, from about 10x to about 100x, and the explanations follow in the rest of this article.

Major Sources of Inspiration

A. Richard Feynman:

Impact of logic/arithmetic and memory/IO

Compiler-generated execution graph

B. Ilya Prigogine:

Impact of energy, entropy, order, and optimization

Compiler-generated data separation

C. Daniel Kahneman:

Impact of approximate computing on precision

Compiler-controlled approx computing

D. Tim Hunt:

Impact of system latency on precision

Compiler-controlled system latency

The Major Axiom of Optimal Computing

A. Whenever the Technology changes,
the Fundamental Paradigm of Computer Architecture
has to change, too.

aSoG (not: FPGA)

B. If several paradigms are available,
the most suitable paradigm for adoption
is the one most effective for modern Applications.

BrontoData (not: ExaBigData)

Is the von Neumann Paradigm still the most effective one?

A. MultiCores?

B. ManyCores?

The Holy Trinity of Generalized Computing

Applications

Architecture

-Size
-Power
-Speedup
-Precision

Technology

The von Neumann Paradigm (1940s)

$$\lim_{i \to \infty} \left(\frac{TALU(i)}{TCOMM(i)} \right) \to \infty$$

Optimal Solution: Finite Automata

The Nobel Laureate Richard Feynman Observations

$$\lim_{i \to \infty} \left(\frac{TALU(i)}{TCOMM(i)} \right) \to 0(t \to \infty)$$

Where is the technology now?

A. Closer to 1940s?

B. Closer to t->∞?

State of the Art in Technology Today

~~The Power Challenge~~

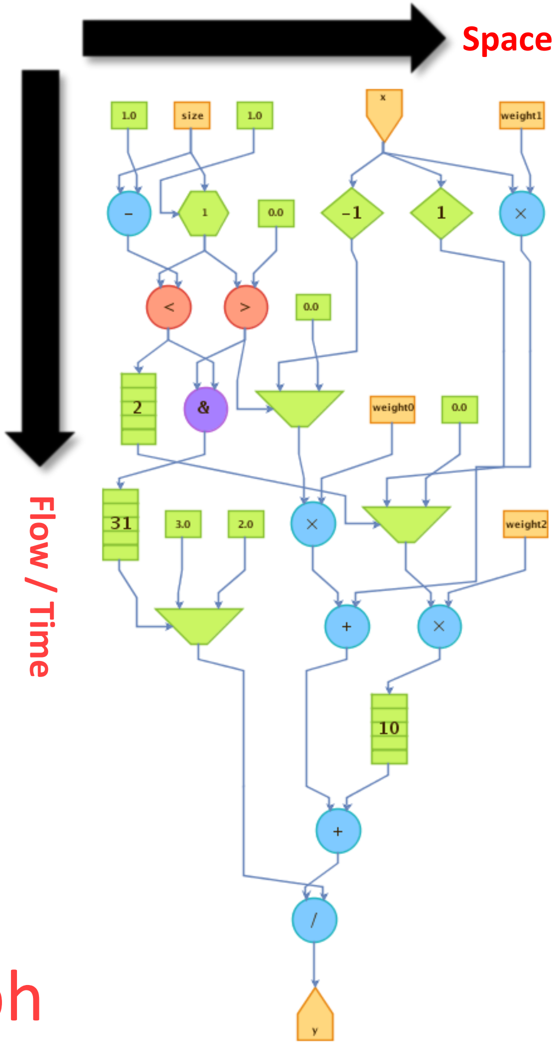
The Data Movement Challenge

	2015	2020
Double precision FLOP	100pj	10pj

- Moving data off-chip will use **200x more energy than computing**!
- Moving data in 1940s was using **1/60x** ...
- Conclusion: We are getting close to the Feynman Asymptote!
- Important: Power and speed could be traded!

The Maxeler Technology Vision: MultiScale DataFlow

- ❑Thinking in space rather than in time
- ❑Difficult change in mindset to overcome
- ❑Transformation of data through flow over time
- ❑Instructions are parallelized across the available space



Optimal Solution: Execution Graph

Comparing the Two Approaches

- The Von-Neumann paradigm resembles an old wall clock

- The Feynman paradigm resembles lightning! Why?

Programming the Two Paradigms

von Neumann:

The Program Moves Data

Feynman:

The Program Configures Hardware

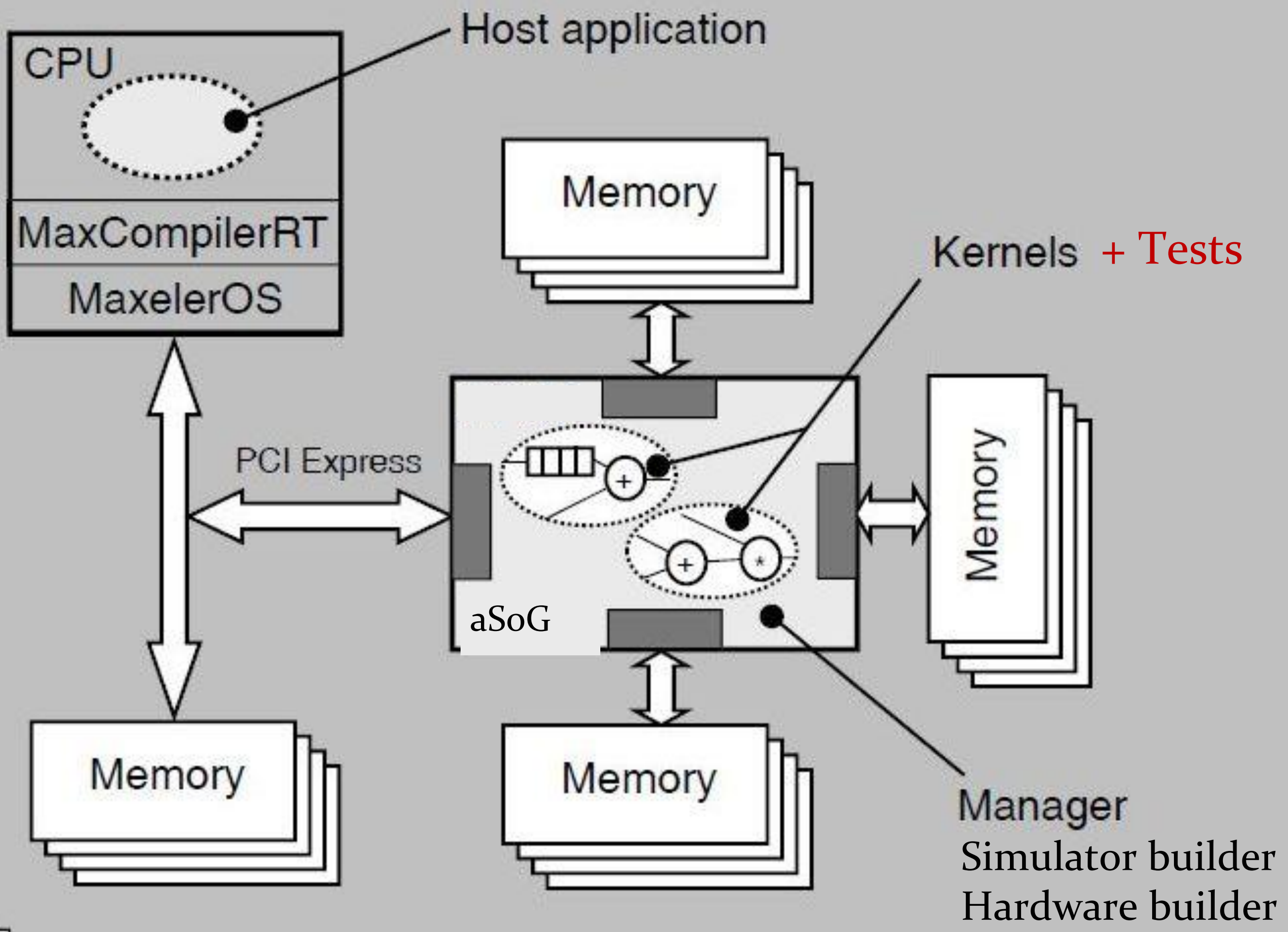
What moves data?

External sources till input.

Voltage difference through aSoG!

Voltage difference moves the important stuff!

The Maxeler Generic Architecture Application



Important: Supporting any CL and any OS!

2n+3

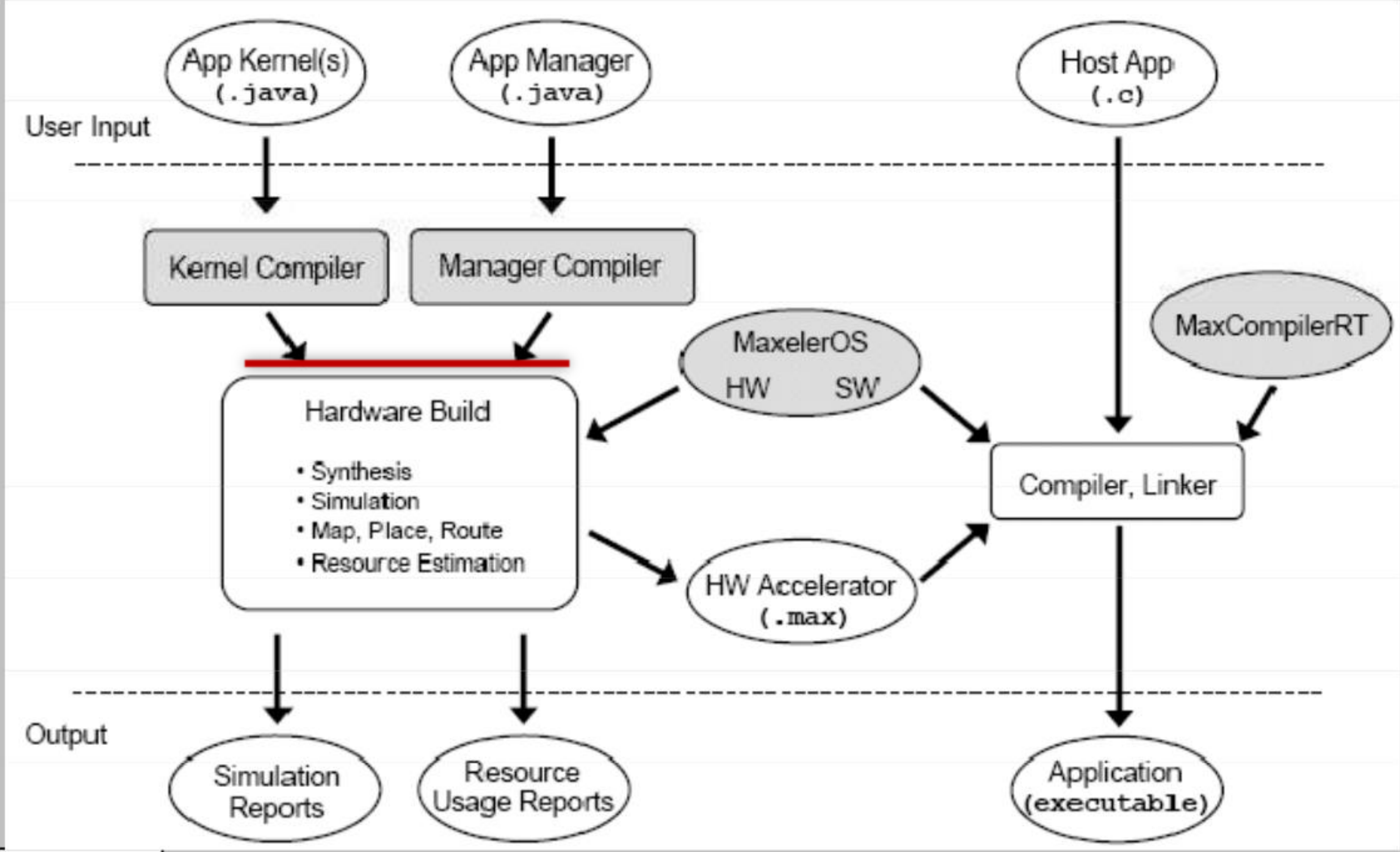
Why The Acceleration Approach?

Nobel Laureate Ilya Prigogine:
Injecting Energy to Decrease Entropy!

Corollary:
Burning energy to split spatial and temporal
decreases the entropy of computing
and enables the DataFlow compiler
to create a maximally effective execution graph.

Final goal:
The execution graph with the minimal length of edges.

MaxCompiler



MAXELER Technologies
MAXIMUM PERFORMANCE COMPUTING

Alliances Being Formed

- Intel acquired Altera
- Qualcomm and IBM teaming up with Xilinx

However:

C	C	C
OpenCL(I)	OpenCL(A)	MaxCompiler
Intel	Altera	Altera@MaxZ
1	X>>1	Y>>X

Nano Accelerators

- Invisible on the DataFlow Concept Level
- Invisible to DataFlow Programmers
- Visible to the MaxCompiler
- The MaxCompiler knows how to utilize them

Best protected by two aSoG (now FPGA) protection levels
and two Vendor (e.g., Maxeler) protection levels!

Publications of Interest for NanoAcceleration

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1. *Flynn, M., Mencer, O., Milutinovic, V., Rakocevic, G., Stenstrom, P., Trobec, R., Valero, M.,*
   Moving from Petaflops (on Simple Benchmarks) to Petadata per Unit of Time and Power (On Sophisticated Benchmarks),
   Communications of the ACM (nano-acceleration), May 2013.

2. *Trobec, R., Vasiljevic, R., Tomasevic, M., Milutinovic, V., Beiveide, M., Valero, M.,*
   Interconnection Networks for SuperComputing,
   ACM Computing Surveys (nano-acceleration) , 2017.

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3. *Milutinovic, V., Tomasevic, M., Markovic, B., Tremblay, M.,*
   The Split Temporal/Spatial Cache: Initial Performance Analysis,
   Proceedings of the SCIzzL-5, Santa Clara, California, USA, March 26, 1996, pp 72-78.

4. *Milutinovic, V., Tomasevic, M., Markovic, B., Tremblay, M.,*
   The Split Temporal/Spatial Cache: Initial Complexity Analysis,
   Proceedings of the SCIzzL-5, Santa Clara, California, USA, September, 1996.

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5. *Milutinovic, V.,*
   A Comparison of Suboptimal Detection Algorithms Applied to the Additive Mix of Orthogonal Sinusoidal Signals,
   IEEE Transactions on Communications, Vol. COM-36, No. 5, May 1988, pp. 538-543.

6. *Milutinovic, V.,*
   Mapping of Neural Networks on the Honeycomb Architectures,
   Proceedings of the IEEE, Vol. 77, No 12, December 1989, pp. 1875-1878.

----

7. *Helbig, W., Milutinovic, V.,*
   The RCA's DCFL E/D MESFET GaAs 32-bit Experimental RISC Machine,
   IEEE Transactions on Computers, Vol. 36, No. 2, February 1989, pp. 263-274.

8. *Jovanovic, Z., Milutinovic, V.,*
   FPGA Accelerator for Floating-Point Matrix Multiplication,
   IEE Computers & Digital Techniques (nano-acceleration), 2012, 6, (4), pp. 249-256.
   The IET 2014 Premium Award for Computing & Digital Techniques.

==

Inspired by:
Feynman

Inspired by:
Prigogine

Inspired by:
Kahneman

Inspired by:
Hunt

Feynman

proceedings OF THE IEEE

THE INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS december 198

98c

$$l = \log_2\left(N_{pe}+1\right)$$

$$N_{total}(l) = (l-1)\left(2^{l-1}+2^{l-2}\right)$$

$$N_{bus}(l) = (l-2)\left(2^{l-2}+2^{l-3}\right)$$

$$N_{idle}(l) = (l-1)\left(2^{l-1}+2^{l-2}\right)-(l-2)\left(2^{l-2}+2^{l-3}\right)-\left(2^{l}-1\right)$$

$$N_{pe}(l) = 2^{l}-1$$

$$N_{total}\left(N_{pe}\right) = \left[\log_2\left(N_{pe}+1\right)-1\right]\left[\frac{N_{pe}+1}{2}+\frac{N_{pe}+1}{2^2}\right]$$

$$N_{total}\left(N_{pe}\right) = \left[\log_2\left(N_{pe}+1\right)-1\right]\left[\frac{3N_{pe}+3}{4}\right]$$

$$N_{total}\left(N_{pe}\right) = \frac{3}{4}\left(N_{pe}+1\right)\left[\log_2\left(N_{pe}+1\right)-1\right]$$

$$N_{bus}(l) = \frac{3}{8}\left(N_{pe}+1\right)\left[\log_2\left(N_{pe}+1\right)-2\right]$$

$$N_{idle}\left(N_{pe}\right) = \frac{3}{4}\left(N_{pe}+1\right)\left[\log_2\left(N_{pe}+1\right)-1\right]-\frac{3}{8}\left(N_{pe}+1\right)\left[\log_2\left(N_{pe}+1\right)-2\right]-N_{pe}$$

$$N_{idle}(l) = \frac{3}{8}\left(N_{pe}+1\right)\left[\log_2\left(N_{pe}+1\right)\right]-\frac{1}{8}\left(11N_{pe}+3\right)$$

F.4.17.1

$$U(l) = \frac{N_{pe}(l)+N_{bus}(l)}{N_{total}(l)} = \frac{2^{l}-1+(l-2)\left(2^{l-2}+2^{l-3}\right)}{(l-1)\left(2^{l-1}+2^{l-2}\right)}$$

$$U\left(N_{pe}\right) = \frac{N_{pe}+\frac{3}{8}\left(N_{pe}+1\right)\left[\log_2\left(N_{pe}+1\right)-2\right]}{\frac{3}{4}\left(N_{pe}+1\right)\left[\log_2\left(N_{pe}+1\right)-1\right]}$$

f.4.18.1

Prigogine

TABLE II
COMPARISON OF PERFORMANCE OF FFT/SIMD AND DFT/MISD

	FFT/SIMD	DFT/MISD
T	$2 \log N + wN$	$\max \{wN + 1, N + w\}$
H	$(3N/2) - 2$	lN
C	$cN/2 + qc(N - 2)$	clN
E	$\frac{(\log 4\, lN) - 2l}{2 \log N + wN}$	$\frac{\log 4lN) - 2l}{2l \max \{wN + 1, N + w\}}$

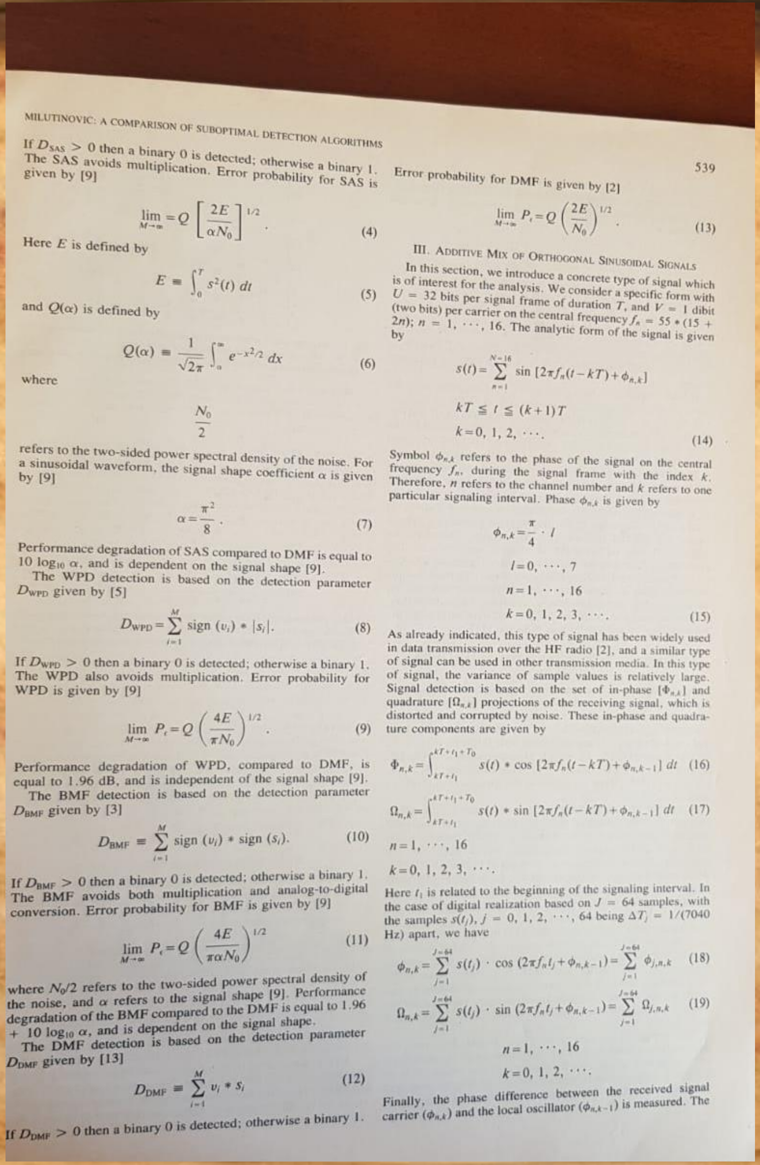
$$\sum_{i=0}^{\log N - 1} c_i = \sum_{i=0}^{X-1} 2^i L + \sum_{i=X}^{\log N - 1} N/2$$

$$= L(2^X - 1) + \frac{N}{2} \log (N - X)$$

$$= (N/2)(\log 4L) - L.$$

The serial execution time of the FFT is then $((N/2) \log 4L - L)$ which is always smaller than LN, the serial execution time of the DFT, for any positive values of N and L. Therefore, it is the best serial execution time. This statement presumes that the units of time are equally defined for both the FFT and DFT algorithms. As we mentioned before, this is not true in our analysis and the resulting error favors the FFT algorithm. Given that the DFT and FFT differ ... efficiency

Kahneman



If $D_{SAS} > 0$ then a binary 0 is detected; otherwise a binary 1. The SAS avoids multiplication. Error probability for SAS is given by [9]

$$\lim_{M\to\infty} = Q\left[\frac{2E}{\alpha N_0}\right]^{1/2}. \quad (4)$$

Here E is defined by

$$E \equiv \int_0^T s^2(t)\,dt \quad (5)$$

and $Q(\alpha)$ is defined by

$$Q(\alpha) \equiv \frac{1}{\sqrt{2\pi}}\int_\alpha^\infty e^{-x^2/2}\,dx \quad (6)$$

where

$$\frac{N_0}{2}$$

refers to the two-sided power spectral density of the noise. For a sinusoidal waveform, the signal shape coefficient α is given by [9]

$$\alpha = \frac{\pi^2}{8}. \quad (7)$$

Performance degradation of SAS compared to DMF is equal to $10 \log_{10} \alpha$, and is dependent on the signal shape [9].

The WPD detection is based on the detection parameter D_{WPD} given by [5]

$$D_{WPD} = \sum_{i=1}^{M} \text{sign}\,(v_i) * |s_i|. \quad (8)$$

If $D_{WPD} > 0$ then a binary 0 is detected; otherwise a binary 1. The WPD also avoids multiplication. Error probability for WPD is given by [9]

$$\lim_{M\to\infty} P_e = Q\left(\frac{4E}{\pi N_0}\right)^{1/2}. \quad (9)$$

Performance degradation of WPD, compared to DMF, is equal to 1.96 dB, and is independent of the signal shape [9].

The BMF detection is based on the detection parameter D_{BMF} given by [3]

$$D_{BMF} \equiv \sum_{i=1}^{M} \text{sign}\,(v_i) * \text{sign}\,(s_i). \quad (10)$$

If $D_{BMF} > 0$ then a binary 0 is detected; otherwise a binary 1. The BMF avoids both multiplication and analog-to-digital conversion. Error probability for BMF is given by [9]

$$\lim_{M\to\infty} P_e = Q\left(\frac{4E}{\pi\alpha N_0}\right)^{1/2} \quad (11)$$

where $N_0/2$ refers to the two-sided power spectral density of the noise, and α refers to the signal shape [9]. Performance degradation of the BMF compared to the DMF is equal to 1.96 + $10 \log_{10} \alpha$, and is dependent on the signal shape.

The DMF detection is based on the detection parameter D_{DMF} given by [13]

$$D_{DMF} \equiv \sum_{i=1}^{M} v_i * s_i \quad (12)$$

If $D_{DMF} > 0$ then a binary 0 is detected; otherwise a binary 1. Error probability for DMF is given by [2]

$$\lim_{M\to\infty} P_e = Q\left(\frac{2E}{N_0}\right)^{1/2}. \quad (13)$$

III. Additive Mix of Orthogonal Sinusoidal Signals

In this section, we introduce a concrete type of signal which is of interest for the analysis. We consider a specific form with $U = 32$ bits per signal frame of duration T, and $V = 1$ dibit (two bits) per carrier on the central frequency $f_n = 55 * (15 + 2n)$; $n = 1, \cdots, 16$. The analytic form of the signal is given by

$$s(t) = \sum_{n=1}^{N=16} \sin\,[2\pi f_n(t - kT) + \phi_{n,k}]$$
$$kT \leqq t \leqq (k+1)T$$
$$k = 0, 1, 2, \cdots. \quad (14)$$

Symbol $\phi_{n,k}$ refers to the phase of the signal on the central frequency f_n, during the signal frame with the index k. Therefore, n refers to the channel number and k refers to one particular signaling interval. Phase $\phi_{n,k}$ is given by

$$\phi_{n,k} = \frac{\pi}{4} \cdot l$$
$$l = 0, \cdots, 7$$
$$n = 1, \cdots, 16$$
$$k = 0, 1, 2, 3, \cdots. \quad (15)$$

As already indicated, this type of signal has been widely used in data transmission over the HF radio [2], and a similar type of signal can be used in other transmission media. In this type of signal, the variance of sample values is relatively large. Signal detection is based on the set of in-phase $[\Phi_{n,k}]$ and quadrature $[\Omega_{n,k}]$ projections of the receiving signal, which is distorted and corrupted by noise. These in-phase and quadrature components are given by

$$\Phi_{n,k} = \int_{kT+t_1}^{kT+t_1+T_0} s(t) * \cos\,[2\pi f_n(t-kT) + \phi_{n,k-1}]\,dt \quad (16)$$

$$\Omega_{n,k} = \int_{kT+t_1}^{kT+t_1+T_0} s(t) * \sin\,[2\pi f_n(t-kT) + \phi_{n,k-1}]\,dt \quad (17)$$

$$n = 1, \cdots, 16$$
$$k = 0, 1, 2, 3, \cdots.$$

Here t_1 is related to the beginning of the signaling interval. In the case of digital realization based on $J = 64$ samples, with the samples $s(t_j)$, $j = 0, 1, 2, \cdots, 64$ being $\Delta T_j = 1/(7040$ Hz) apart, we have

$$\phi_{n,k} = \sum_{j=1}^{J=64} s(t_j) \cdot \cos\,(2\pi f_n t_j + \phi_{n,k-1}) = \sum_{j=1}^{J=64} \phi_{j,n,k} \quad (18)$$

$$\Omega_{n,k} = \sum_{j=1}^{J=64} s(t_j) \cdot \sin\,(2\pi f_n t_j + \phi_{n,k-1}) = \sum_{j=1}^{J=64} \Omega_{j,n,k} \quad (19)$$

$$n = 1, \cdots, 16$$
$$k = 0, 1, 2, \cdots.$$

Finally, the phase difference between the received signal carrier $(\phi_{n,k})$ and the local oscillator $(\phi_{n,k-1})$ is measured. The

Hunt

MILUTINOVIĆ *et al.*: GaAs-BASED MICROPROCESSOR ARCHITECTURE FOR REAL-TIME

In the case of DO looping, as indicated in Fig. 5, the computation of the new value for the loop control variable is followed by both writing the new value back into the multiport data memory, and testing the condition. The longer of these two operations will determine the total execution time.

$$T_{\mathrm{DO}} = \begin{cases} T_{\mathrm{GOTO}} + T_{A/C}(K_{A/C}=1,\, K_B;\, N_{\mathrm{OPER}} = N^{\mathrm{eff}}_{\mathrm{OPER}}); \\ \qquad (N_{G/\mathrm{RELA}} + N_{G/BR}) * T_G > T_{MM} \\ T_{\mathrm{GOTO}} + T_{A/C}(K_{A/C}=0;\, N_{\mathrm{OPER}} = N^{\mathrm{eff}}_{\mathrm{OPER}}); \\ \qquad (N_{G/\mathrm{RELA}} + N_{G/BR}) * T_G \le T_{MM} \end{cases} \tag{15}$$

where $N_{G/\mathrm{RELA}}$ is the number of gate delays for the unit which determines the value of the HLL relation, and $N^{\mathrm{eff}}_{\mathrm{OPER}} = \max\{(N_{OPER/I} + 1), N_{\mathrm{OPER}/F}, (N_{\mathrm{OPER}/S} + 1)\}$. Symbols I, F, and S refer to INITIAL, FINAL, and STEP expressions in the primary control statement. By this we conclude the derivation of execution-time formulas for assignments and control constructs.

Now we concentrate on execution-time formulas for call/return and low-level I/O. We assume the same number of input and output ports of the multport data memory (N_{MM}), and certain number of input and output subroutine parameters (N_{PARA}; $N_{\mathrm{PARA}} \neq N_{MM}$). Consequently,

$$\begin{aligned} T_{\mathrm{CALL/RETURN}} = T_{\mathrm{CALL}} + T_{\mathrm{RETURN}} = 2 \\ * \Bigg[T_{PM} + \left(\max \left\{ 1, \left\lceil \frac{N_{\mathrm{PARA}}}{N_{MM}} \right\rceil \right\} - 1 \right) \\ * \max\{T_F,\, 2 * T_{MM} + T_G\} \\ + \max\, [N_{G/BR} * T_G],\, [\mathrm{sgn}\,(N_{\mathrm{PARA}}) \\ * (2 * T_{MM} + T_G)]\}\Bigg] \end{aligned} \tag{16}$$

https://www.maxeler.com/moving-from-petaflops-to-petadata/

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Moving from Petaflops to Petadata

May 5, 2013

VIEWPOINT: Moving from Petaflops to Petadata

M. Flynn[‡‖], *O. Mencer*[‡Ŧ], *V. Milutinovic*[†], *G. Rakocevic*[§], *P. Stenstrom*[ϐ], *R.Trobec*[þ], *M. Valero*[Ⅎ]

[‡]Maxeler Technologies, [‖]Stanford University, [Ŧ] Imperial College London, [†]University of Belgrade, [§]Mathematical Institute of the Serbian Academy of Sciences and Arts in Belgrade, [ϐ] Chalmers University of Technology, [þ]Jožef Stefan Institute, [Ⅎ]Barcelona Supercomputing Centre

Communications of the ACM, Vol. 56 No. 5 May 2013, doi: 10.1145/2447976.2447989

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IF(2012)=3.80

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ACM Computing Surveys

From Wikipedia, the free encyclopedia

ACM Computing Surveys (CSUR) is a peer reviewed scientific journal published by the Association for Computing Machinery. The journal publishes survey articles and tutorials related to computer science and computing. It was founded in 1969; the first editor-in-chief was William S. Dorn.[1]

In ISI Journal Citation Reports, *ACM Computing Surveys* has the highest impact factor among all computer science journals.[2] In a 2008 ranking of computer science journals, *ACM Computing Surveys* received the highest rank "A*".[3]

ACM Computing Surveys

Abbreviated title (ISO 4)	*ACM Comput. Surv.*
Discipline	Computer science
Language	English
Edited by	Sartaj K Sahni
Publication details	
Publisher	ACM (United States)
Publication history	1969–present
Frequency	Quarterly
Indexing	
ISSN	0360-0300 (print) 1557-7341 (web)
Links	

- Journal homepage
- Online access
- Online archive

See also [edit]

- *ACM Computing Reviews*

References [edit]

1. ^ Dorn, William S. (1969). "Editor's Preview...". *ACM Computing Surveys*. **1** (1): 2–5. doi:10.1145/356540.356542.
2. ^ "Journal Citation Reports". *ISI Web of Knowledge*. Retrieved 2009-10-03. "JCR Science Edition 2008"; subject categories "COMPUTER SCIENCE, ...".
3. ^ "Journal Rankings". *CORE: The Computing Research and Education Association of Australasia*. July 2008. Archived from the original on 29 March 2010. Retrieved 2010-03-19..

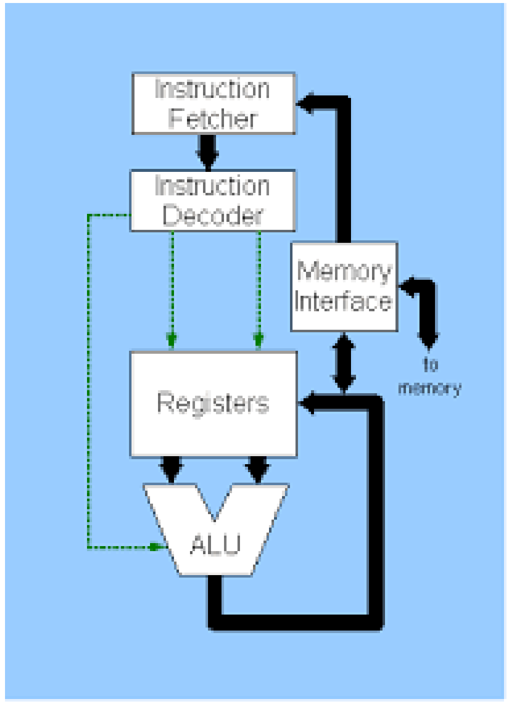
External links [edit]

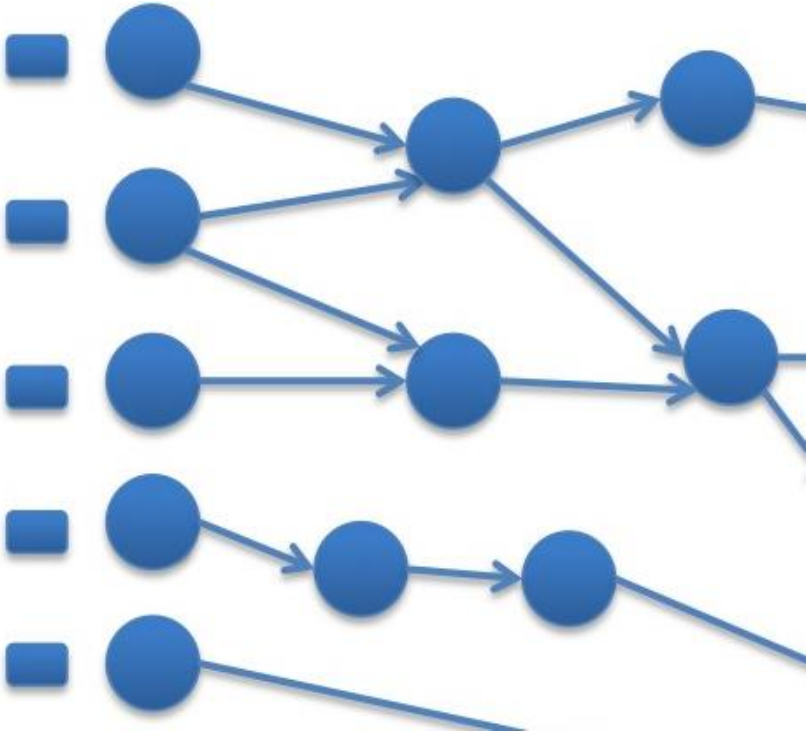
- *ACM Computing Surveys* home page.
- *ACM Computing Surveys* in ACM Digital Library.
- *ACM Computing Surveys* in DBLP.

IF (2007) = 15

Essence: Feynman Enabled by Prigogine

- TALU possible at zero power (Arithmetic+Logic)
- TCOMM not possible at zero power (MEM+MPS)





Essence: Feynman

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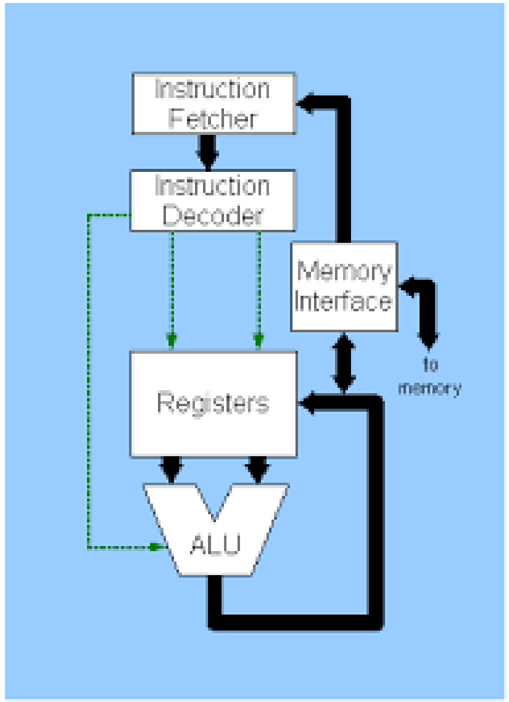

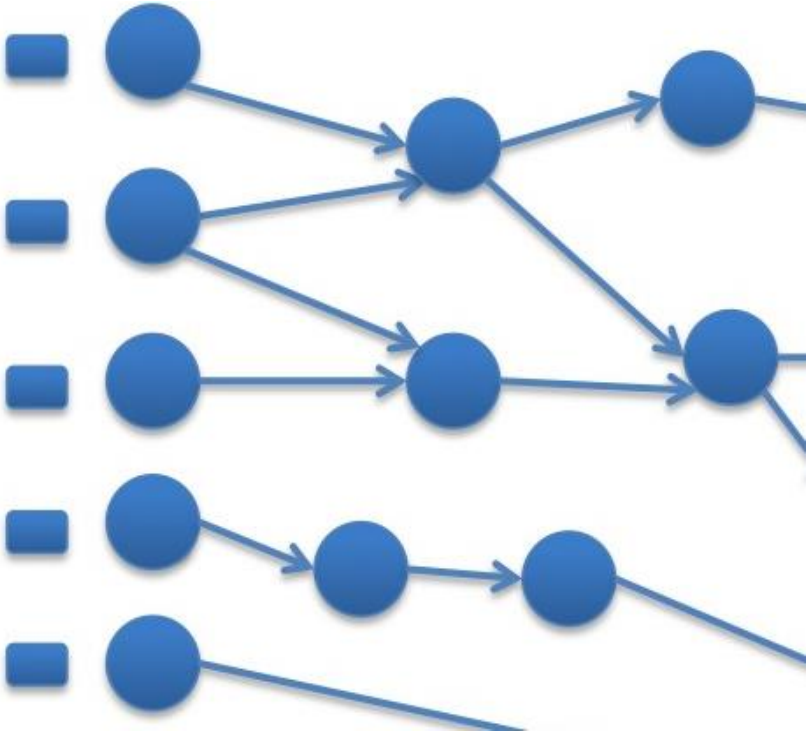

Essence: Feynman

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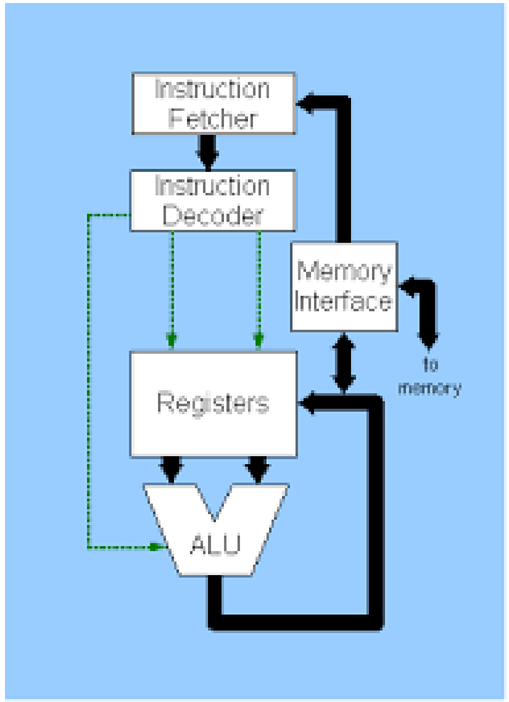
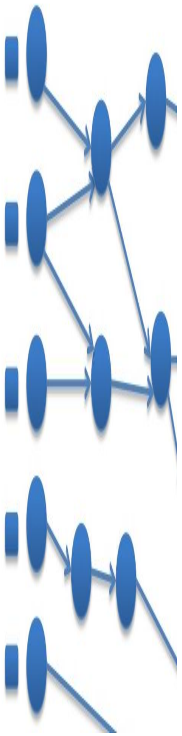




Programming the Maxeler Technology
Generic Acceleration Architecture

MaxJ, the Maxeler Java,

a DSL acting as a SuperSet of classical Java:
- A. A vector of built-in domain-specific classes
- B. Two sets of variables: SW + HW

MaxJ is a SubSet of OpenSPL,
created by the Imperial-Stanford-Tokyo-Tsinghua consortium.

Possible Future Mutations of OpenSPL:
- MaxPython and/or MaxR
- (lower Kolmogorov complexity)
- MaxHaskel and/or MaxScala
- (easier extension to approximate computing).

Approximate Computing for Better Precision: Kahneman

Note: Small approximations in one domain
may bring large benefits in another domain

Example: Weather forecast

A 15-bit computational precision (rather than the 64-bit precision)
may decrease the forecast precision for only 2%,
and at the same time,
may increase the grid precision 25 times,
and the forecast precision at grid intersections up to 10^4.

Easily doable in DataFlow, difficult to do in ControlFlow.

Delayed Decision for Better Precision: Hunt

Note: Small latencies in time domain
may bring large benefits in precision domains

Example: Optimal utilization of internal DataFlow pipelines

Compiler optimizations create internal pipelines
that experienced DataFlow programmers know how to utilize

BigDataAnalytics

Existing Maxeler-based publications:

20 [Size]
20 [Power]
20, 200 [Speedup]
20 [Precision]

Ultimate aSoG-based future:

20-200 [Size]
20-200 [Power]
20, 200, 2000, 20000 [Speedup]
20+ [Precision]

Applications

Architecture

Technology

Maxeler Dataflow Appliance

40U: Good for Cloud

- Software Based Solution
- Dataflow Computing in the Datacentre

MPC C Series

The CPU

Conventional CPU cores and up to 6 DFEs with 288GB of RAM

1U: Good for Fog

MPC X Series

The Dataflow Appliance

Dense compute with 8 DFEs, 768GB of RAM and dynamic allocation of DFEs to CPU servers with zero-copy RDMA access

MPC N Series

The Networking Appliance

Intel Xeon CPUs and 4 DFEs with direct links to up to twelve 40Gbit Ethernet connections

MicroMAX.5: Good for Dew (Edge Processing of IoT Data)

MAXELER Technologies

The Major Application Successes

- **Finances:**
  - Credit derivatives
  - Risk assessment
  - Stability of economical systems
  - Evaluation of econo-political mechanisms
- **GeoPhysics:**
  - Oil&Gas
  - Weather forecast
  - Astronomy
  - Climate changes
- **Science:**
  - Physics
  - Chemistry
  - Biology
  - Genomics
- **Engineering: Synergy of all the Above (ML, etc...)**

Technology ignites our business. Be the spark.

J.P.Morgan

Innovation in Investment Banking Technology

Field Programmable Gate Arrays (FPGAs)

A Field Programmable Gate Array (FPGA) is **a silicon chip containing a matrix of configurable logic blocks (CLBs) that are connected through programmable interconnects.** By combining optimized use of available silicon with fine-grained parallelism, sustained acceleration improvements of over 300x can be achieved across a range of vanilla and complex mathematical models. The current work is the first time that FPGA technology has been employed at this scale to accelerate computational performance anywhere in the finance industry.

Power and Versatility

- Can accelerate performance by between 100 and 1,000x across a range of mathematical models, with the ability to perform a task in less than a second
- Can be reprogrammed and precisely configured to compute exact algorithm(s) at the desired level of numerical accuracy required by any given application, unlike normal microprocessors whose design is fixed by the manufacturer
- Can be deeply pipelined to achieve maximum parallelism from arithmetic, algorithms and data streaming

Key Business Challenges

- Reduce the execution time of existing applications to meet business and regulatory demands
- Decrease cost of running existing applications and developing new ones
- Provide fast, cost-effective extra computational capacity to address problems that are currently inextricable
- Achieve a step-change improvement in price-performance and end-to-end compute time across many applications

Key Benefits (Business/Clients)

- Competitive advantage to valuation, execution, risk management and complex scenario analyses by speeding up existing applications
- Lower cost of existing applications as hardware costs can be reduced by a factor between 100 and 1,000
- Ability to perform previously difficult calculations, such as complex trading strategies or risk evaluations of global portfolio simulations.

Technology Overview

- Low clock speed chips
- Maximal usage of available silicon resources
- Acceleration through use of fine-grained parallelism
- Reconfigurable hardware
- Silicon configurable to fit algorithm

LOB/Function(s) Impacted

- Credit & interest rates
- Equities & commodities
- Loan & mortgage modeling
- Finance & accounting
- High frequency trading
- Risk management & VaR

Industry/External Recognition

- Used by Cisco in all routers
- Simulation of real and theoretical systems
- Geophysics for oil and gas exploration
- Astrophysics & hydrodynamics
- Defense for cryptography
- Video games
- Genotyping

Technology ignites our business. Be the spark.

J.P.Morgan

Functionality Overview

Double precision floating point-capable FPGAs became commercially available in 2002, but it was the arrival of the Virtex 5 and 6 series chips from market leader Xilinx that really provided the scale required for the development of production-grade accelerated solutions. Using FPGAs in high performance compute solutions provides distinct advantages over conventional CPU clusters.

Operational Advantages

- Significantly increases performance for two main types of applications: those based around highly complex mathematical models and those using simpler algorithms that can be massively parallelized
- Enables a dramatic increase in compute density per cubic meter by using FPGAs as computational accelerators
- Consumes around 1% of the power of a single CPU core

Performance Improvements

- Performance improvements in the range 200-300x faster than the existing CPU cores used on the Compute BackBone (CBB) have been achieved in credit and interest rates hybrids businesses
- In equities, direct market access can run risk and loan stock at wire speed (3.5 micro secs) using a low-latency FPGA solution
- Benchmarked average throughput for J.P. Morgan's existing 40-node hybrid FPGA machine of 984MFlops/watt/cubic meter
- Potential standing at the top of the Green-500 ecological global supercomputer performance table

Development/Delivery

Timeline

- Initial porting of an algorithm can vary from one to three months depending on complexity.
- Production capabilities then depend on the scale of the application and the scope and intensity of the testing and reconciliation cycle

Partners

- London-based Applied Analytics group: includes three technology and business specialists with extensive experience in developing and delivering high performance solutions across a range of asset classes, models and lines of business
- Maxeler Technologies: external consultants trained in Imperial College, Stanford and MIT research labs

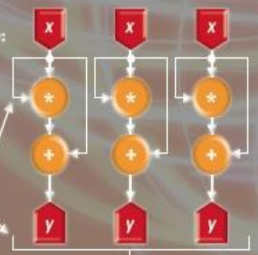
FPGAs at Work

- An algorithm is implemented as a special configuration of a general purpose electric circuit
- Connections between prefabricated wires are programmable
- Function of calculating elements is itself programmable
- FPGAs are two dimensional matrix-structures of configurable logic blocks (CLBs) surrounded by input/output blocks that enable communication with the rest of the environment

A very simple sample:

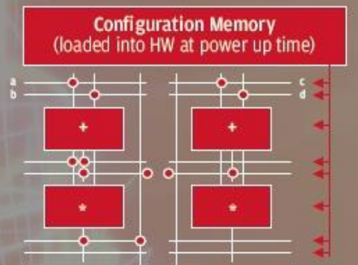
$f(x) = 2x + x$

Moving from a single calculation to a fine grained parallelism



A slightly more complex example:

$e = (a+b)*(c+d)$



Migrating algorithms from C++ to FPGAs involves doing a Fourier Transform from time domain execution to spatial domain execution in order to maximize computational throughput. **It's a paradigm shift to stream computing that provides acceleration of up to 1,000x compared to an Intel CPU.**

The know-how needed for deep security!

DSF Analytics – CME Group

www.cmegroup.com/trading/interest-rates/dsf-analytics.html

Designed for educational use only using Maxeler Technologies' curve construction methodology. This tool uses delayed data and displayed results are indicative representations only.

Please hover your mouse pointer over column titles and links for further information.

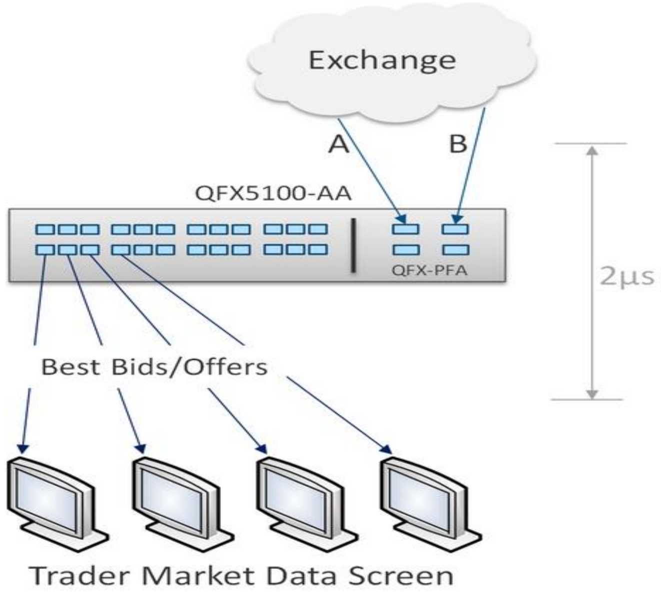
CME Ticker	Bloomberg Ticker	DSF Pricing					Timestamp
		Price	Coupon	PV01	NPV	Implied Rate	
T1UM4 2Y	CTPM4	100'057	0.750%	$19.97	$179.69	0.6600%	4:00:03 PM CT 4/4/2014
F1UM4 5Y	CFPM4	100'115	2.000%	$48.49	$359.38	1.9259%	4:00:03 PM CT 4/4/2014
N1UM4 10Y	CNPM4	100'225	3.000%	$90.16	$703.12	2.9220%	4:00:03 PM CT 4/4/2014
B1UM4 30Y	CBPM4	102'270	3.750%	$195.07	$2,843.75	3.6042%	4:00:03 PM CT 4/4/2014
T1UU4 2Y	CTPU4	100'085	1.000%	$19.93	$265.62	0.8668%	4:00:03 PM CT 4/4/2014
F1UU4 5Y	CFPU4	100'110	2.250%	$48.27	$343.75	2.1788%	4:00:03 PM CT 4/4/2014
N1UU4 10Y	CNPU4	101'125	3.250%	$89.55	$1,390.62	3.0948%	4:00:03 PM CT 4/4/2014
B1UU4 30Y	CBPU4	106'020	4.000%	$193.47	$6,062.50	3.6868%	4:00:03 PM CT 4/4/2014

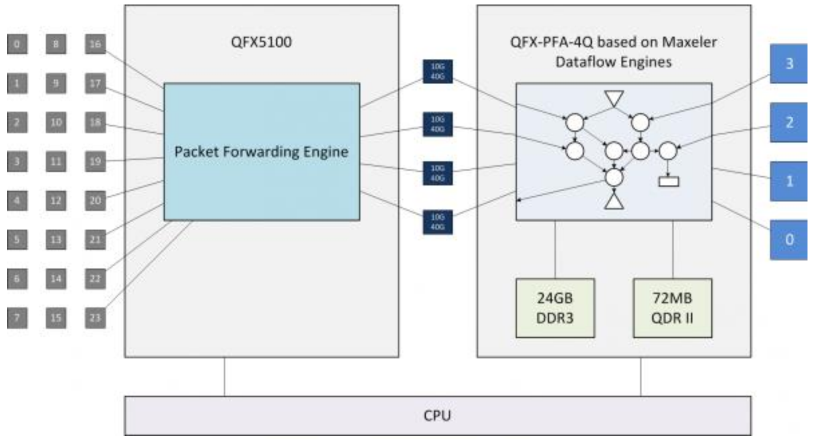
Quotes and analytics are updated every 15 minutes.

Analytics powered by Maxeler Technologies©

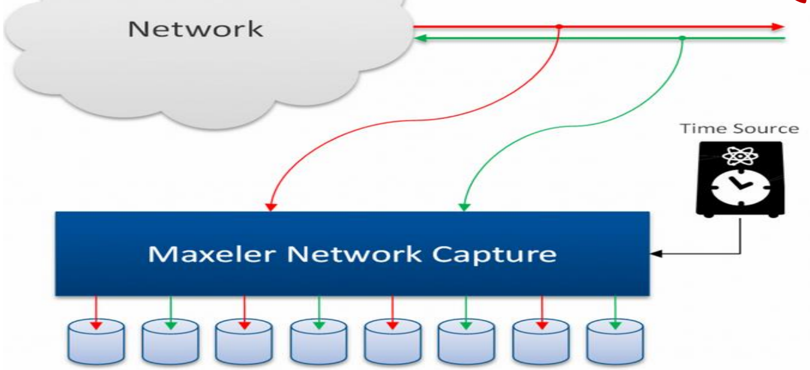
Instrument	CPU 1U-Node	Max 1U-Node	Comparison
European Swaptions	848,000	35,544,000	42x
American Options	38,400,000	720,000,000	19x
European Options	32,000,000	7,080,000,000	221x
Bermudan Swaptions	296	6,666	23x
Vanilla Swaps	176,000	32,800,000	186x
CDS	432,000	13,904,000	32x
CDS Bootstrap	14,000	872,000	62x

Juniper for Online Trading

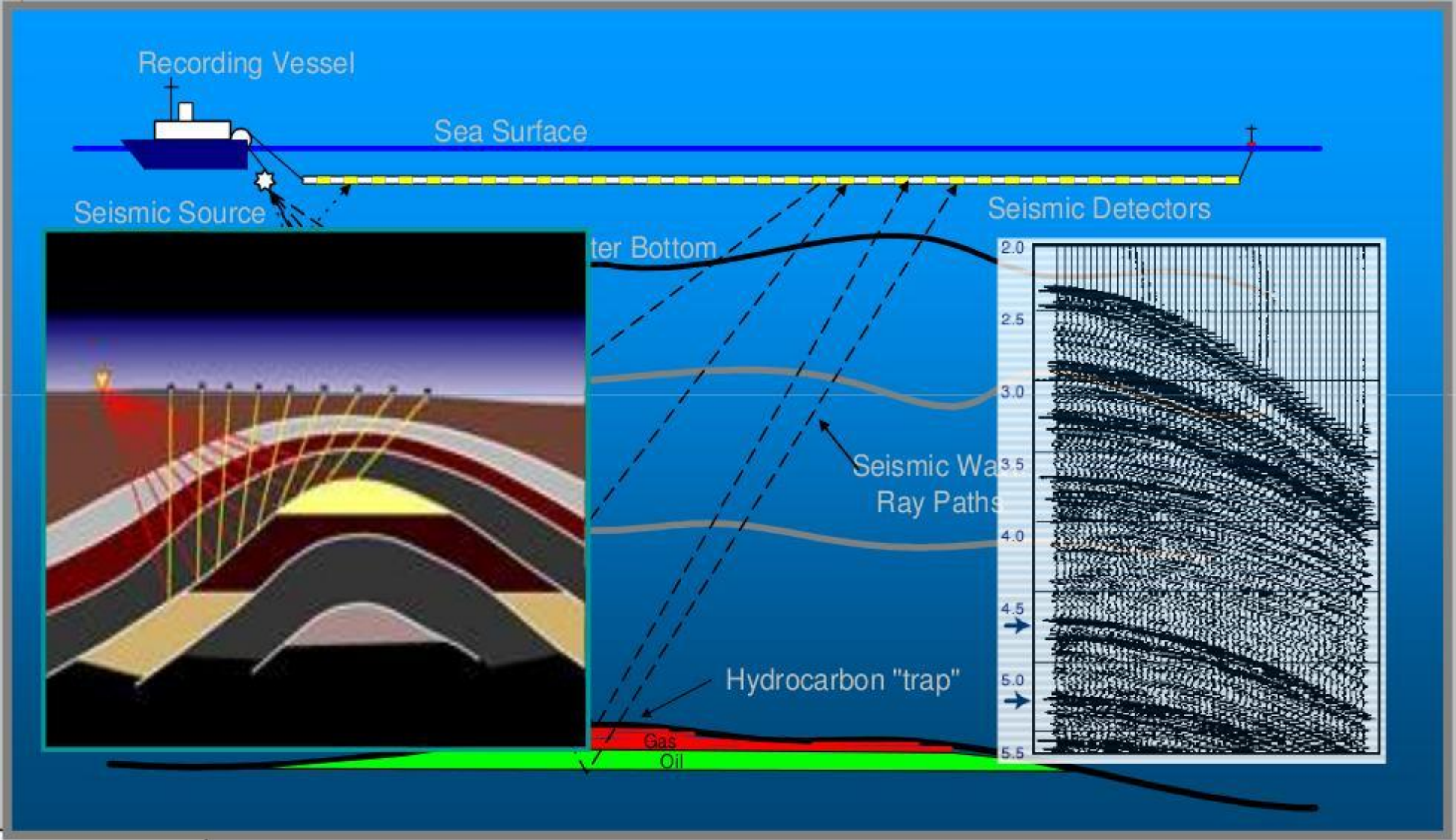




NASDAQ



Seismic Data Acquisition



Courtesy of Schlumberger

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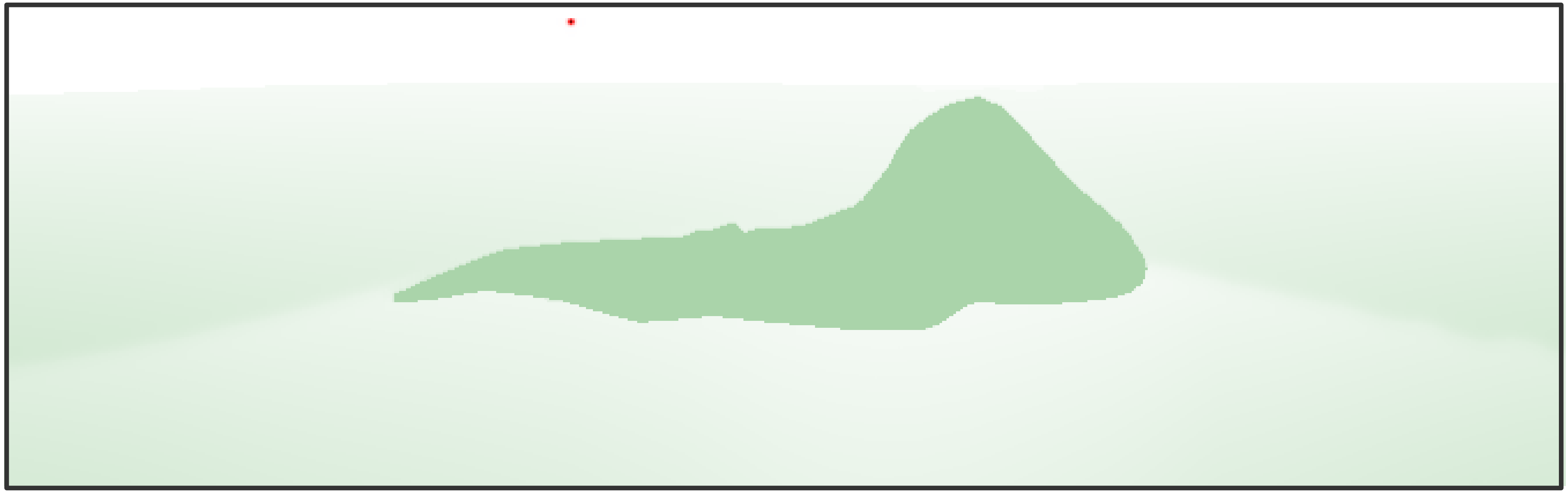
MAXIMUM PERFORMANCE COMPUTING

Seismic Imaging



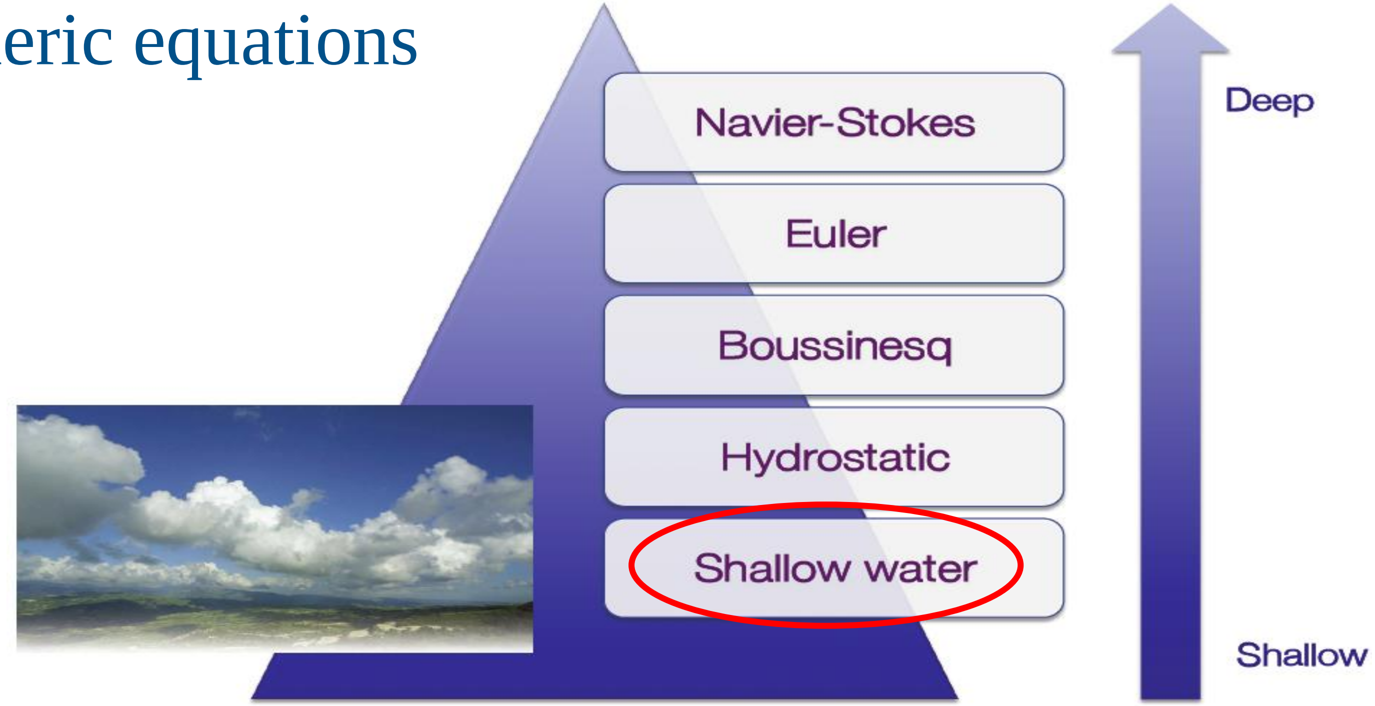
- Running on MaxNode servers
  - 8 parallel compute pipelines per chip
  - 10x less power: 150MHz vs 1.5GHz
  - 30x faster than microprocessors

An Implementation of the Acoustic Wave Equation on FPGAs

T. Nemeth†, J. Stefani†, W. Liu†, R. Dimond‡, O. Pell‡, R.Ergas§
†Chevron, ‡Maxeler, §Formerly Chevron, SEG 2008

Global Weather Simulation: Size is Relevant

- Atmospheric equations



- Equations: Shallow Water Equations (SWEs)

$$\frac{\partial Q}{\partial t} + \frac{1}{\Lambda}\frac{\partial(\Lambda F^1)}{\partial x^1} + \frac{1}{\Lambda}\frac{\partial(\Lambda F^1)}{\partial x^2} + S = 0$$

[L. Gan, H. Fu, W. Luk, C. Yang, W. Xue, X. Huang, Y. Zhang, and G. Yang, Accelerating solvers for global atmospheric equations through mixed-precision data flow engine, FPL2013] Tsinghua

Weather Model – Performance Gain

Platform	Performance ()	Speedup
6-core CPU	4.66K	1
Tianhe-1A node	110.38K	23x
MaxWorkstation	468.1K	**100x**
MaxNode	1.54M	**330x**

14x

Meshsize: 1024 × 1024 × 6

MaxNode speedup over Tianhe node: 14 times

Weather Model -- Power Efficiency

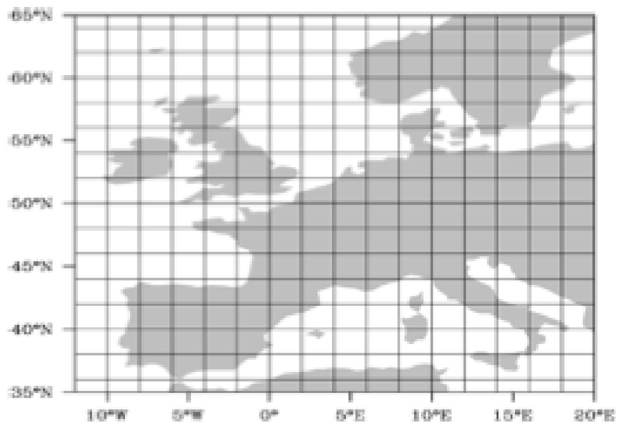
Platform	Power Efficiency ( )	Speedup
6-core CPU	20.71	1
Tianhe-1A node	306.6	14.8x
MaxWorkstation	2.52K	**121.6x**
MaxNode	3K	**144.9x**

Meshsize: $1024 \times 1024 \times 6$

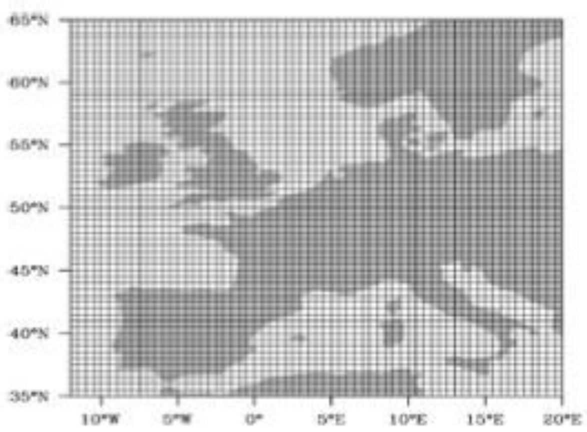
9 x

MaxNode is 9 times more power efficient

Weather and Climate Models: Precision

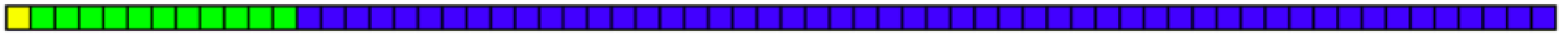


Which one is better?

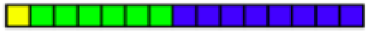


Finer grid and higher precision are obviously preferred
but the computational requirements will increase ❾ Power usage → $$

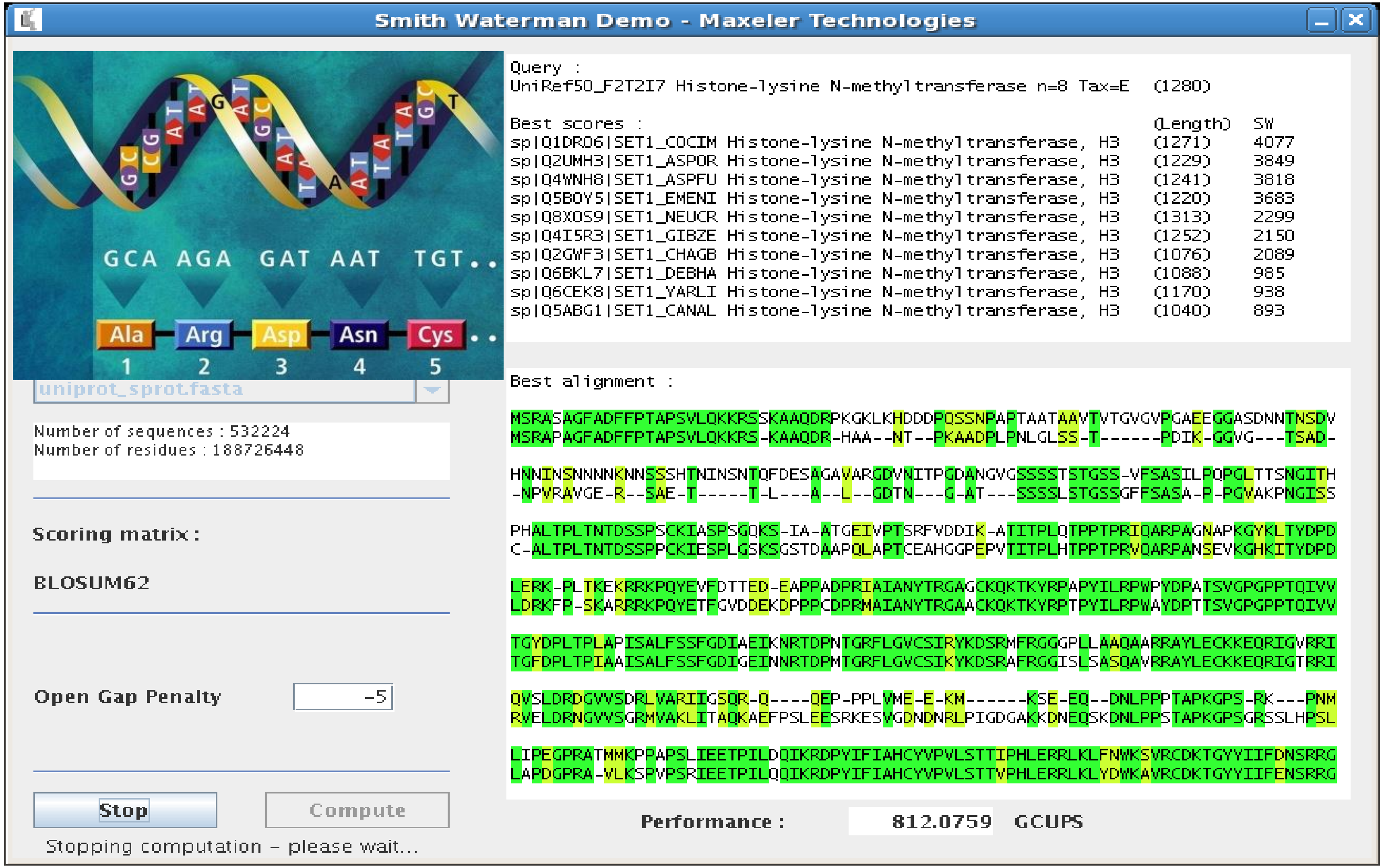
What about using reduced precision? (15 bits instead of 64 double precision FP)



We use only **15 bits** for 98% of the computation:



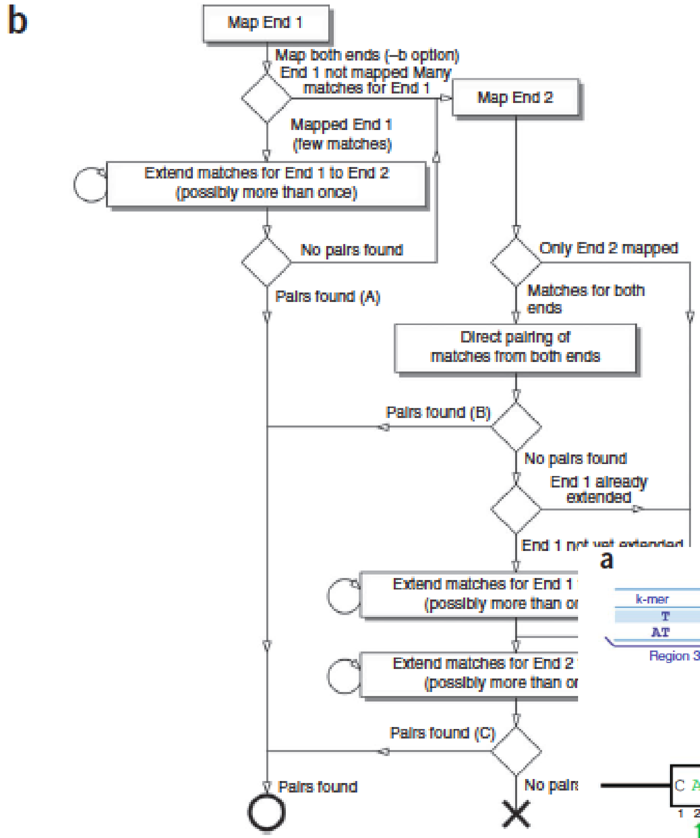
Maxeler Running Smith Waterman

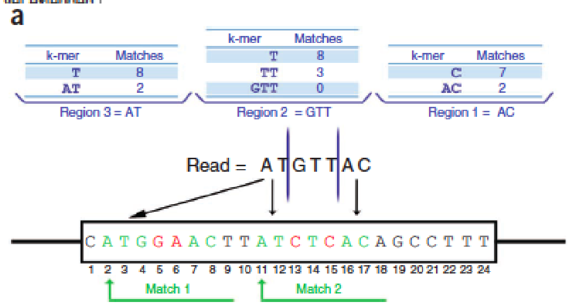


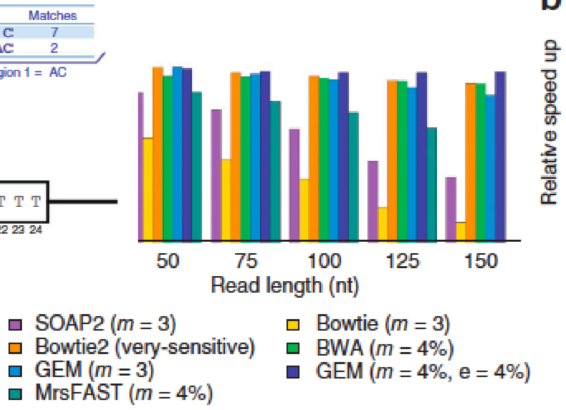
The GEM mapper: fast, accurate and versatile alignment by filtration

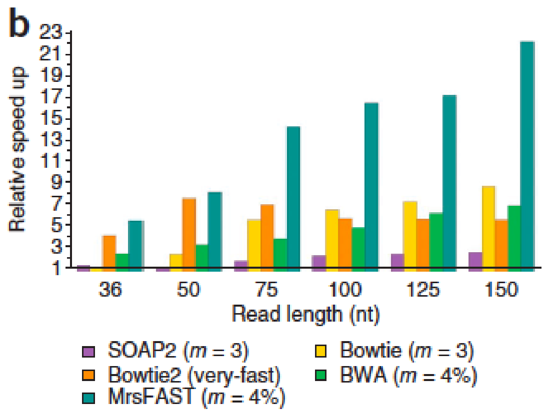
Niederman-Wuensch

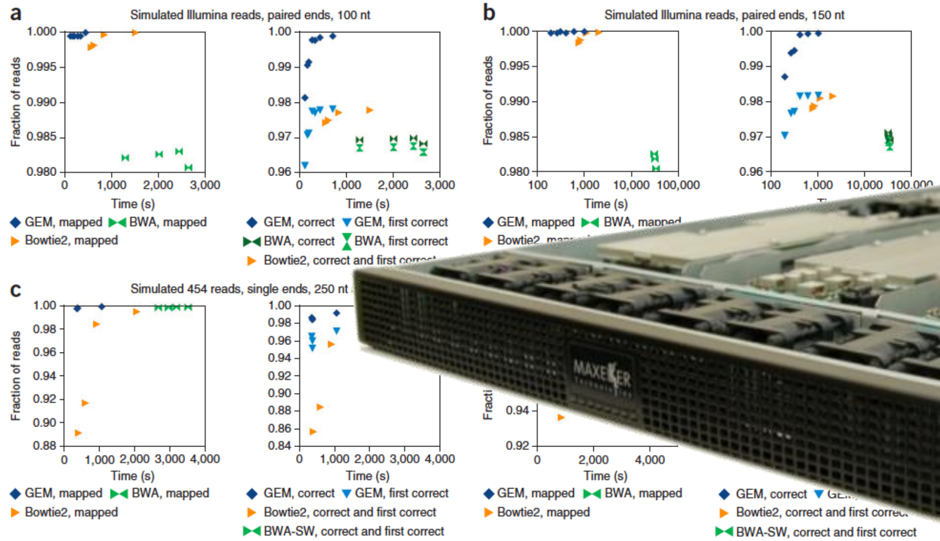
Santiago Marco-Sola[1], Michael Sammeth[1], Roderic Guigó[2] & Paolo Ribeca[1,3]

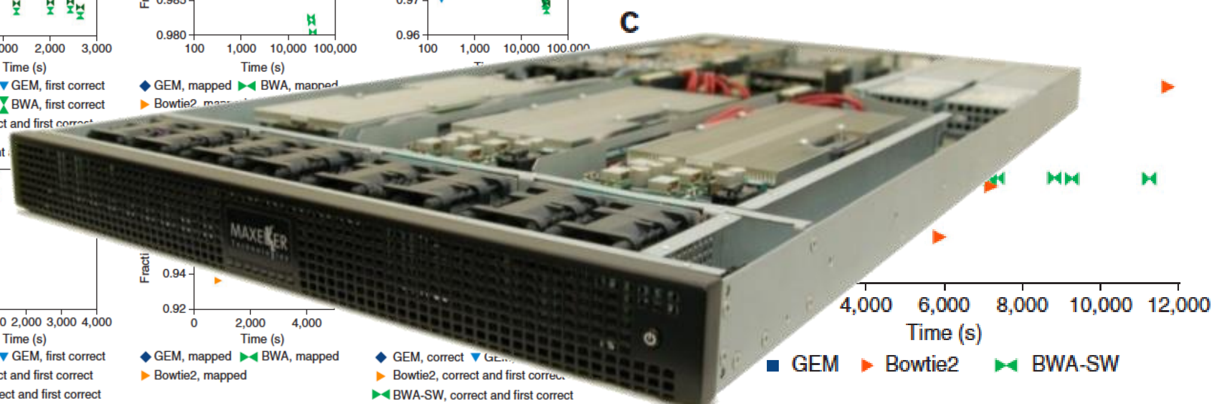


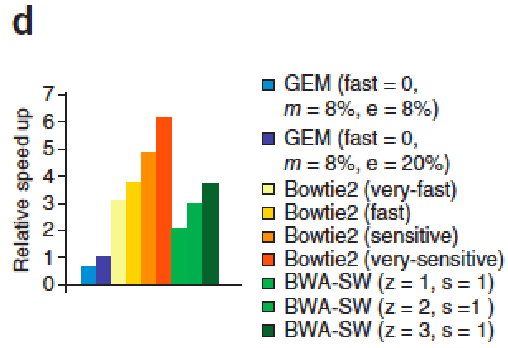




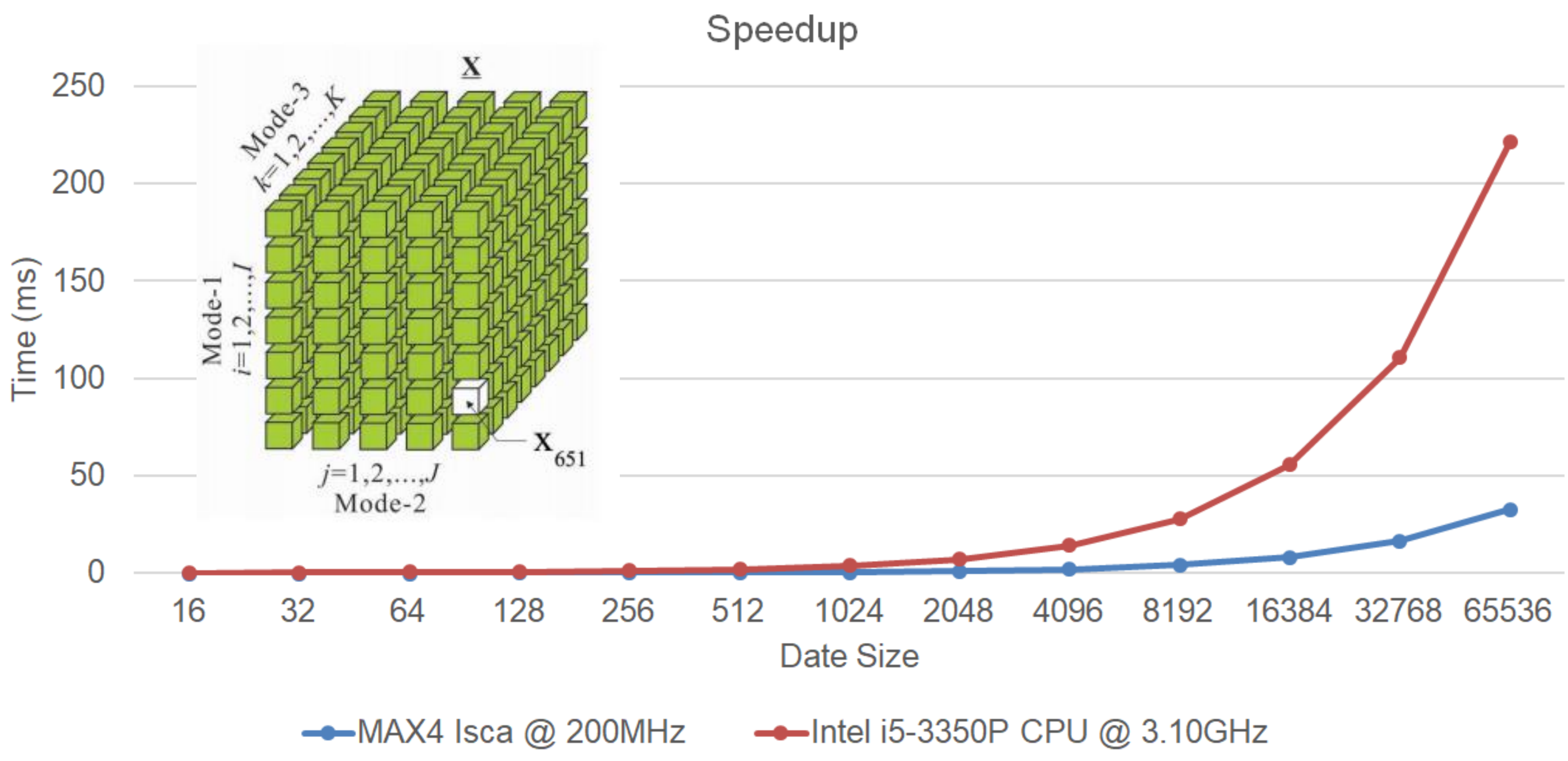








Analysis of the Tensor Calculus Operations on DataFlow
(PhD Thesis by Miloš Kotlar, on DataFlow-based Machine Learning)



The speedup of **6.75x** achieved as early as for KiloData (Perceptron), with **10x** less on-chip transistors and the power savings of **4.6x**

Conditions for the Y-Chart-Based "Kernelization" of Loops @ML
(PhD Thesis by Nenad Korolija, on the Mapping of Algorithms onto DataFlow)

1.	BigData (RAM vs. STREAM)	$O(n^2)$
2.	Code reusability (WORO vs. WORM)	+
3.	Overall application tolerance to latency	+
4.	Over 95% of run time in loops	++
5.	Reusability of the data in loops	++
6.	Potential for utilization of pipes	$O(n)$

Essentials for speedup:
algorithmic modifications,
pipeline utilization,
data choreography,
decision making on precision

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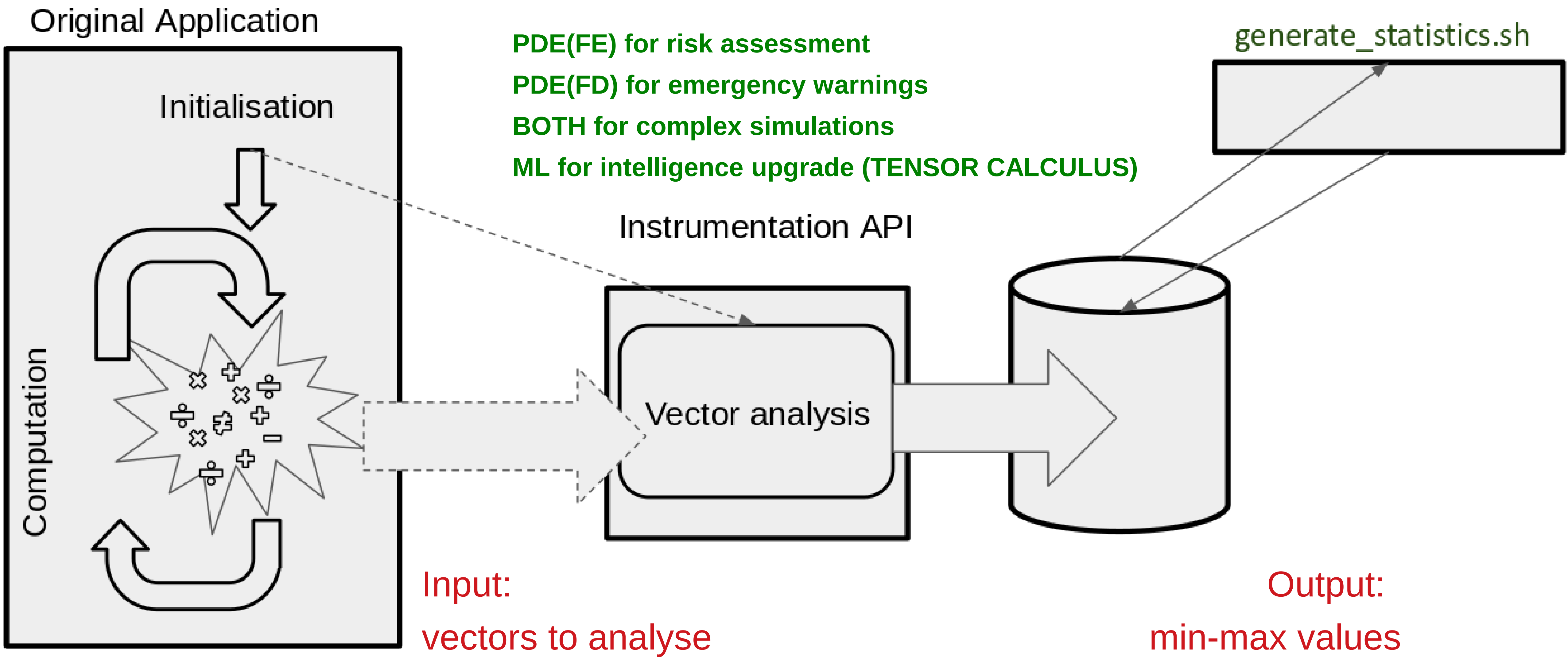
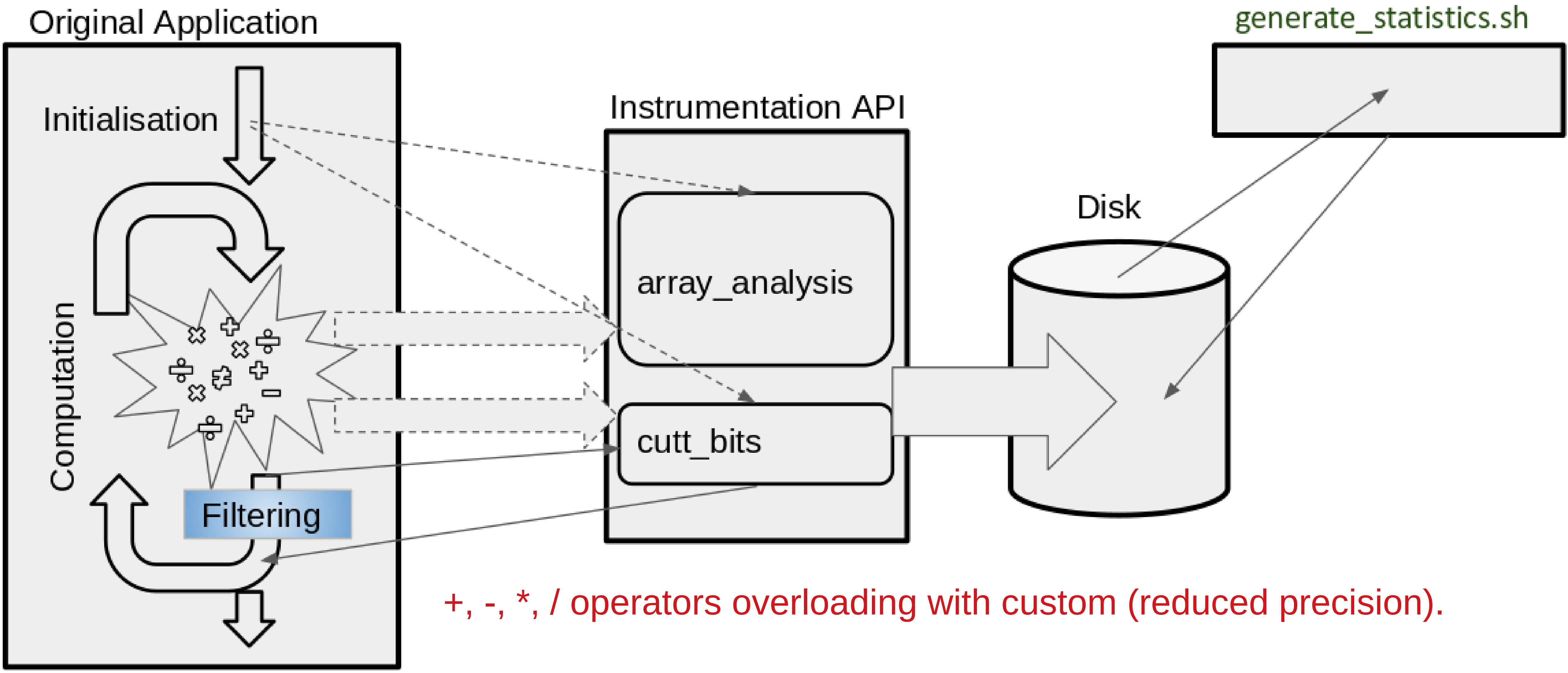
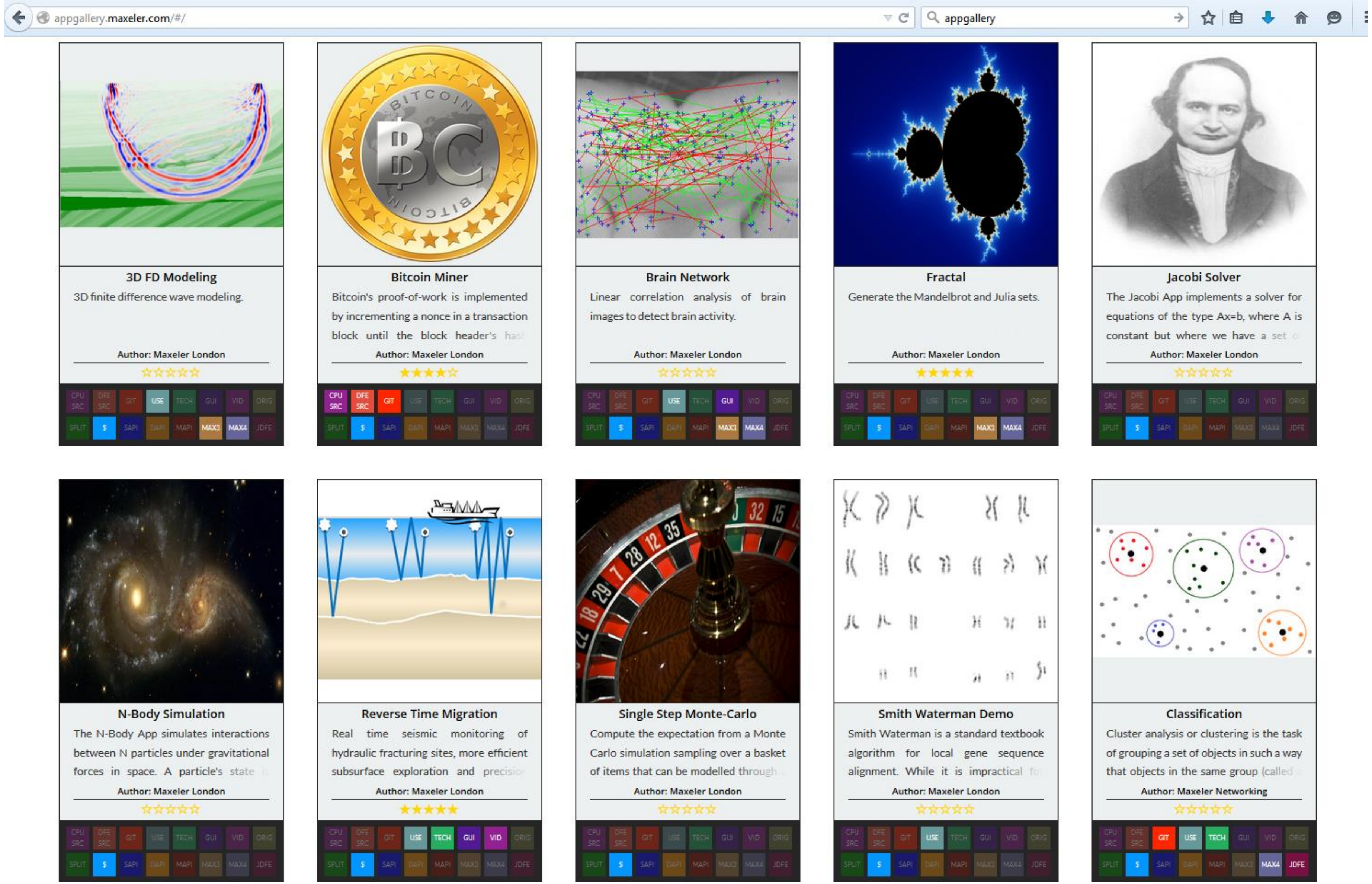


Original Application
Initialisation
Computation
PDE(FE) for risk assessment
PDE(FD) for emergency warnings
BOTH for complex simulations
ML for intelligence upgrade (TENSOR CALCULUS)
Instrumentation API
Vector analysis
generate_statistics.sh
Input:
vectors to analyse
Output:
min-max values


Original Application
Initialisation
Computation
Filtering
Instrumentation API
array_analysis
cutt_bits
Disk
generate_statistics.sh
+, -, *, / operators overloading with custom (reduced precision).



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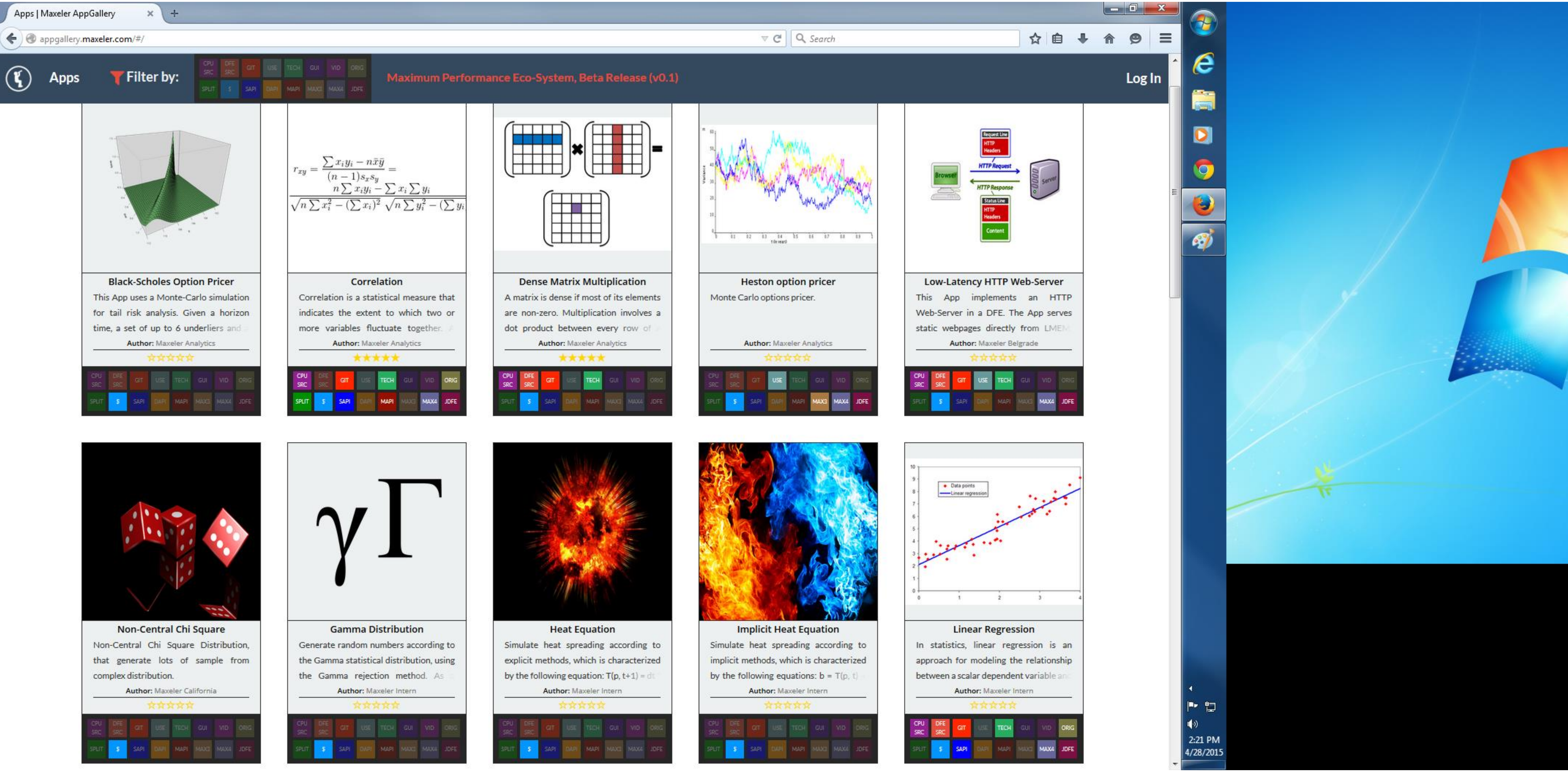
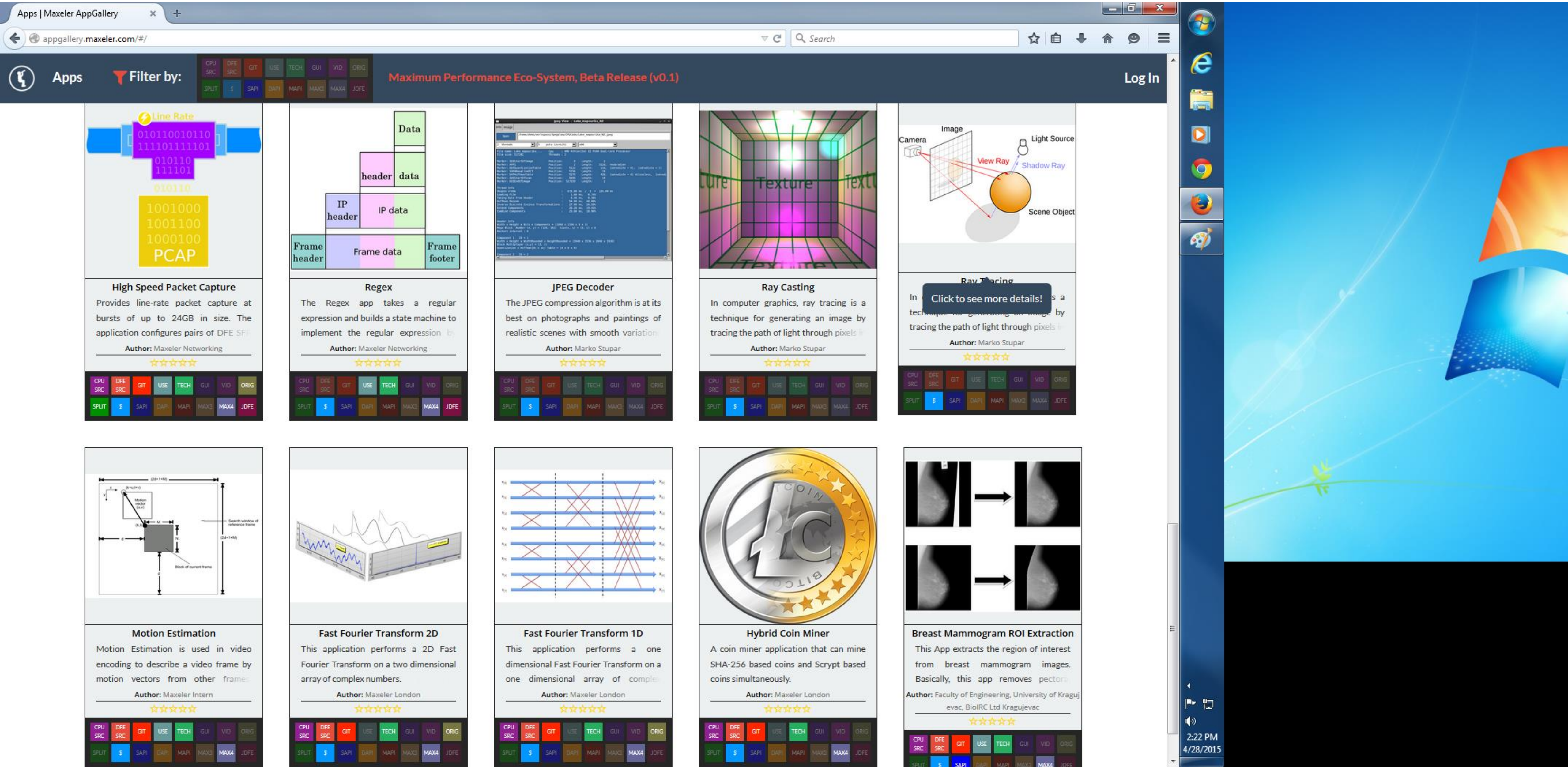
http://www.mi.sanu.ac.rs/~appgallery.maxeler/


Apps | Maxeler AppGallery
appgallery.maxeler.com/#/
Search
Apps
Filter by:
Maximum Performance Eco-System, Beta Release (v0.1)
Log In
Black-Scholes Option Pricer
This App uses a Monte-Carlo simulation for tail risk analysis. Given a horizon time, a set of up to 6 underliers and
Author: Maxeler Analytics
Correlation
Correlation is a statistical measure that indicates the extent to which two or more variables fluctuate together.
Author: Maxeler Analytics
Dense Matrix Multiplication
A matrix is dense if most of its elements are non-zero. Multiplication involves a dot product between every row of
Author: Maxeler Analytics
Heston option pricer
Monte Carlo options pricer.
Author: Maxeler Analytics
Low-Latency HTTP Web-Server
This App implements an HTTP Web-Server in a DFE. The App serves static webpages directly from LMEM
Author: Maxeler Belgrade
Non-Central Chi Square
Non-Central Chi Square Distribution, that generate lots of sample from complex distribution.
Author: Maxeler California
Gamma Distribution
Generate random numbers according to the Gamma statistical distribution, using the Gamma rejection method. As
Author: Maxeler Intern
Heat Equation
Simulate heat spreading according to explicit methods, which is characterized by the following equation: T(p, t+1) = dt
Author: Maxeler Intern
Implicit Heat Equation
Simulate heat spreading according to implicit methods, which is characterized by the following equations: b = T(p, t)
Author: Maxeler Intern
Linear Regression
In statistics, linear regression is an approach for modeling the relationship between a scalar dependent variable and
Author: Maxeler Intern
CPU SRC
DFE SRC
GIT
USE
TECH
GUI
VID
ORIG
SPLIT
$
SAPI
DAPI
MAPI
MAX3
MAX4
JDFE
2:21 PM
4/28/2015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Apps
Filter by:
Maximum Performance Eco-System, Beta Release (v0.1)
Log In
High Speed Packet Capture
Provides line-rate packet capture at bursts of up to 24GB in size. The application configures pairs of DFE
Author: Maxeler Networking
Regex
The Regex app takes a regular expression and builds a state machine to implement the regular expression
Author: Maxeler Networking
JPEG Decoder
The JPEG compression algorithm is at its best on photographs and paintings of realistic scenes with smooth variation
Author: Marko Stupar
Ray Casting
In computer graphics, ray tracing is a technique for generating an image by tracing the path of light through pixels
Author: Marko Stupar
Ray Tracing
Click to see more details!
tracing the path of light through pixels
Author: Marko Stupar
Motion Estimation
Motion Estimation is used in video encoding to describe a video frame by motion vectors from other frames
Author: Maxeler Intern
Fast Fourier Transform 2D
This application performs a 2D Fast Fourier Transform on a two dimensional array of complex numbers.
Author: Maxeler London
Fast Fourier Transform 1D
This application performs a one dimensional Fast Fourier Transform on a one dimensional array of comple
Author: Maxeler London
Hybrid Coin Miner
A coin miner application that can mine SHA-256 based coins and Scrypt based coins simultaneously.
Author: Maxeler London
Breast Mammogram ROI Extraction
This App extracts the region of interest from breast mammogram images. Basically, this app removes pector
Author: Faculty of Engineering, University of Kragujevac, BioIRC Ltd Kragujevac
2:22 PM
4/28/2015

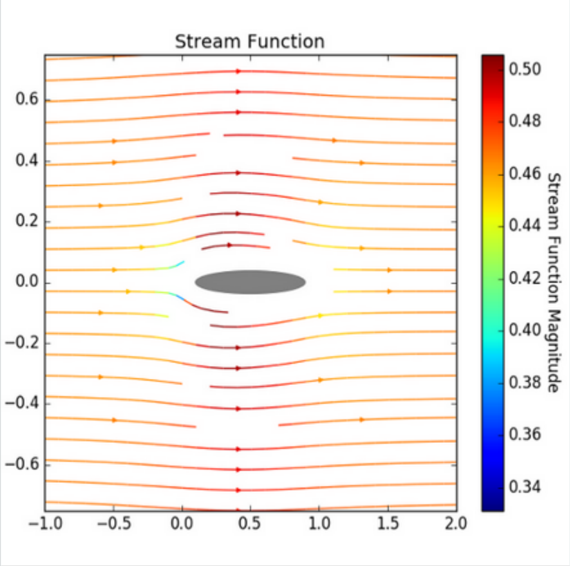
Lz-77

An implementation of Lz77 compression algorithm. Performance: Throughput: 8 input bytes per cycle clock Compression

Authors: Bisheng Huang ,Xinyu Niu

☆ ☆ ☆ ☆ ☆

CPU SRC | DFE SRC | GIT | USE | TECH | GUI | VID | ORIG
SPLIT | $ | SAPI | DAPI | MAPI | MAX3 | MAX4 | JDFE



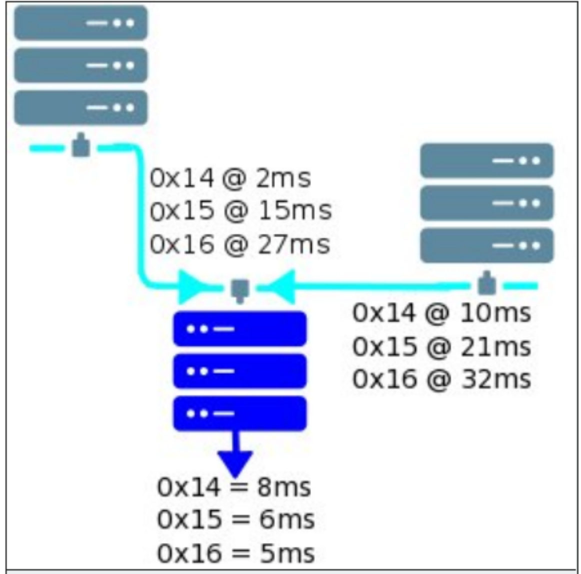
Unstructured Finite Volume

The Airfoil application demonstrates the capability of Maxeler Dataflow Engines to perform Unstructured Mesh

Author: David Packwood

★ ★ ★ ★ ★

CPU SRC | DFE SRC | GIT | USE | TECH | GUI | VID | ORIG
SPLIT | $ | SAPI | DAPI | MAPI | MAX3 | MAX4 | JDFE



Network Latency Measurement

An application to measure the latency between two network connections by sending in identical packets and

Author: Kyle Wallpe

☆ ☆ ☆ ☆ ☆

CPU SRC | DFE SRC | GIT | USE | TECH | GUI | VID | ORIG
SPLIT | $ | SAPI | DAPI | MAPI | MAX3 | MAX4 | JDFE

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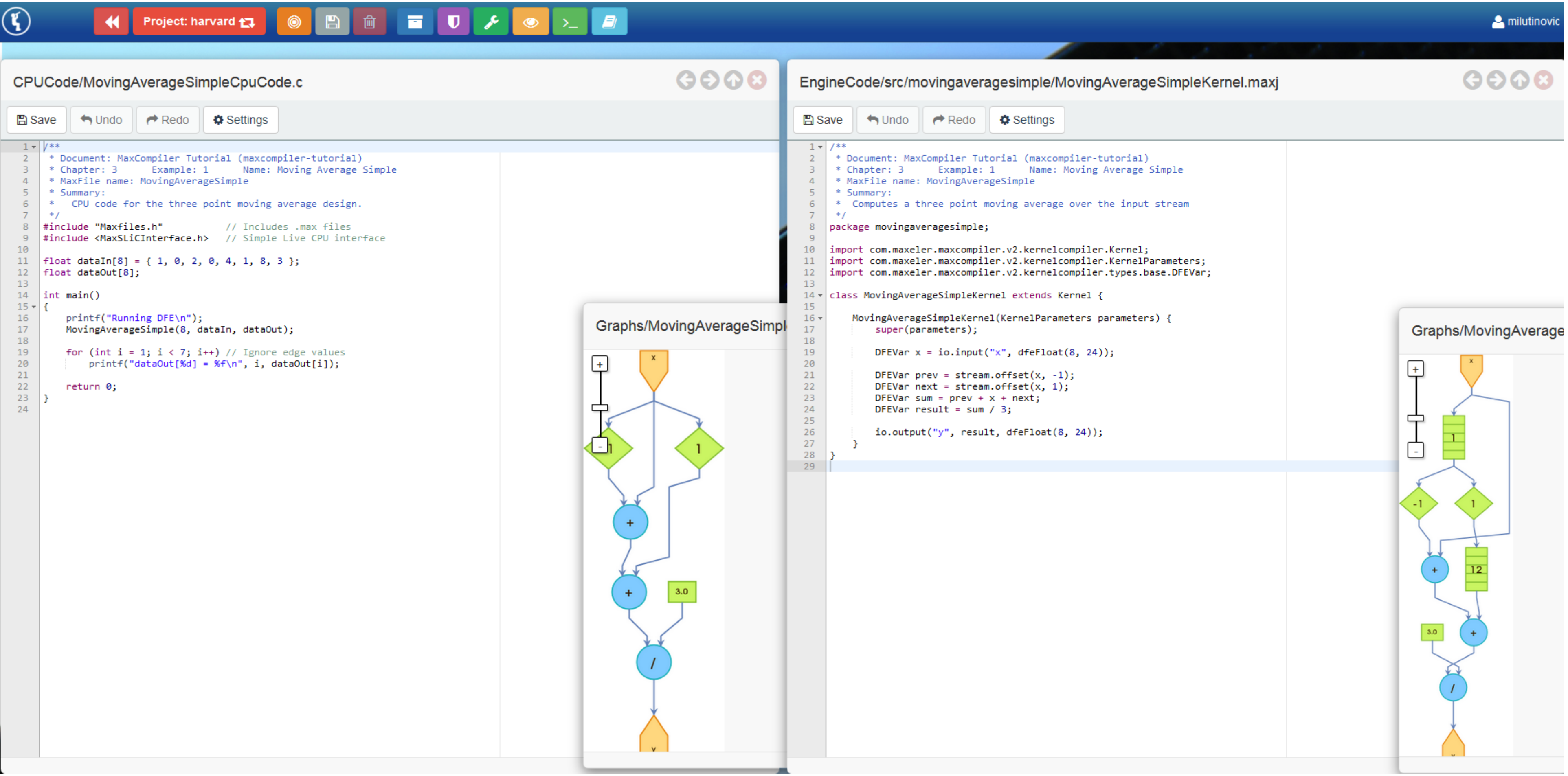

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MAXIMUM PERFORMANCE COMPUTING
HPC
Cloud


MAXE
Techno
spective

Which way are the horses going?

DualCore?

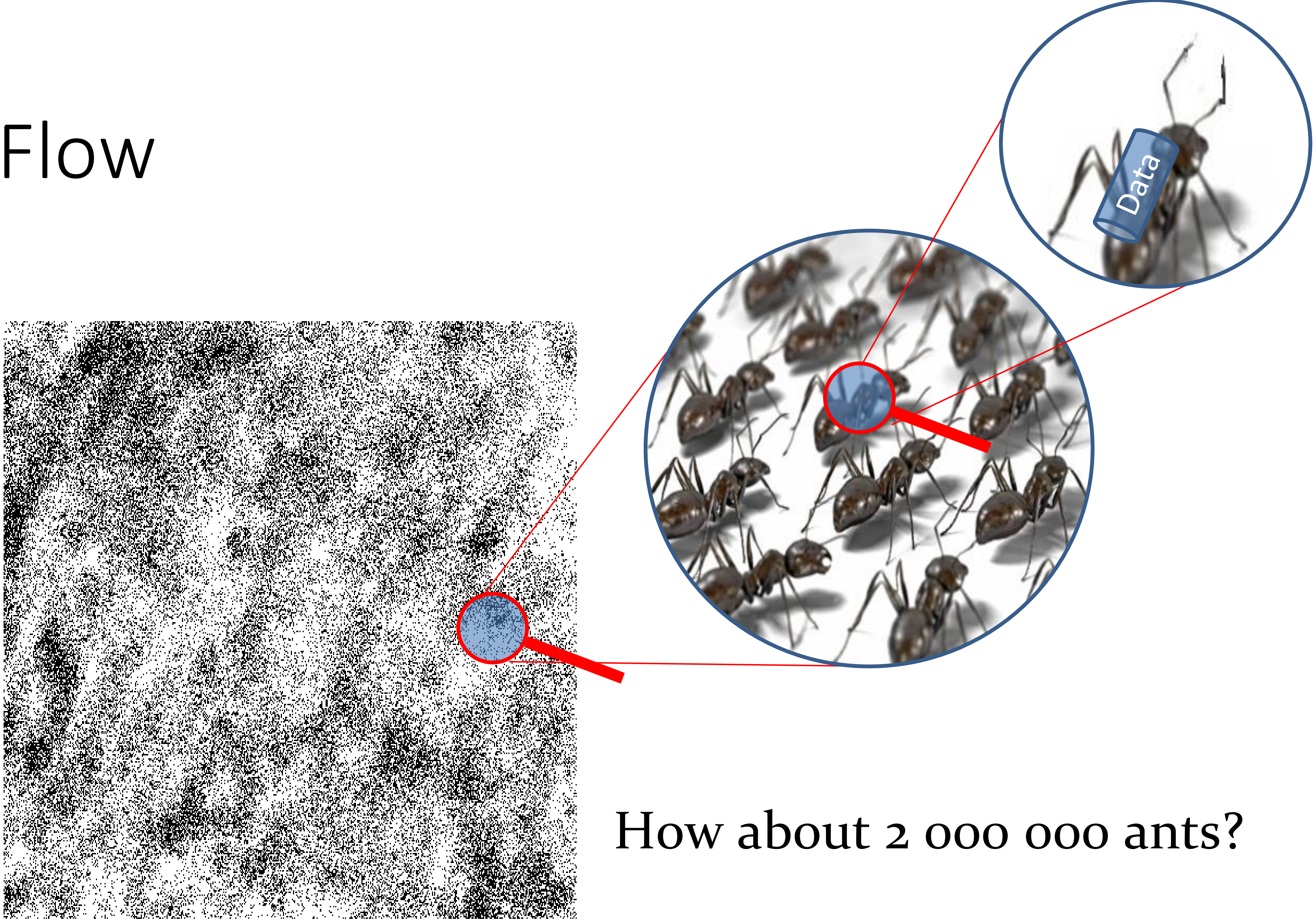
ManyCore

- Is it possible
  to use 2000 chicken instead of two horses?

?
==

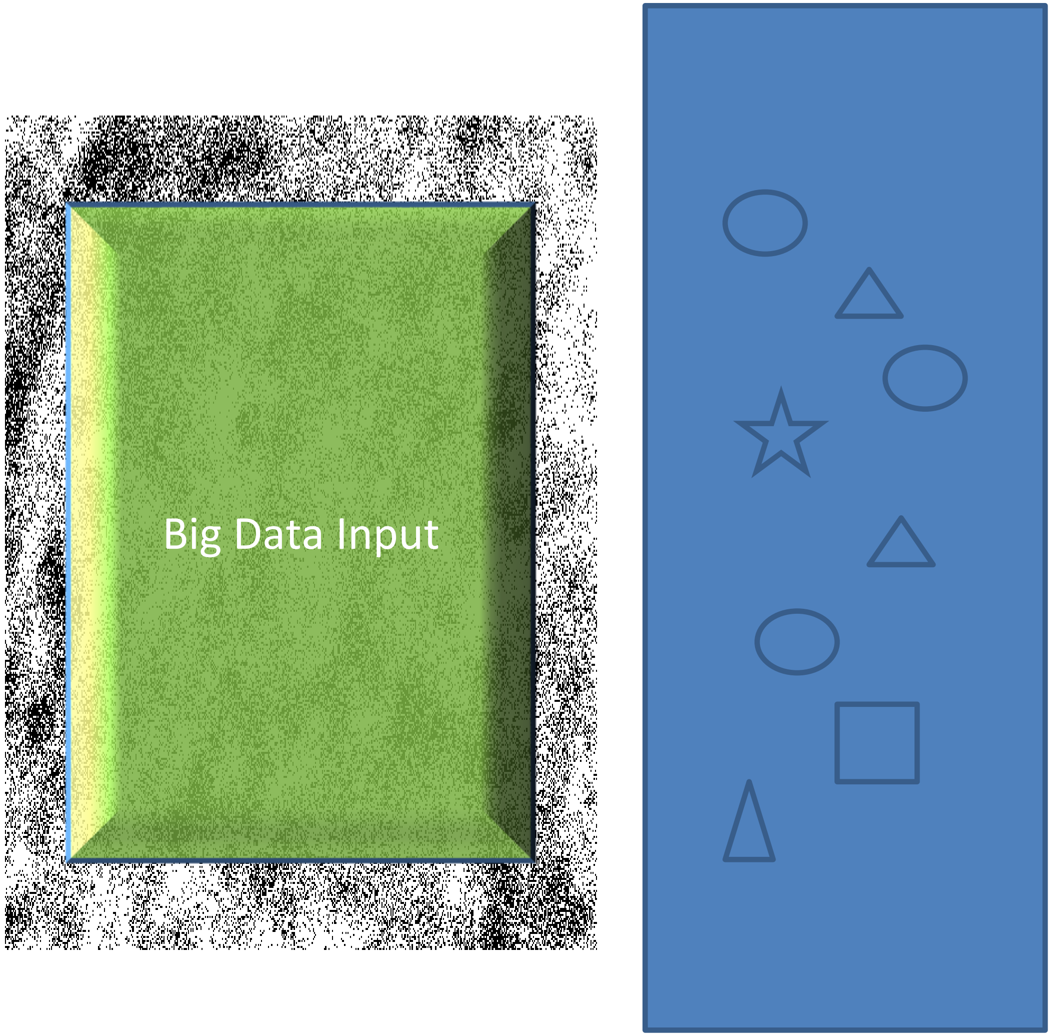
What is better, real and anecdotic?

DataFlow

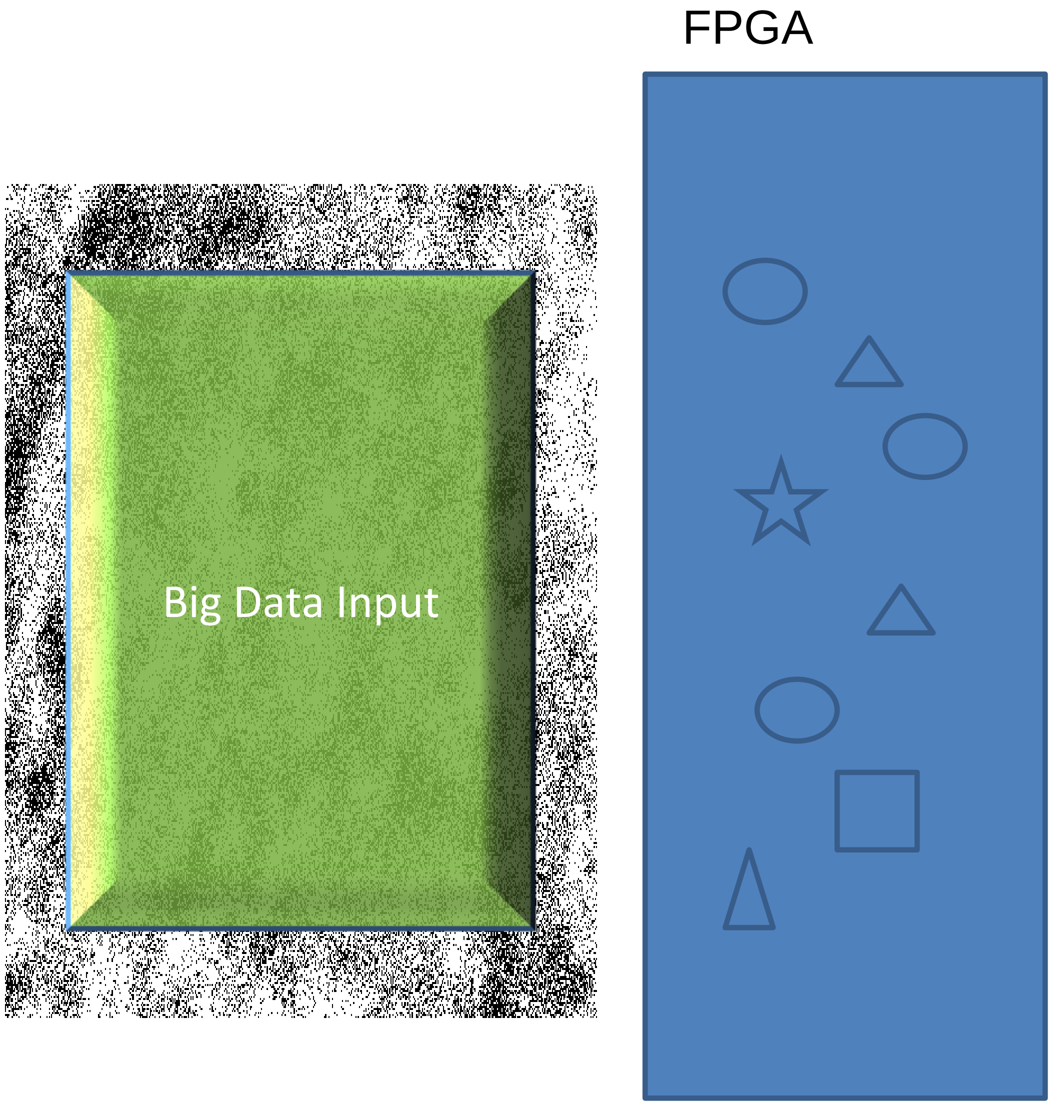


How about 2 000 000 ants?

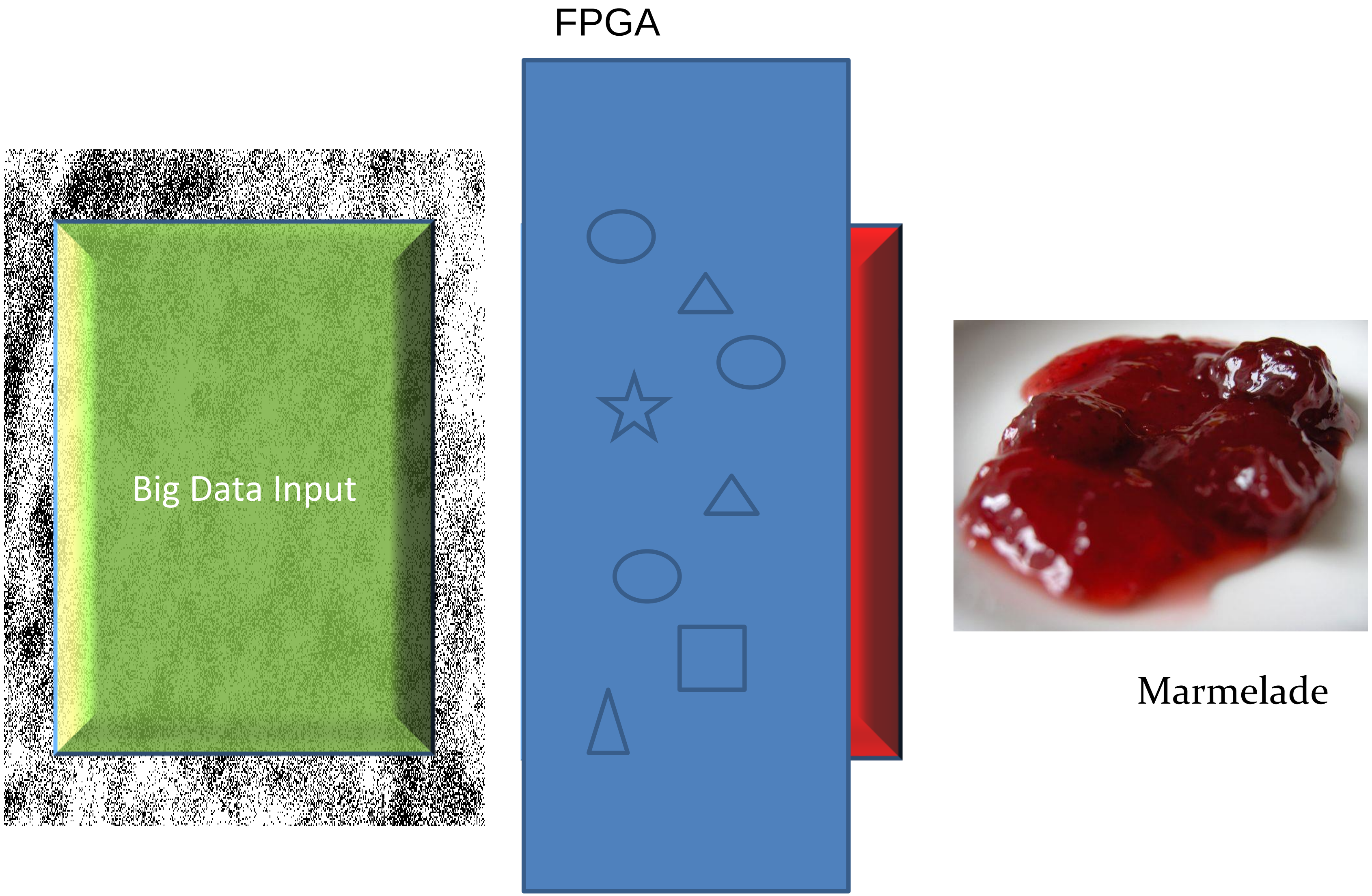
DataFlow



DataFlow



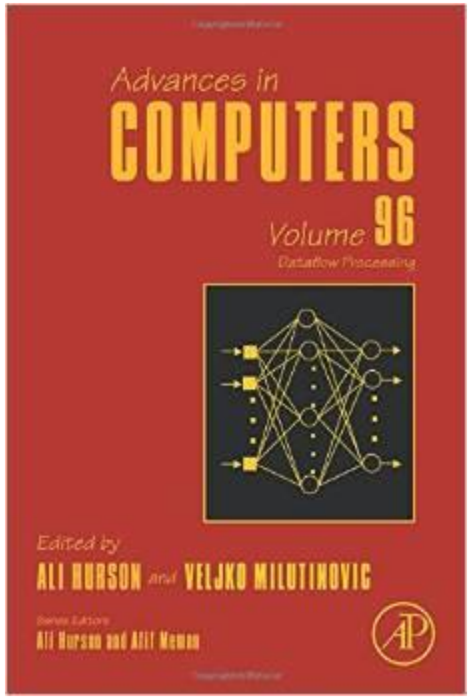
DataFlow



An Edited Book Covering the Applications

- http://www.amazon.com/Dataflow-Processing-Volume-Advances-Computers/dp/0128021349
- http://www.elsevier.com/books/dataflow-processing/milutinovic/978-0-12-802134-7

Indexed by: WoS (SCI)

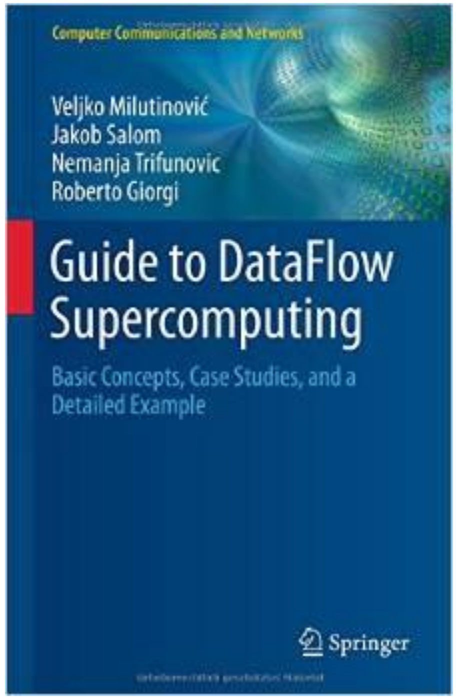


Contributions welcome for the follow-ups: Vol. 102 + Vol. 104 + Vol.126 + etc…

An Original Book Covering the Essence

- http://www.amazon.com/Guide-DataFlow-Supercomputing-Concepts-Communications/dp/3319162284

- http://www.springer.com/gp/book/9783319162287



The first source to use the term the Feynman Paradigm in contrast with the Von Neumann Paradigm

NEWS
6/27/2014
10:09 AM

Google I/O: Hello Dataflow, Goodbye MapReduce

Google introduces Dataflow to handle streams and batches of big data, replacing MapReduce and challenging other public cloud services.

Charles Babcock
News

Connect Directly

4 COMMENTS
COMMENT NOW

Google I/O this year was overwhelmingly dominated by consumer technology, the end user interface, and extension of the Android universe into a new class of mobile devices, the computer you wear on your wrist.

Hadoop Jobs: 9 Ways To Get Hired

(Click image for larger view and slideshow.)

At the same time, there were one or two enterprise-scale data handling and cloud computing gems scattered among all the end user announcements.

Alibaba recently did the same!

newelectronics
The site for electronic design engineers

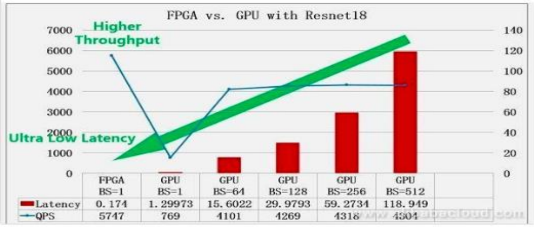
Intel says logic is faster than GPUs

Home | News | In Depth | Products & Suppliers | Magazine | Videos | Blogs | Events | About Us

Home > Technology > Article

Alibaba recently claimed the same!

Comment | Share 15 | Tweet 34 | Share 1 | Share 622 | share 994



Intel's Programmable Systems Group takes its first step towards FPGA based system in package portfolio

Speaking in 2012, Danny Biran – then Altera's senior VP for corporate strategy – said he saw a time when the company would be offering 'standard products' – devices featuring an FPGA, with different dice integrated in the package. "It's also possible these devices may integrate customer specific circuits if the business case is good enough," he noted.

Jordan Inkeles, Altera's director of product marketing for high end FPGAs

There was a lot going on behind the scenes then; already, Altera was talking with Intel about using its foundry service to build 'Generation 10' devices, eventually being acquired by Intel in 2015.

Now the first fruit of that work has appeared in the form of Stratix 10 MX. Designed to meet the needs of those developing high end communications systems, the device integrates stacked memory dice alongside an FPGA die, providing users with a memory bandwidth of up to 1Tbyte/s.

28 June 2016

QoL

Maxeler is one of the Top 10 HPC projects
to impact QoL in the World :)

Scientific Computing
[www.scientificcomputing.com/articles/2014/11]

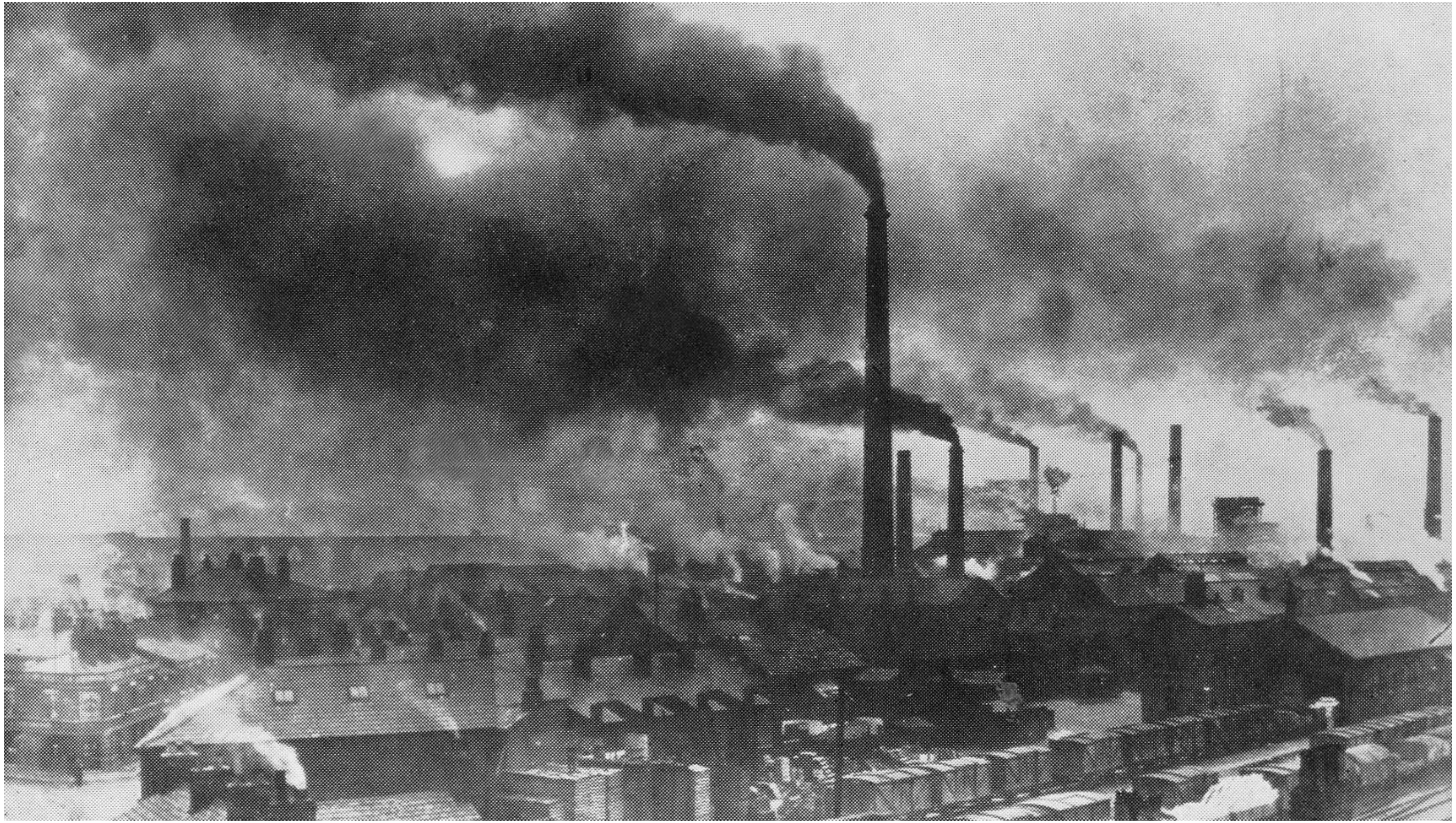
by

Don Johnson

of

Lawrence Livermore National Labs
[editor@ScientificComputing.com]

How About QoL?



DataFlow

SW

HW

AW

Essence of the Paradigm:

For Big Data algorithms
and for the same hardware price as before,
achieving:

a) speed-up, 20-200

b) monthly electricity bills, reduced 20 times

c) size, 20 times smaller

d) precision, X times better

The major issues of engineering are: design cost and design complexity.

Remember, economy has its own rules: production count and market demand!

Why is DataFlow so Much Faster?

- Factor: 20 to 200

MultiCore/ManyCore

Machine Level Code

DataFlow

Gate Transfer Level

Why are Electricity Bills so Small?

- Factor: 20

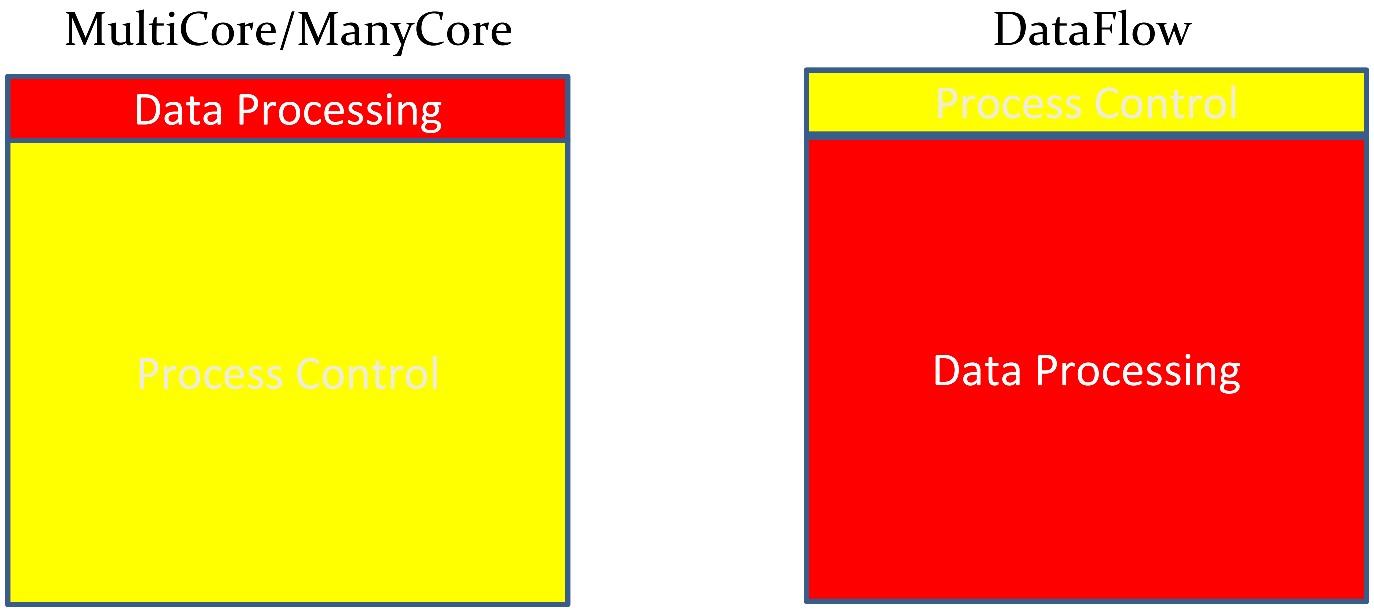
MultiCore/ManyCore

DataFlow

$$P = kfU^2$$

Why is the Cubic Foot so Small?

- Factor: 20



Why is the Precision Better?

- Factor: X

M

N

Bits	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	50	52	54
18	1	1	1	1	2	2	2	2	2	2	2	2	2	3	3	3	3	3	3
20	1	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3
22	1	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3
24	1	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3
26	2	2	2	2	4	4	4	4	4	4	4	4	4	6	6	6	6	6	6
28	2	2	2	2	4	4	4	4	4	4	4	4	4	6	6	6	6	6	6
30	2	2	2	2	4	4	4	4	4	4	4	4	4	6	6	6	6	6	6
32	2	2	2	2	4	4	4	4	4	4	4	4	4	6	6	6	6	6	6
34	2	2	2	2	4	4	4	4	4	4	4	4	4	6	6	6	6	6	6
36	2	3	3	3	4	4	4	4	4	5	5	5	5	6	6	6	6	6	7
38	2	3	3	3	4	4	4	4	4	5	5	5	5	6	6	6	6	6	7
40	2	3	3	3	4	4	4	4	4	5	5	5	5	6	6	6	6	6	7
42	2	3	3	3	4	4	4	4	4	5	5	5	5	6	6	6	6	6	7
44	3	3	3	3	6	6	6	6	6	6	6	6	6	9	9	9	9	9	9
46	3	3	3	3	6	6	6	6	6	6	6	6	6	9	9	9	9	9	9
48	3	3	3	3	6	6	6	6	6	6	6	6	6	9	9	9	9	9	9
50	3	3	3	3	6	6	6	6	6	6	6	6	6	9	9	9	9	9	9
52	3	3	3	3	6	6	6	6	6	6	6	6	6	9	9	9	9	9	9
54	3	4	4	4	6	6	6	6	6	7	7	7	7	9	9	9	9	9	10

Q&A

vm@etf.rs

US 20180189063A1

(19) **United States**

(12) **Patent Application Publication**

FLEMING et al.

(10) **Pub. No.: US 2018/0189063 A1**

(43) **Pub. Date: Jul. 5, 2018**

(54) **PROCESSORS, METHODS, AND SYSTEMS WITH A CONFIGURABLE SPATIAL ACCELERATOR**

(71) Applicant: **Intel Corporation**, Santa Clara, CA (US)

(72) Inventors: **KERMIN FLEMING**, Hudson, MA (US); **KENT D. GLOSSOP**, Merrimack, NH (US); **SIMON C. STEELY, Jr.**, Hudson, NH (US)

(21) Appl. No.: **15/396,395**

(22) Filed: **Dec. 30, 2016**

Publication Classification

(51) **Int. Cl.**

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G06F 13/42	(2006.01)

(52) **U.S. Cl.**

CPC ***G06F 9/3016*** (2013.01); ***G06F 13/4221*** (2013.01)

(57) **ABSTRACT**

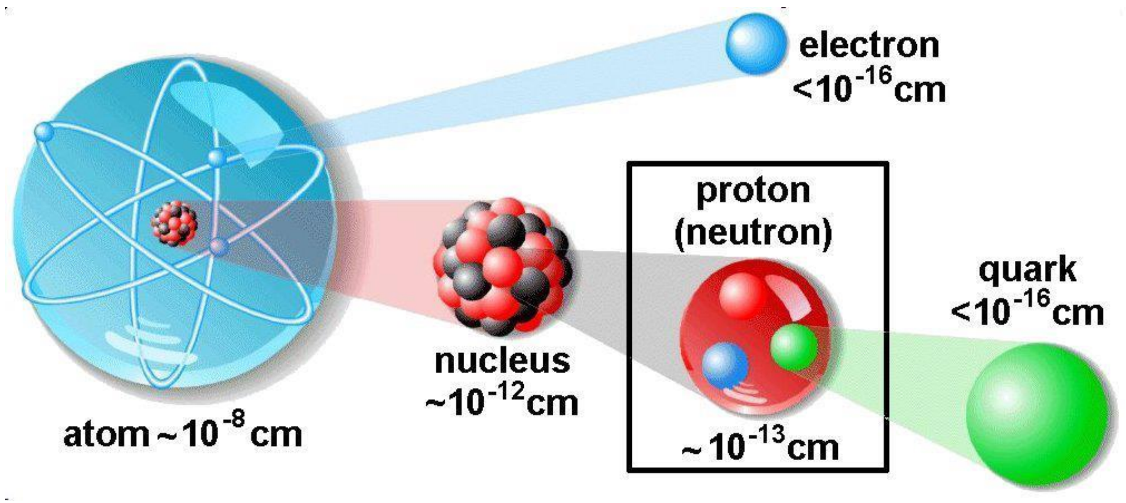
Systems, methods, and apparatuses relating to a configurable spatial accelerator are described. In one embodiment, a processor includes a core with a decoder to decode an instruction into a decoded instruction and an execution unit to execute the decoded instruction to perform a first operation; a plurality of processing elements; and an interconnect network between the plurality of processing elements to receive an input of a dataflow graph comprising a plurality of nodes, wherein the dataflow graph is to be overlaid into the interconnect network and the plurality of processing elements with each node represented as a dataflow operator in the plurality of processing elements, and the plurality of processing elements are to perform a second operation by a respective, incoming operand set arriving at each of the dataflow operators of the plurality of processing elements.


BRITISH AIRWAYS

BQCD on a Maxeler Dataflow Computer

MAXELER Technologies
MAXIMUM PERFORMANCE COMPUTING

BQCD on a Dataflow Computer



- Quantum Chromodynamics (QCD): models interactions of subatomic particles
- Lattice QCD (LQCD): its discretisation, suitable for numerical computation
- Berlin QCD (BQCD): most popular implementation of the LQCD algorithm
- Conjugate Gradient (CG): Majority of the compute time (benchmark: 68%)
  - CG iteratively solves linear algebra problem of form $M\mathbf{x} = \mathbf{b}$
  - Operator M contains Wilson-dslash and Clover operators
- **PROJECT TARGET 40x speedup of CG part of BQCD, followed by speedup of the entire application by 20x comparing same size boxes Dataflow vs BlueGene/Q**

Maxeler QCD - Deployment

Maxeler QCD solution is deployed at Jülich Supercomputing Center, running on a Maxeler Dataflow system (TEV = 682.38).

	2 racks of Jülich BlueGene/Q machine	On-premise Maxeler Dataflow system: scale to 1PF equivalent	Factor
Volume	6.75 m^3	0.87 m^3	7.76
Overall Time to Solution	1576.60 s	689 s	2.29
Overall Energy to Solution	169.6 kWh	4.42 kWh	**38.4**
Volume x TTS	10,642.05 m^3s	599.43 m^3s	**17.8**

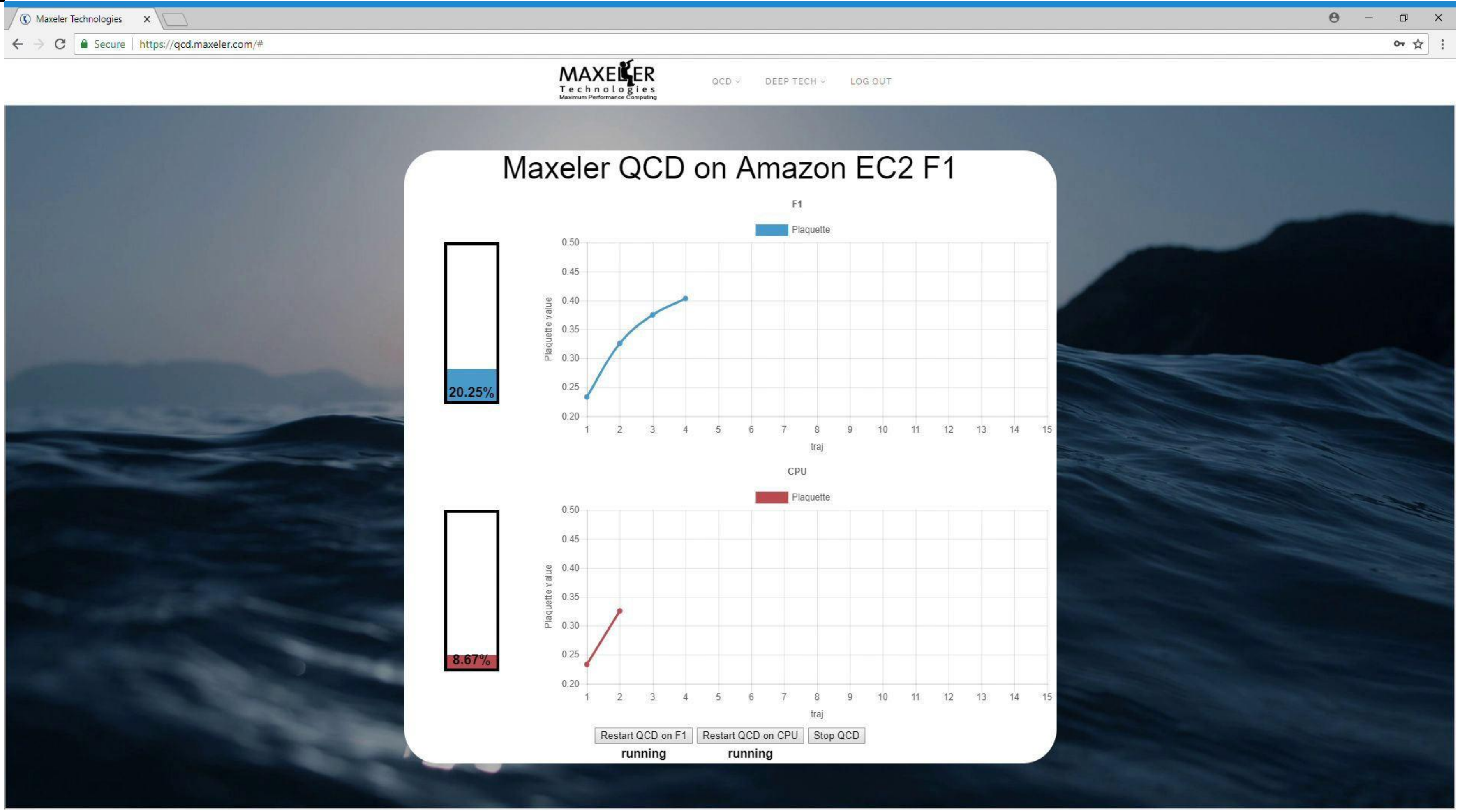
PRACE

Maxeler QCD on Amazon EC2 F1

powered by aws

74

QCD Demo



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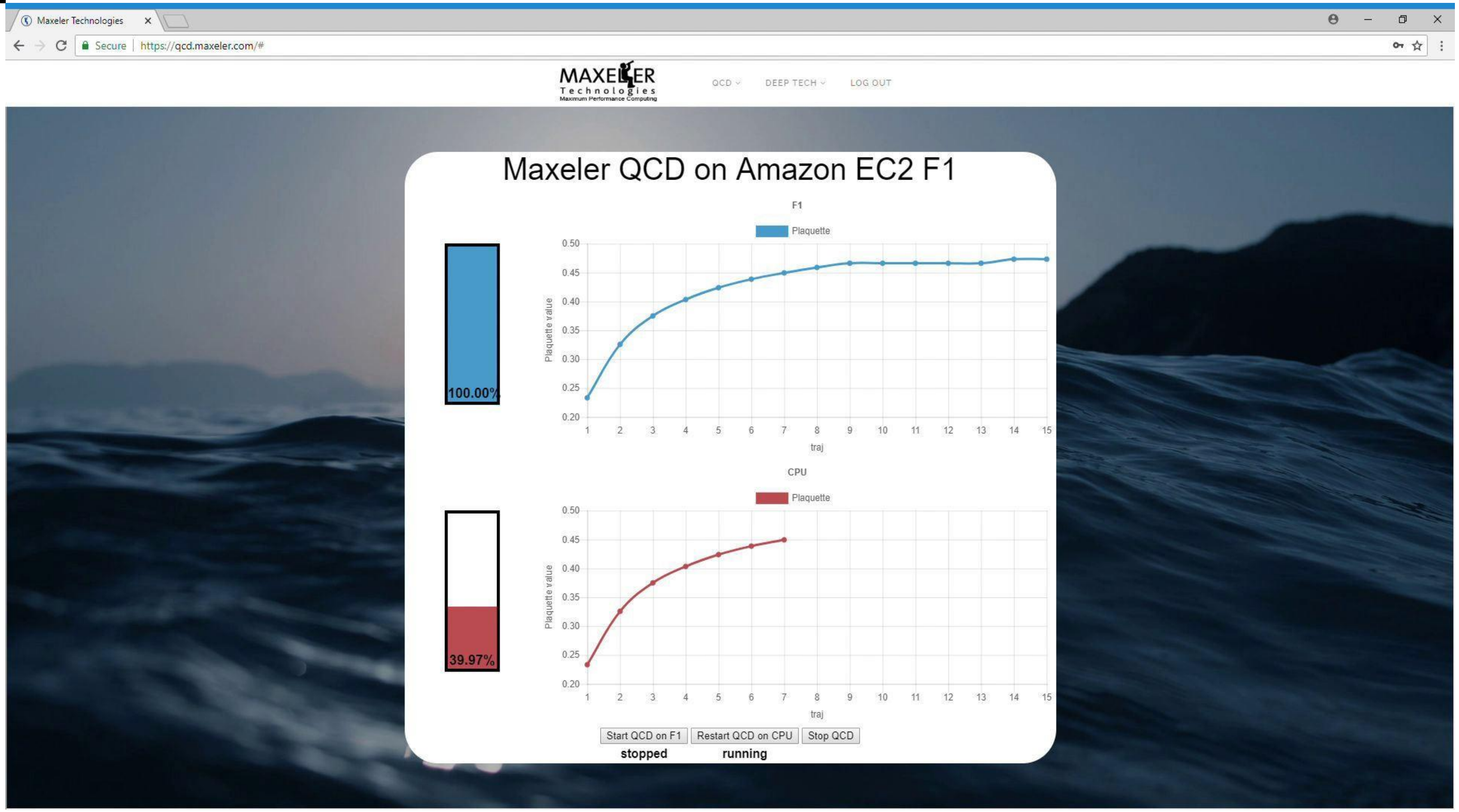
QCD Demo



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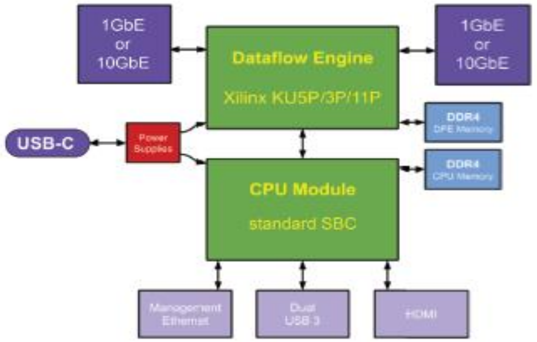
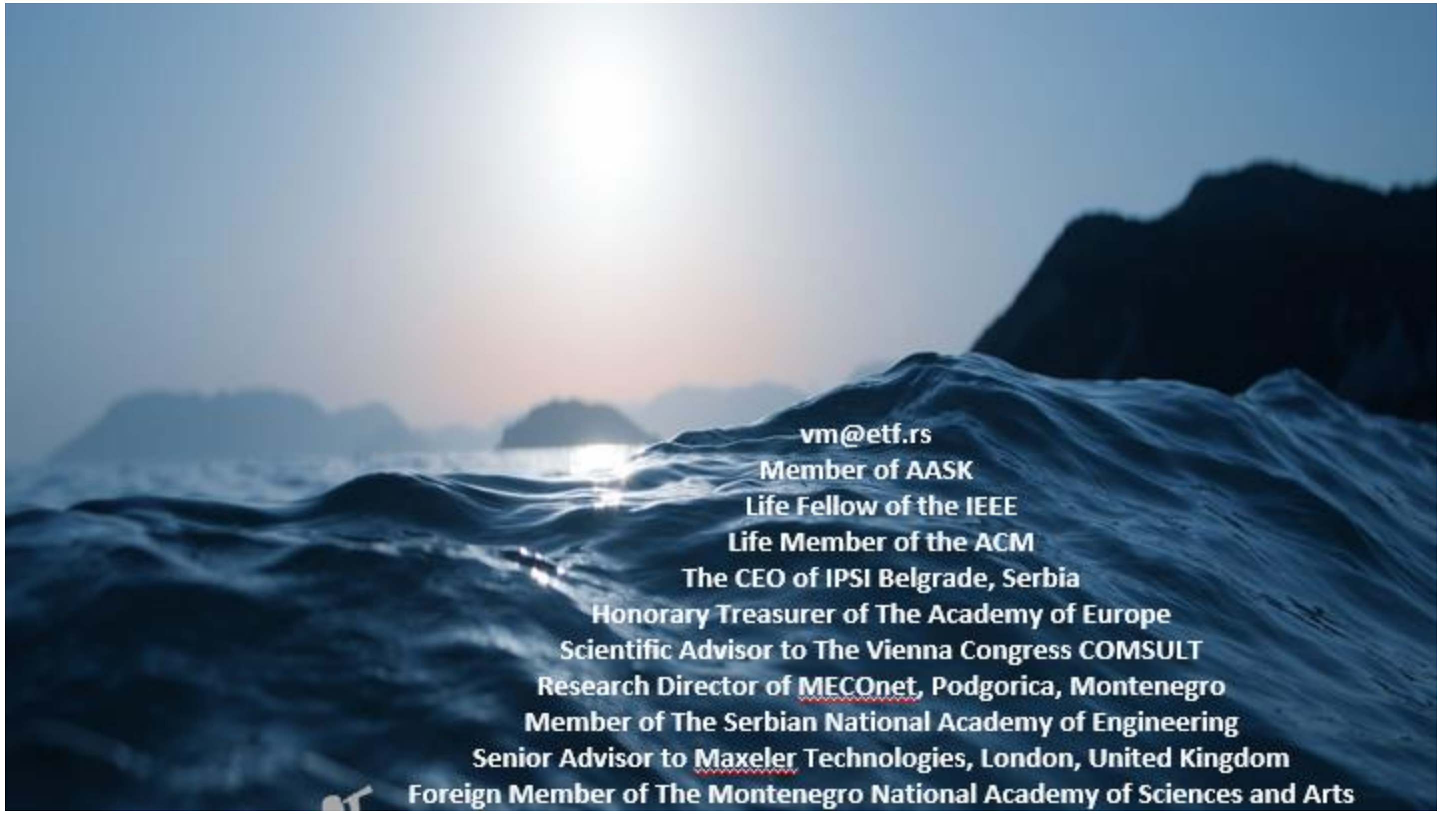


Dimensions	5.7 in (144mm) Wide x 5.7 in Deep (144mm) x 2.2 in High (57mm), excluding power supply		
Form factor	Desktop enclosure, fanless design. Wall or rail mount options available		
Weight	34 oz (950g), excluding power supply		
Power Supply	Separate wall plug unit, providing 60W of USB-PD power from 100-240V, 50-60Hz mains		
Input and Output (Standard Ports)	**Ethernet**	1GbE or 10GbE (copper or fibre)	SFP+ Cage
	USB-C	Power input over USB-PD (Min 15V 3A supply required) USB-3 SuperSpeed II I/O on same connector supporting DisplayPort Alternate mode	
Input and Output (Optional Ports)	**Management LAN**	1GbE	RJ45
	USB	Dual USB-3 Type A ports	
	Video Output	HDMI Type A	
Controlflow Engine	**CPU**	AMD 3rd Generation R- or G-Series - choose from - Quad Core Merlin Falcon RX-416GD @1.6GHz - Dual Core Brown Falcon GX-217I @1.7GHz	Other SBC options available on request
	Memory	2x 4GB DDR4-2133 SODIMM, total 8GB	Higher or lower capacities as required
	Storage	64GB Solid State Memory	
	Operating System	Linux - CentOS 7	Other OS options available on request
Dataflow Engine	**FPGA**	Xilinx Kintex Ultrascale Plus series - choose from - KU5P (217K LUTs, 544 BRAMs, 1,824 DSPs) - KU3P (163K LUTs, 408 BRAMs, 1,368 DSPs) - KU11P (299K LUTs, 680 BRAMs, 2,928 DSPs)	KU5P fitted as standard
	Memory	1x 16GB DDR4-2400 SODIMM	Or 8GB or 32GB

subEdge: microMAX5

Purdue, IU, MIT, Harvard, Boston, NEU, Dartmouth, U of Massachusetts at Amherst, USC, UCLA, Columbia, NYU, Princeton, NJIT, CMU, Temple U, UIUC, Michigan, Wisconsin, Minnesota, FAU, FIU, Miami, Central Florida, U of Alabama, U of Kentucky, GeorgiaTech, Ohio State, Imperial, King's, Manchester, Huddersfield, Cambridge, Oxford, Dublin, Cork, Cardiff, Edinburgh, EPFL, ETH, TUWIEN, UNIWIE, Graz, Linz, Karlsruhe, Stuttgart, Bonn, Frankfurt, Heidelberg, Aachen, Darmstadt, Dortmund, KTH, Uppsala, Karlskrona, Karlstad, Napoli, Salerno, Siena, Pisa, Barcelona, Madrid, Valencia, Oviedo, Ankara, Bogazici, Koc, Istanbul, Technion, Haifa, BerSheba, Eilat, Belgrade, Podgorica, Koper, Ljubljana, Maribor, Nova Gorica, etc, etc. Also at the World Bank in Washington DC, IMF, the Telenor Bank of Norway, the Raiffeisen Bank of Austria, Brookhaven National Laboratory, Lawrence Livermore National Laboratory, IBM TJ Watson, HP Encore Labs, Intel, Qualcomm VP, NCR, RCA, Fairchild, Honeywell, Yahoo NY, Google CA, Microsoft, Finsoft, ABB Zurich, Oracle Zurich, and many other industrial labs, as well as at Tsinghua University, Shandong, NIS of Singapore, NTU of Singapore, Tokyo, Sendai, Seoul, Pusan, Sydney University of Technology, University of Sydney, Hobart, Auckland, Toronto, Montreal, Durango, MontereyTech, Cuernavaca, UNAM, Kragujevac, Novi Sad...

MAXELER Technologies
MAXIMUM PERFORMANCE COMPUTING


vm@etf.rs
Member of AASK
Life Fellow of the IEEE
Life Member of the ACM
The CEO of IPSI Belgrade, Serbia
Honorary Treasurer of The Academy of Europe
Scientific Advisor to The Vienna Congress COMSULT
Research Director of MECOnet, Podgorica, Montenegro
Member of The Serbian National Academy of Engineering
Senior Advisor to Maxeler Technologies, London, United Kingdom
Foreign Member of The Montenegro National Academy of Sciences and Arts

Design of
an SIMD Multimicroprocessor
for RCA

GaAs Systolic Array Based on 4096 Node Processor Elements

Adaptive signal processing is of crucial importance for advanced radar and communications systems. In order to achieve real-time throughput and latencies, one is forced to use advanced semiconductor technologies (e.g., gallium arsenide, or similar) and advanced parallel architectures (e.g., systolic arrays, or similar).

The systolic array described here was designed to support two important applications: (a) adaptive antenna array beam forming, and (b) adaptive Doppler spectral filtering. In both cases, in theory, the system output is calculated as the product of the signal vector **x** (complex N-dimensional vector) and the weight vector **w** (optimal N-dimensional vector).

Complex vector **x** is obtained by multiplying N input samples with the corresponding window weighting function consisting of N discrete values. Optimal vector **w** is obtained as:

$$w = R^{-1} s^{*} = M^{-1} v^{*}.$$

Symbol **R** refers to the N-by-N inverse covariance matrix of the signal with the (i,j)-th component defined as:

$$r_{i,j} = x_i^* x_j^T,$$

and symbol s^* refers to N-dimensional vector which defines the antenna direction
(in the case of adaptive antenna beamforming)
or Doppler peak (in the case of adaptive Doppler spectral filtering).

Symbols **M** and **v** represent scaled values of **R** and **s**, respectively.
In practice, the scaled values **M** and **v** may be easier to obtain,
and consequently the remaining explanation is adjusted.

The core of the processing algorithm is the inversion of a N-by-N matrix in real time.
This problem can be solved in a number of alternative ways which are computationally less complex.
The one chosen here includes the operations explained in ***Figure abcd***.

Positive semi definite matrix **M** can be defined as:

$$M = U\,D\,U^T$$

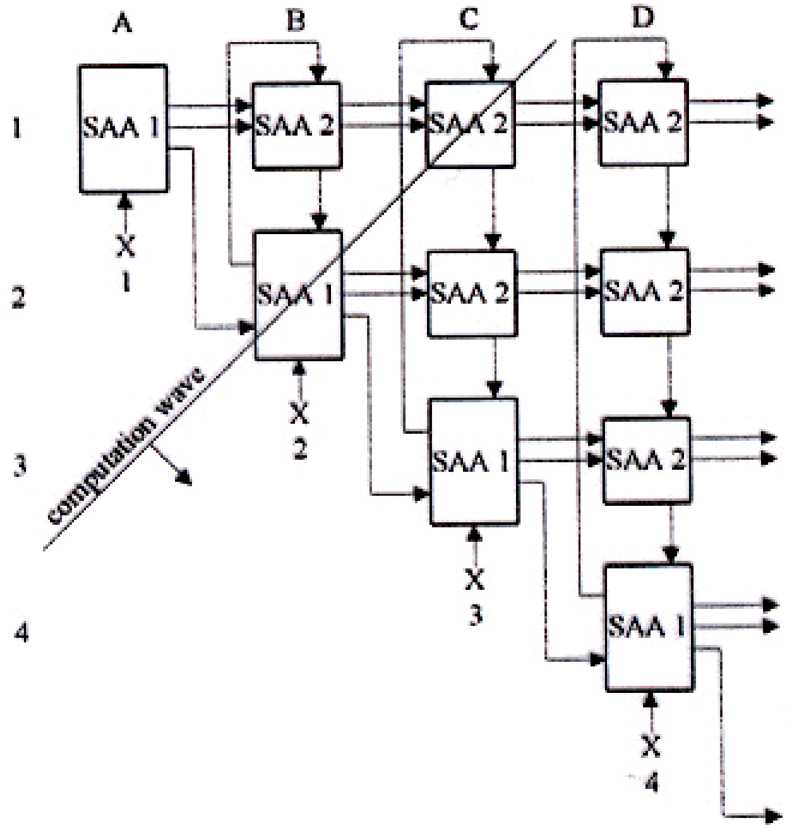
Matrices U and **D** are defined using the formula:

$$M_K = U_K D_K U_K^T = \sum_{i=1}^{K} x_i x_i^T$$

which is recursively updated using the formula:

$$U_K D_K U_K^T = U_{K-1} D_{K-1} U_{K-1}^T + x_K x_K^T$$

a.



b.

Legend:

SAA1 – Cells involved in root covariance update, and the first step of back substitution;

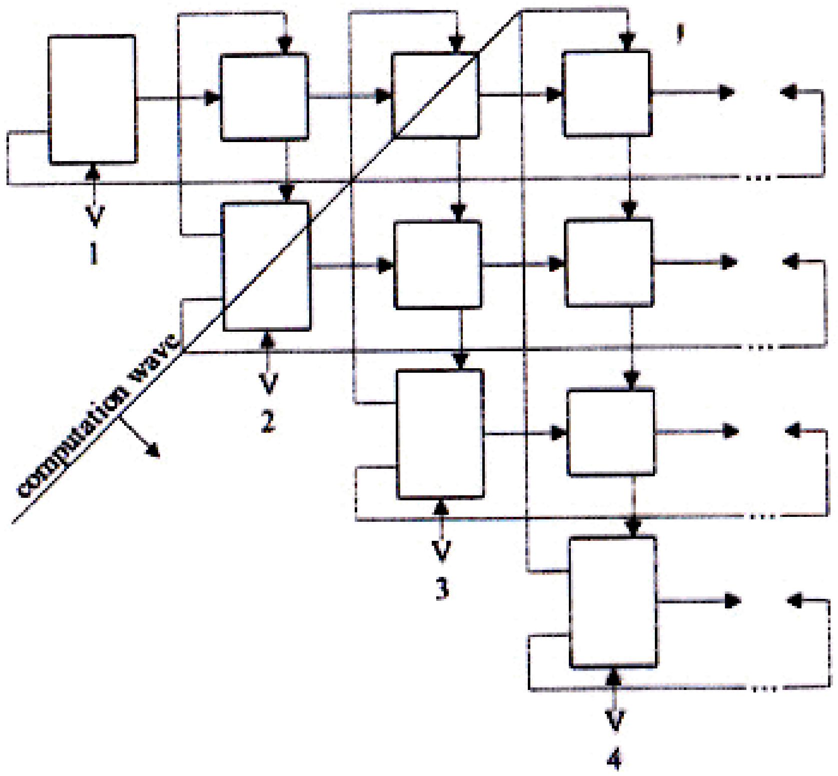
SAA2 - Cells involved in root covariance update, and in both steps of back substitution;

U – Lower triangular matrix with unit diagonal elements

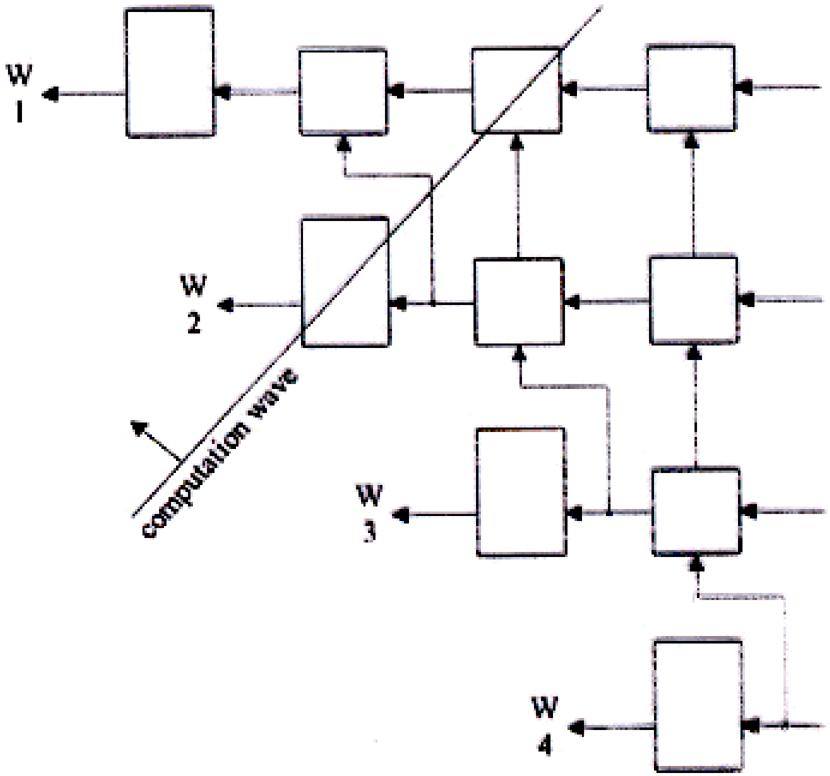
D – Diagonal matrix with positive or zero diagonal elements

b – A scalar initially set to 1

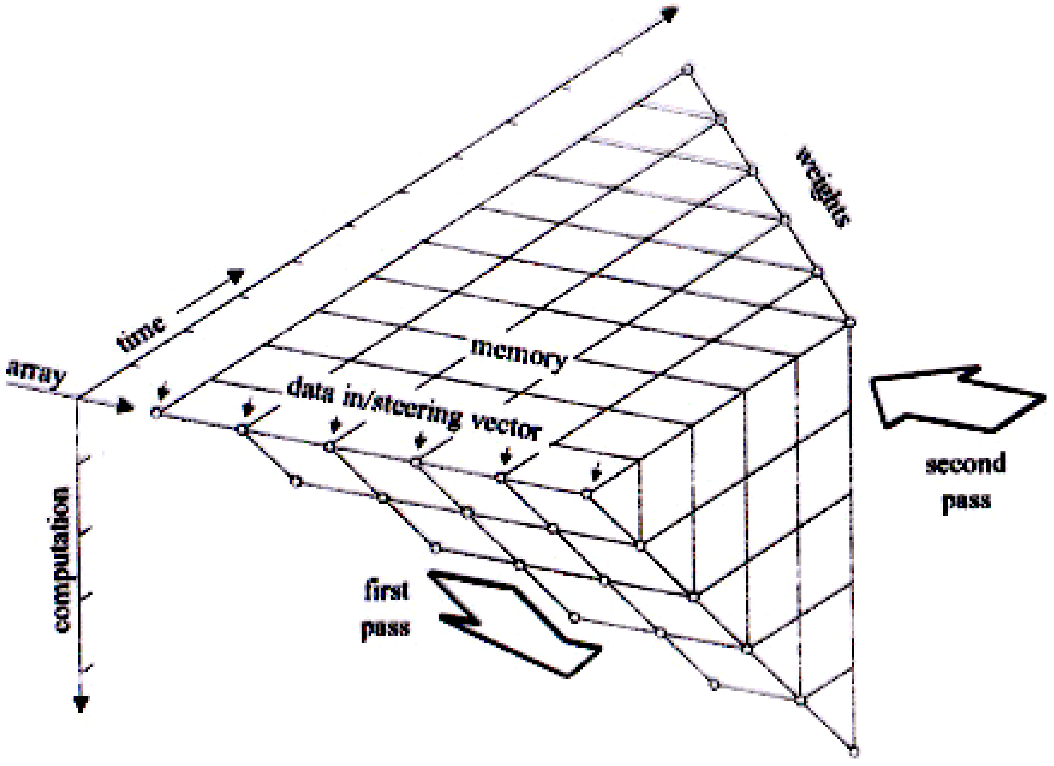
K – Iteration count.



c.

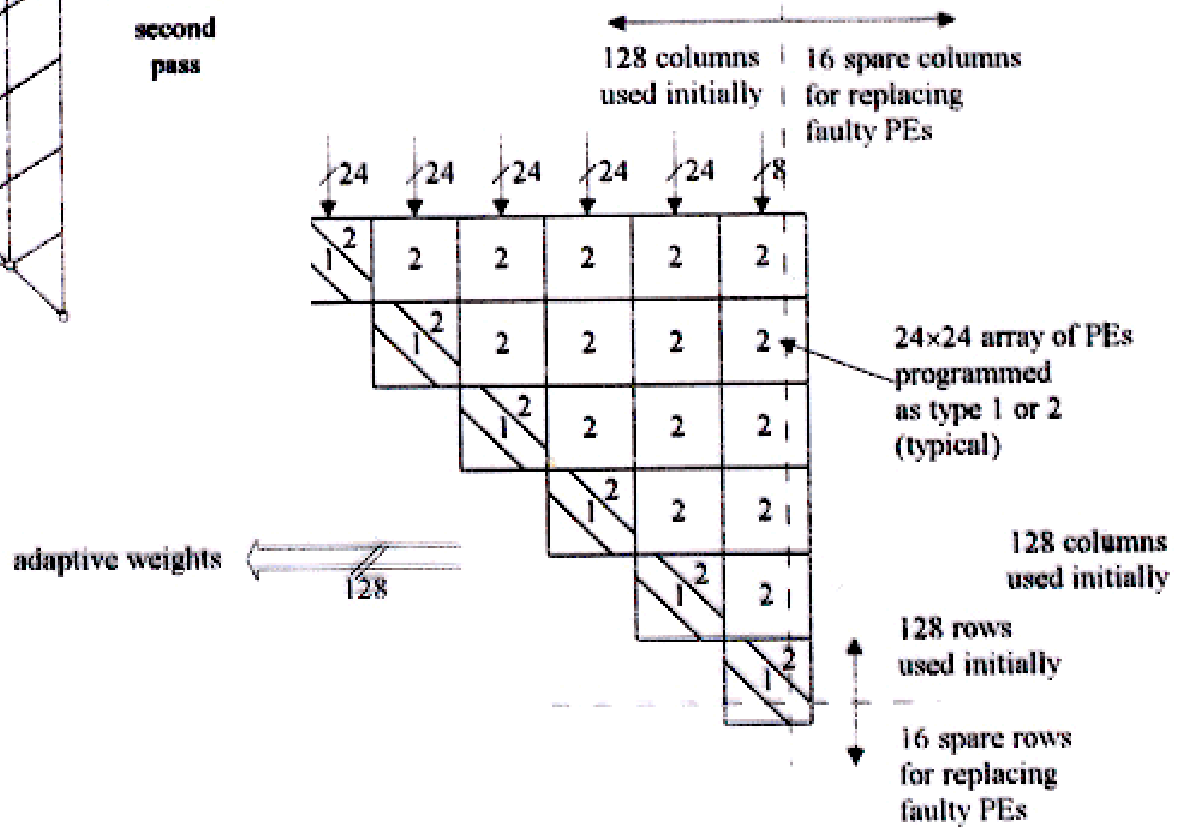


d.



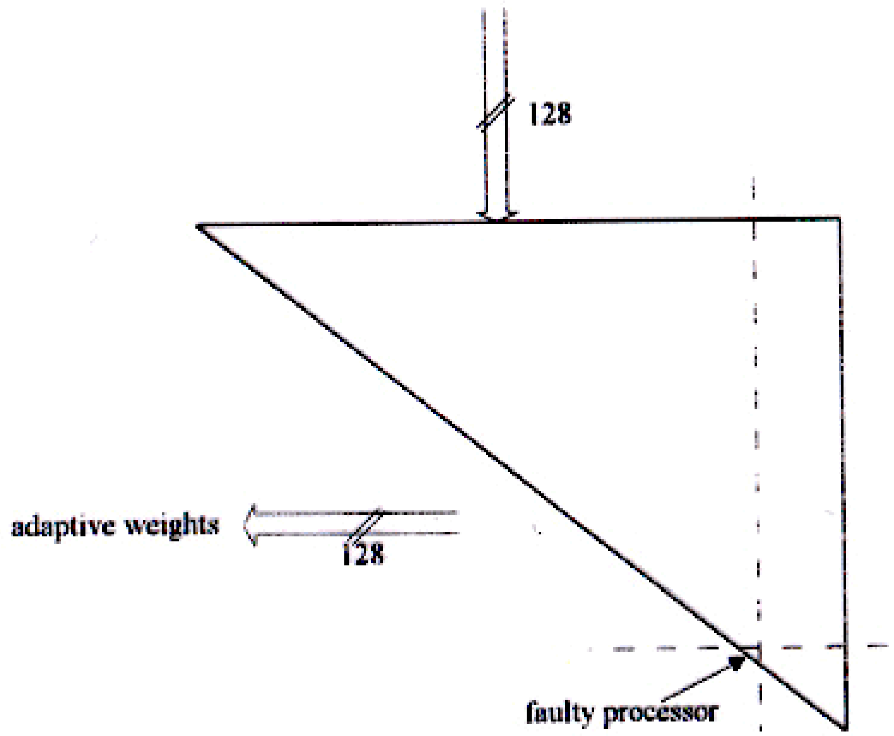
array
time
weights
memory
data in/steering vector
second pass
first pass
computation

e.

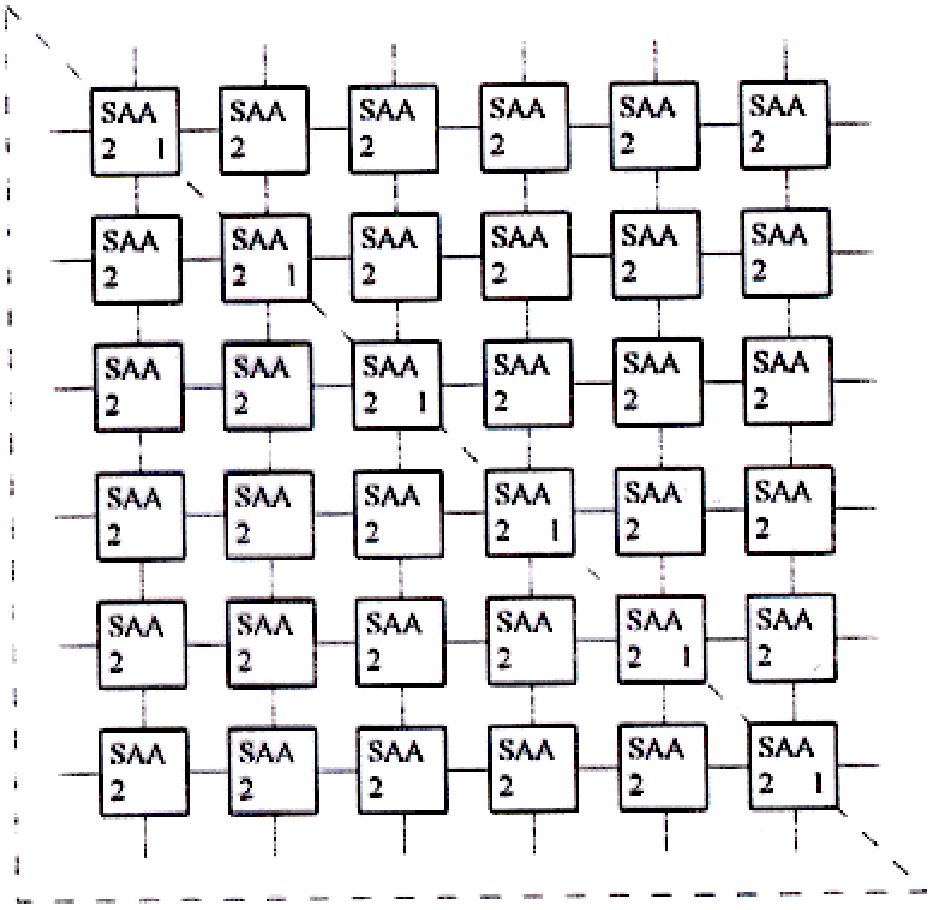


128 columns used initially
16 spare columns for replacing faulty PEs
24
24
24
24
24
8
24×24 array of PEs programmed as type 1 or 2 (typical)
adaptive weights
128
128 columns used initially
128 rows used initially
16 spare rows for replacing faulty PEs

f.



g.



h.

Veljko Milutinović

VLSI for SuperComputing: From Applications and Algorithms to Masks and Chips

World Top Foundries in VLSI for SuperComputing

- #1?

World Top Foundries in VLSI for SuperComputing

- #1? Qualcomm

World Top Foundries in VLSI for SuperComputing

- #1? Qualcomm
- #2?

World Top Foundries in VLSI for SuperComputing

- #1? Qualcomm
- #2? Intel

World Top Foundries in VLSI for SuperComputing

- #1? Qualcomm
- #2? Intel
- #3?

World Top Foundries in VLSI for SuperComputing

- #1? Qualcomm
- #2? Intel
- #3? TSM (Taiwan Semiconductor Manufacturing)

World Top Foundries in VLSI for SuperComputing

- #1? Qualcomm
- #2? Intel
- #3? TSM
- The Qualcomm VP/TD started from this course (first as my PhD student and later as the course TA)

World Top Foundries in VLSI for SuperComputing

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- An Intel VP/TD took the IR4RVL in-a-nut-shell (at my past IEEE/ACM tutorial)

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- #2? Intel
- #3? TSM
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- Maybe, a Next Gen VP/TD comes from ETF

World Top Foundries in VLSI for SuperComputing

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- Maybe,
  a next TSM VP/TD comes from ETF
  or MAT

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- Maybe,
  a next TSM VP/TD comes from ETF
  or MAT or FFH

World Top Foundries in VLSI for SuperComputing

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  (at my past IEEE/ACM-HICSS conference tutorial)
- Maybe,
  a next TSM VP/TD comes from ETF
  or MAT or FFH or FON

Who works here?

The Holistic Foundry (R&DFab) in VLSI for SuperComputing

- Phase#1: From Applications to Algorithms
  Phase#2: From Algorithms to Masks
  Phase#3: From Masks to Chips
  Phase#4: From Chips to Applications

The Holistic Foundry (R&DFab) in VLSI for SuperComputing

- Phase#1: From Applications to Algorithms
  Phase#2: From Algorithms to Masks
  Phase#3: From Masks to Chips

- Verification is crucial
  in each one of these phases,
  and related teaching is done in coop
  with ELSYS or HDLDH!

The Holistic Foundry (R&DFab) in VLSI for SuperComputing

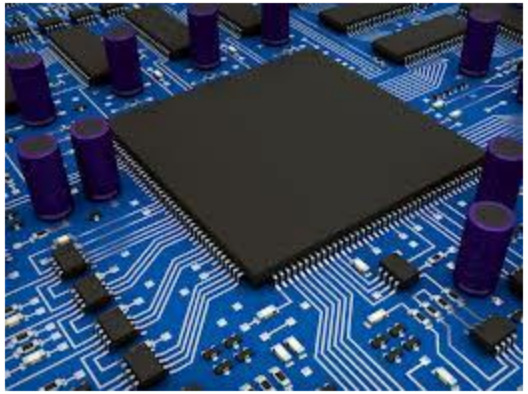
- Phase#1: From Applications to Algorithms
  Phase#2: From Algorithms to Masks
  Phase#3: From Masks to Chips
  Phase#4: From Chips to Applications

- Verification is crucial
  in each one of these phases,
  and related teaching is done in coop with ELSYS or HDLDH!

- Management issues of importance for an R&D Fab
  are covered in the accompanying course: IR4USP
  (including 12 related homework assignments)!

Contents:
From Algorithms to Masks

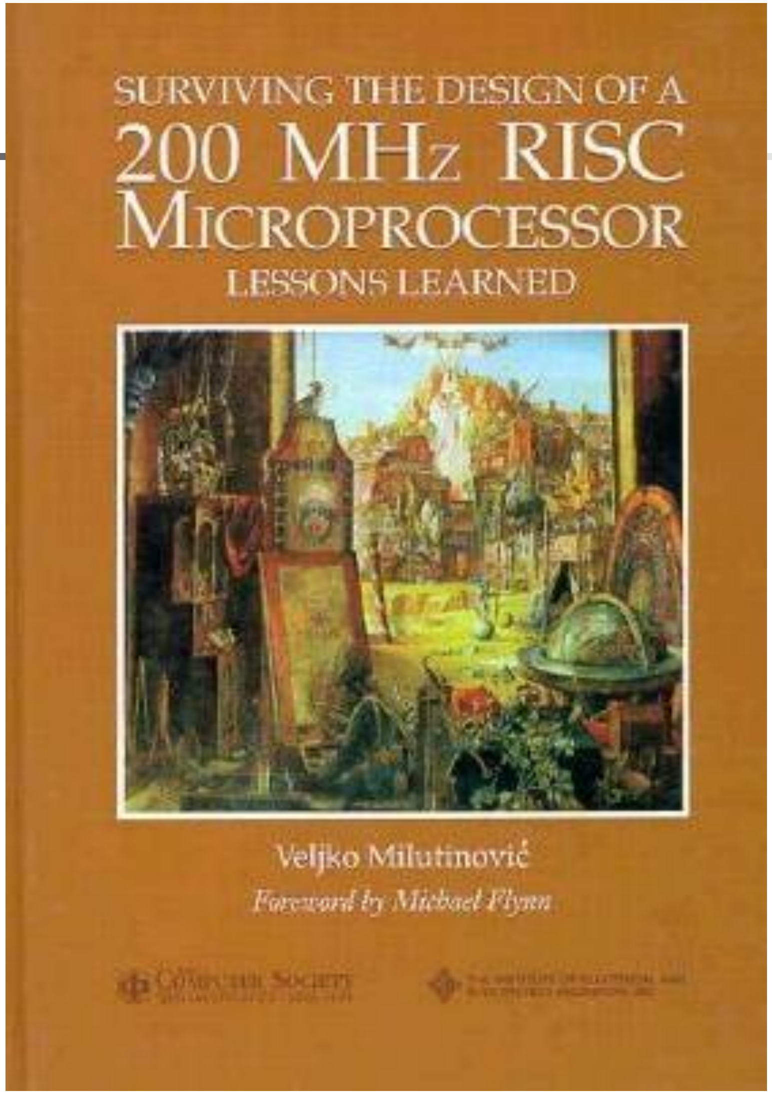
- Part#1: VLSI for ControlFlow SuperComputing
- Part#2: VLSI for DataFlow SuperComputing
- Part#3: VLSI for WirelessFlow SuperComputing
- Part#4: VLSI for EnergyFlow SuperComputing (QMOC)

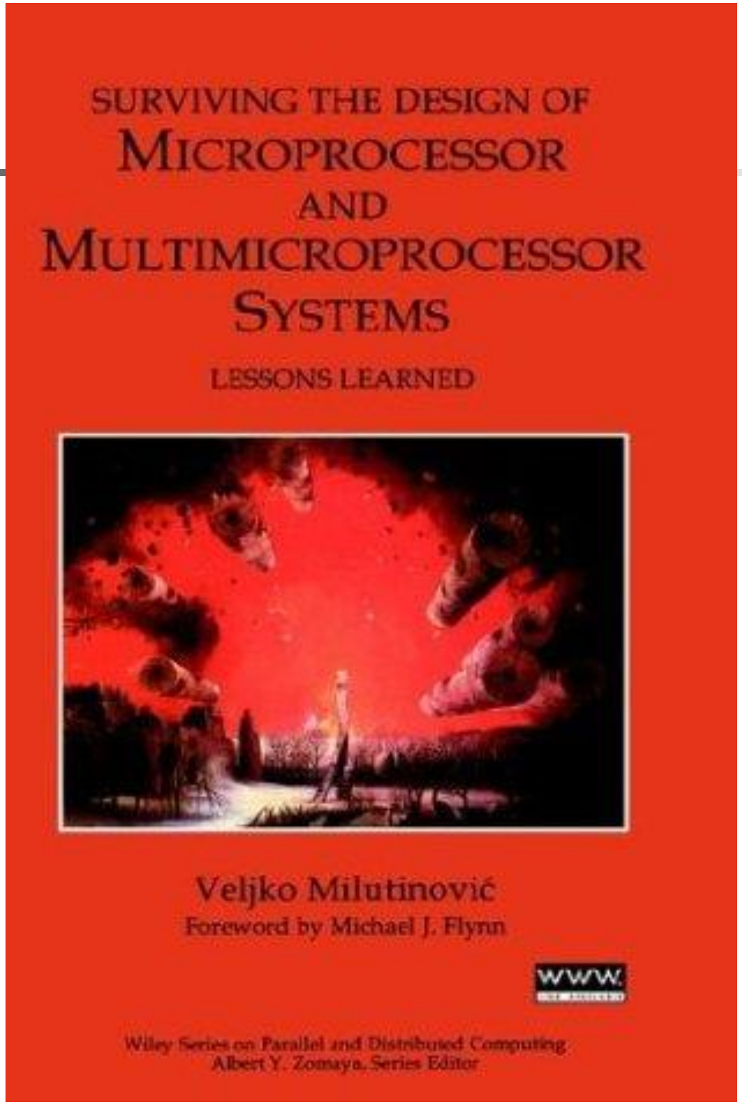


VLSI for ControlFlow SuperComputing

ManyCore Systems:
- Enabler Technology: VHDL vs Verilog (0.5 weeks)
- Design and Programming of a 200MHz RISC Microprocessor (2.5 weeks) + HW#1

MultiCore Systems:
- Enabler Technology: Verification by Elsys (2 weeks) + Lab#1
- Design of MicroProcessor and MultiMicroProcessor Systems by Wiley (1 week)


SURVIVING THE DESIGN OF A
200 MHz RISC
MICROPROCESSOR
LESSONS LEARNED
Veljko Milutinović
Forward by Michael Flynn


SURVIVING THE DESIGN OF
MICROPROCESSOR
AND
MULTIMICROPROCESSOR
SYSTEMS
LESSONS LEARNED
Veljko Milutinović
Foreword by Michael J. Flynn
WWW.
Wiley Series on Parallel and Distributed Computing
Albert Y. Zomaya, Series Editor

VLSI for DataFlow SuperComputing

FineGrain DataFlow:
- Enabler Technology: Altera vs Xilinx (0.5 weeks)
- Design and Programming of the 200MHz Maxeler Machine (3.5 weeks) + HW#2

SystolicArray DataFlow:
- Enabler Technology: Systolic Array Architectures (0.5 weeks)
- Design of DARPA Systolic Architectures (0.5 weeks) + Lab#2

On Flexible DataFlow:

- **Advances in Computer Architecture**
  (North Holland)
  by Veljko M. Milutinovic
  with a contribution from John Hennessy

On Fixed DataFlow:

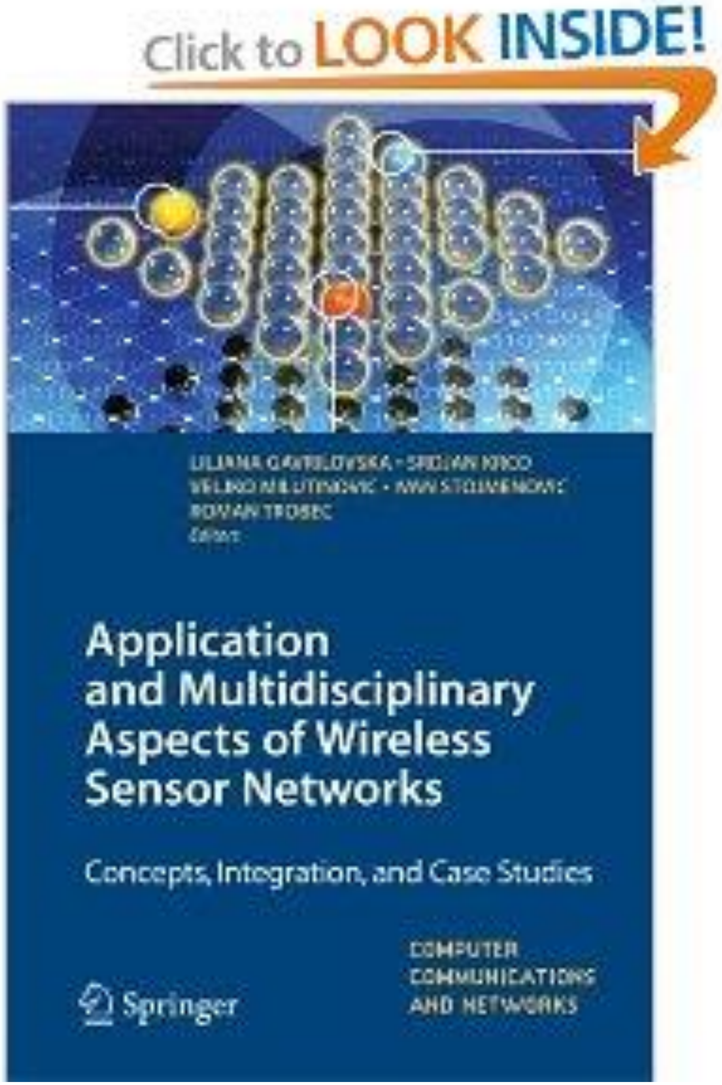
- **High-Level Language Computer Architecture**
  (Elsevier Computer Science Press)
  by Veljko M. Milutinovic
  with a contribution from Michael Flynn

VLSI for WirelessFlow SuperComputing

WSNs: Part#1
- Hardware (0.25 weeks)
- Software (0.25 weeks)

WSNs: Part#2
- Systems (SUN+Slimmer) (0.25 weeks)
- Applications (UbiComputing@WSN+DataMining@WSN) (0.25 weeks)

The stress is on integration of WSNs, IoT, Ethernet, and Internet:
Z. Babovic et al., "Web Performance Evaluation of Internet of Things Applications," IEEE Access 2016.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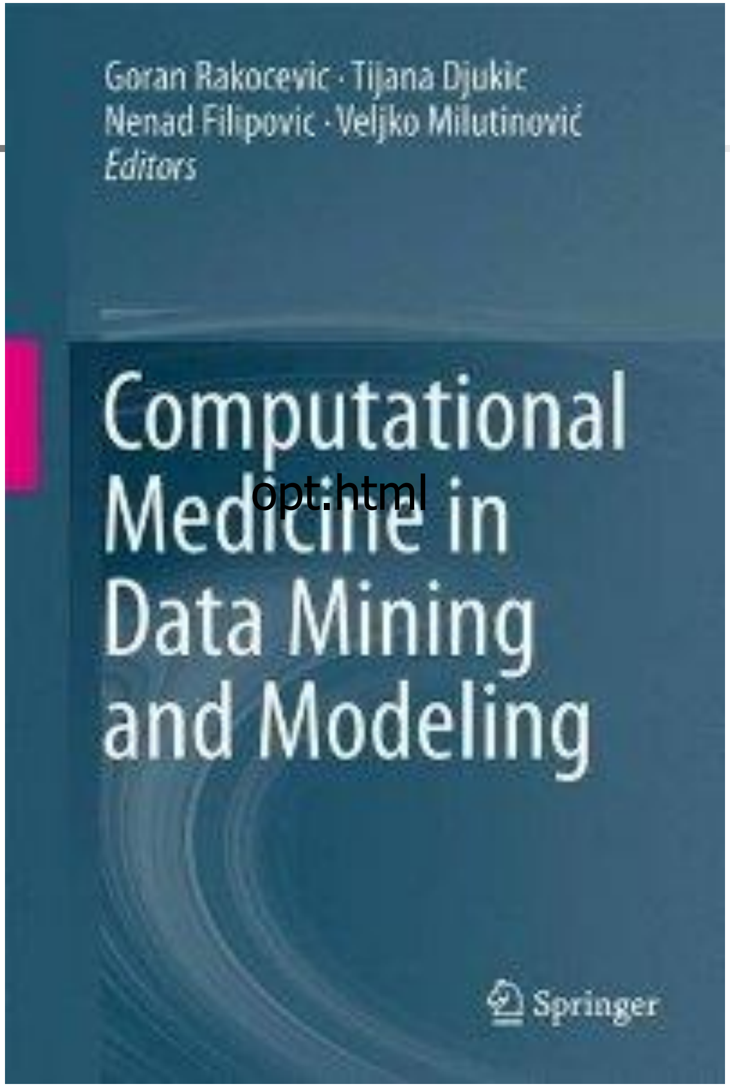
Goran Rakocevic · Tijana Djukic
Nenad Filipovic · Veljko Milutinović
Editors
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VLSI for Quantum, Molecular, Optical, and Chemical SuperComputing

Basics:
- Hardware (N weeks)
- Software (N weeks)

Advances:
- Systems (N weeks)
- Applications (N weeks)

Optional class projects for extra points!

Quantum Computing

- A quantum computer **uses qubits to run multidimensional quantum algorithms**.
- Their processing power increases exponentially as qubits are added.
- A classical processor uses bits to operate various programs.
- Their power increases linearly as more bits are added.

Potentials:

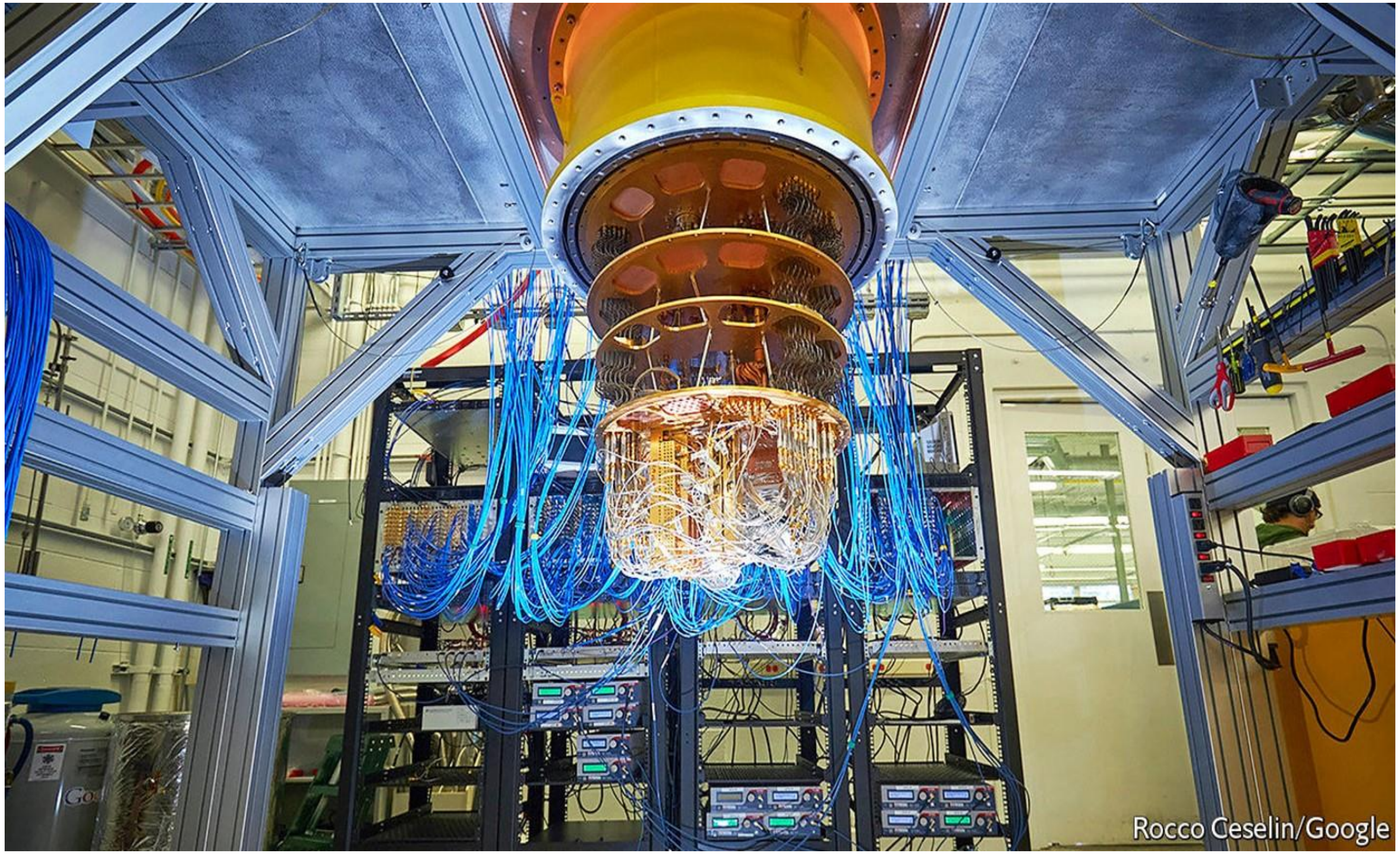
- Quantum computing is **158 million times faster than the most sophisticated supercomputer we have in the world today**.
- It is a device that could do in four minutes what it would take a traditional supercomputer 10,000 years to accomplish.

Reality:

The First Eight Companies in Quantum Computing:

- Atom Computing
- Xanadu
- IBM
- ColdQuanta
- Zapata Computing
- Azure Quantum
- D-Wave
- Strangeworks

Beginnings:



The Best Quantum Computing Stocks:

- IBM (NYSE: IBM)
- Alphabet (Nasdaq: GOOG, Nasdaq: GOOGL)
- Intel (Nasdaq: INTC)
- Microsoft (Nasdaq: MSFT)
- Amazon (Nasdaq: AMZN)
- Quantum Computing (OTC: QUBT)

Libraries:

- TensorFlow Quantum (TFQ)
  is a quantum machine learning library
  for rapid prototyping
  of hybrid quantum-classical ML models.
- Interfacing for control-flow.

Tensor Flow Quantum:

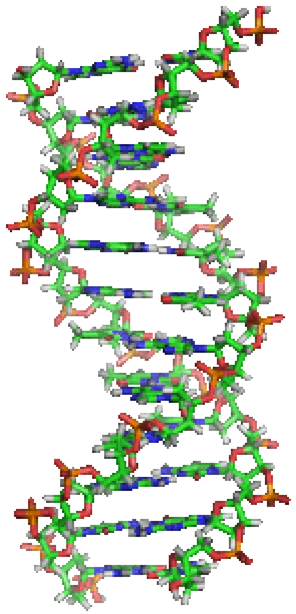
- TensorFlow Quantum focuses on *quantum data* and building *hybrid quantum-classical models*.
- It integrates quantum computing algorithms and logic designed in Cirq, and provides quantum computing primitives compatible with existing TensorFlow APIs, along with high-performance quantum circuit simulators.

A Programming Example:

```
# A hybrid quantum-classical model.
      model = tf.keras.Sequential

([

# Quantum circuit data comes in inside of tensors.
      tf.keras.Input(shape=(), dtype=tf.dtypes.string),

# Parametrized Quantum Circuit (PQC) provides output

# data from the input circuits run on a quantum computer.
      tfq.layers.PQC(my_circuit, [cirq.Z(q1), cirq.X(q0)]),

 # Output data from quantum computer passed through model.

      tf.keras.layers.Dense(50)
])
```

Molecular Computing

- Molecular computing is **a branch of computing that uses DNA, biochemistry, and molecular biology hardware, instead of traditional silicon-based computer technologies, to achieve dense computing!**
- Applications for dense computing are many!



Who invented DNA computing?

- **Leonard Adleman**,
  professor of computer science and molecular biology at the University of Southern California, USA, who pioneered the field when he built the first DNA based computer.

- L. M. Adleman, Science 266, 1021–102; 1994.

Essence:

- With DNA,
  the way the molecules can be triggered
  to bind with each other
  can be used to create a circuit
  of logic gates in test tubes.
- Compatible with nature based computing!

Techniques:

- Many gates can be combined in a circuit:
- Each output DNA will bind to the next logic gate until some predictable terminal output strand is liberated, to produce an intermediate or final result.

N.B.:

- In ***DNA computing***, information is represented using the four-character genetic alphabet: A [adenine], G [guanine], C [cytosine], and T [thymine].
- The core advantage of molecular computing is its potential to pack vastly more circuitry onto a microchip than silicon will ever be capable of—and to do it cheaply.

Miniaturization:

- Molecules are only a few nanometers in size.
- Making it possible to manufacture chips that contain billions, even trillions, of switches and components.

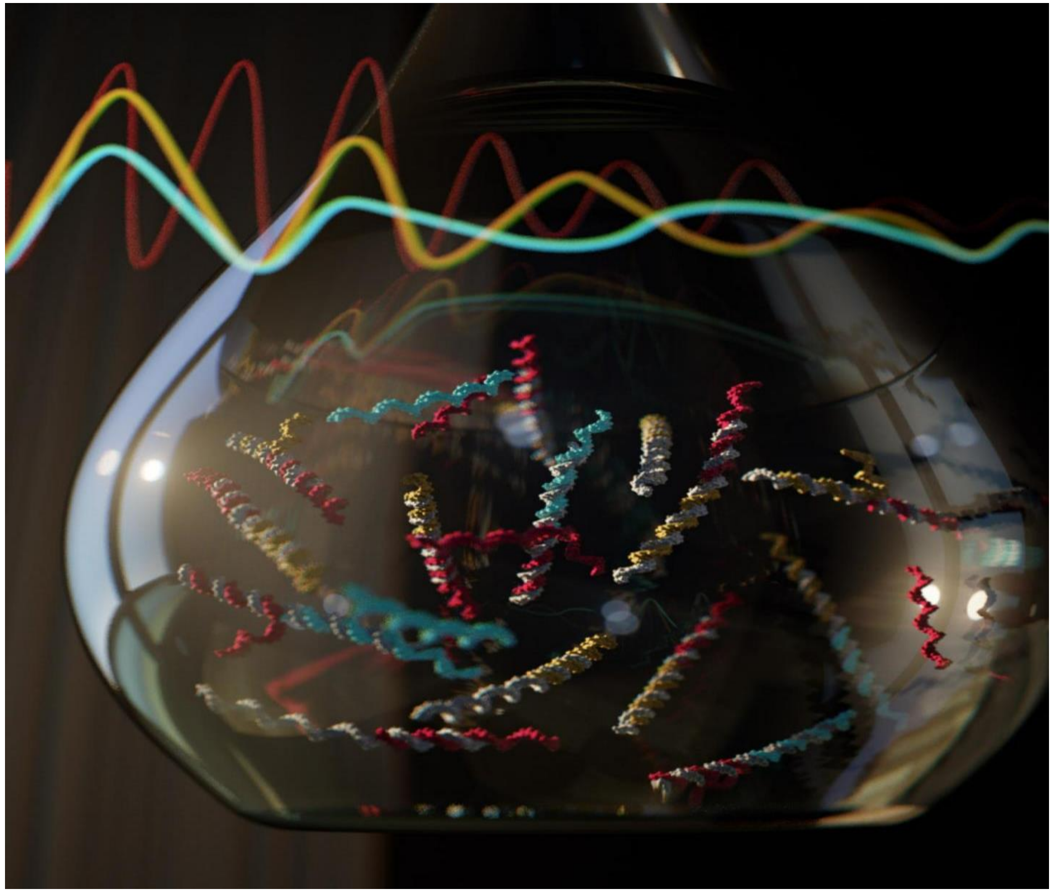
Can DNA be programmed?

- **Researchers at The University of Texas at Austin have programmed DNA molecules**
  to follow specific instructions
  to create sophisticated molecular machines
  that could be capable of communication,
  signal processing, problem-solving, decision-making,
  and control of motion in living cells.
- Libraries under construction at MIT.

How does DNA computing work?

- In one method, called DNA strand displacement, the input of DNA that binds to a DNA logic gate displaces a strand of DNA that serves as the output.
- Many gates can be combined in a circuit:
  Each output DNA will bind to the next logic gate until some predictable terminal output strand is liberated.

An Experiment:



Flexibility:

- With the flexible molecular algorithms on the rise, one might be able to assemble a complex entity on the nanoscale with the reprogrammable tile set.
- Flexibility is the issue for a great number of applications!

Do DNA computers exist?

- The DNA Computing Technology is in the research phase.
- **DNA computers can't be found at your local electronics store yet**.
- The technology is still in development; it didn't even exist as a concept a decade ago.

Connecting Living Neurons Through Synapses

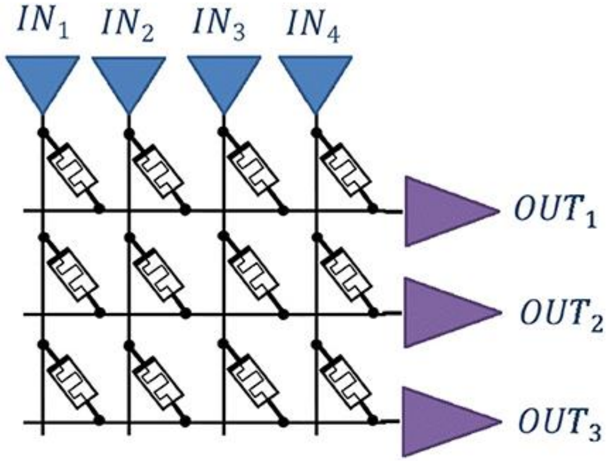
- Systolic array connected through resistors
(varied electrical conductivity represents the strength of the connection between the neurons/resistors - adaptable based learning)

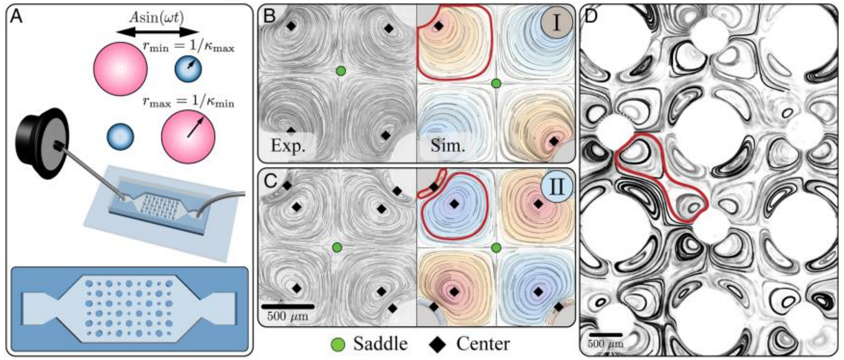
- To match the world of computing to the needs of AI, we must build computers with a structure inspired by the brain. These computers should have no separation between storage and computation, but instead use units capable of calculating and saving the values simultaneously. Electrical circuits inspired by the brain are called neuromorphic circuits

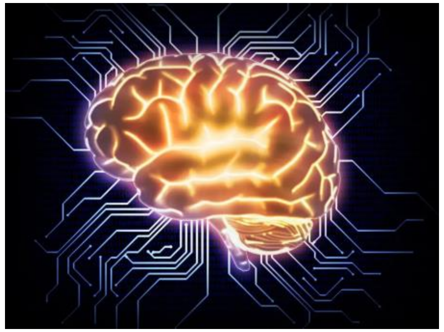
- Electronic circuits that mimic the behavior of neurons in the human body. Neuromorphic circuits can be combined to create neuromorphic computers.

- In neuromorphic circuits, the same electronic components are used to both perform a calculation and to store the result locally. The structure of such a computer is completely different from a von Neumann machine, and it looks like a neural network in the brain.

- Researchers from John Hopkins University and Cortical Labs suggest that it's time to create a new type of computer that uses biological components. They believe that biological computers could outperform electronic computers in certain applications and use significantly less electricity.







Courtesy of Prof. Josep Torrellas, Prof. Shahar Kvatinsky, and Researchers from John Hopkins University and Cortical Labs

Optical Computing

- **Optical computing** or **photonic computing** uses light waves produced by lasers or incoherent sources for data processing, data storage or data communication for computing.
- For decades, photons have shown promise to enable a higher bandwidth than the electrons used in conventional computers.

Essence:

- "Computation-by-propagation,
  where the computation takes place
  as the wave propagates through a medium,
  can perform computation at the speed of light!"
- Speed of light = 300.000km/sec

Applications:

- Application-specific devices,
  such as synthetic-aperture radar (SAR)
  and optical correlators,
  have been designed
  to use the principles of optical computing.
- Correlators can be used, for example,
  to detect and track objects,
  and to classify serial time-domain optical data.

Techniques:

- Techniques have been developed that can perform continuous Fourier Transform optically, by utilizing the natural Fourier transforming property of lenses.
- The input is encoded using a liquid crystal spatial light modulator.

Artificial Neural Network with Optical Components:

- Early optical neural networks
used a photorefractive Volume Hologram
to interconnect arrays of input neurons
to arrays of output.
- With synaptic weights in proportion
to the multiplexed hologram's strength.

Optical Neural Networks:

- Some artificial neural networks
  that have been implemented
  as optical neural networks
  include the Hopfield neural network.
- Also the Kohonen map
  with liquid crystal spatial light modulators.

Do Photonic Chips Exist?

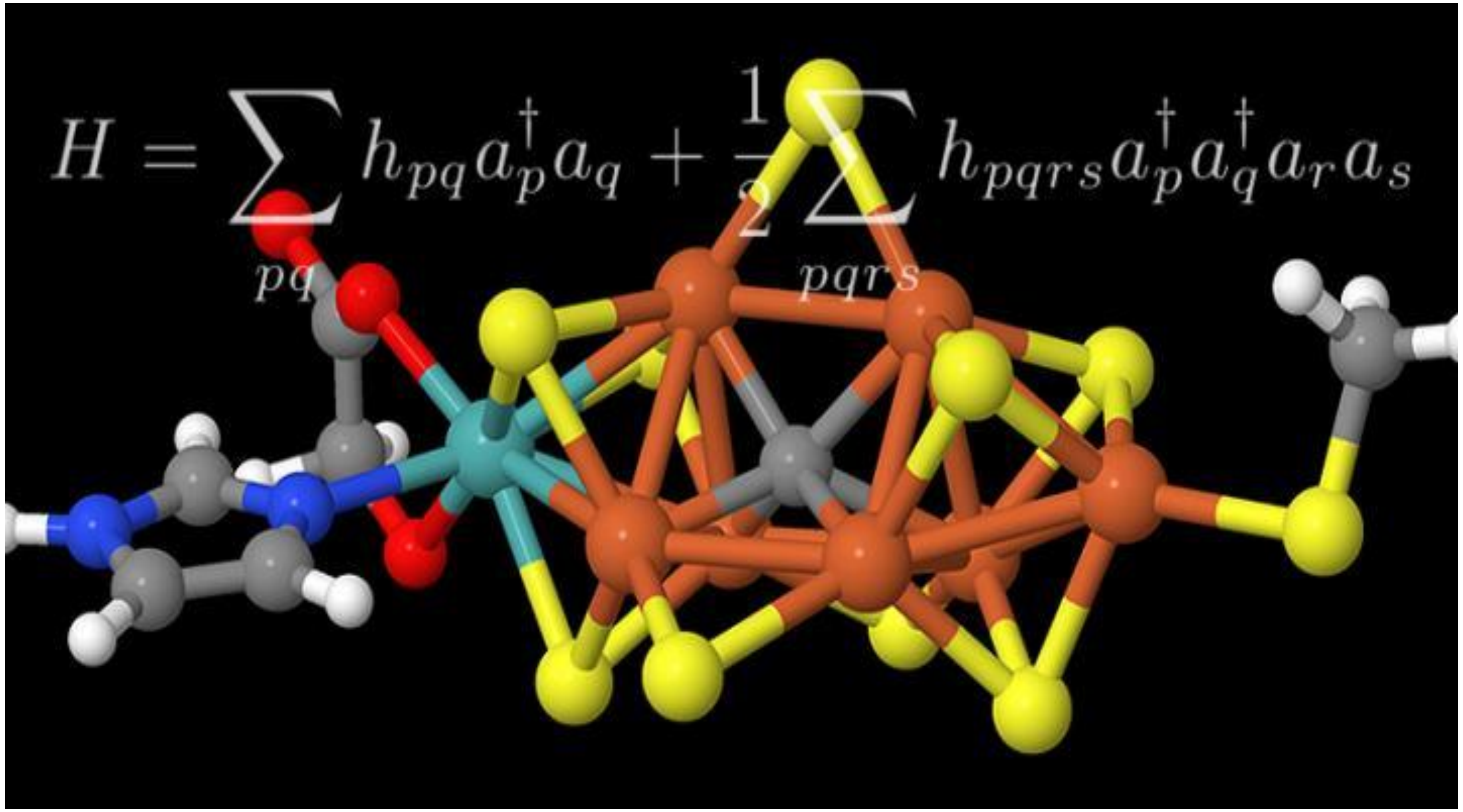
- Now **scientists have developed a deep neural network on a photonic microchip** that can classify images in less than a nanosecond, roughly the same amount of time as a single tick of the kind of clocks found in state-of-the-art electronics.
- A lot more effective compared to microprocessor based design!

An Application:

Chemical Computing

- **Computing with real molecules like programming electronic devices, but using principles taken from chemistry and appropriate chemical processes.**
- Effective for simulations in chemistry and physical chemistry!

An Example:


$H = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{pqrs} h_{pqrs} a_p^\dagger a_q^\dagger a_r a_s$

A **chemical computer**, also called a **reaction-diffusion computer**:

- **Belousov–Zhabotinsky** (**BZ**) **computer**, or **gooware computer**, is an unconventional computer based on a semi-solid chemical "soup" where data are represented by varying concentrations of chemicals.
- The computations are performed by naturally occurring chemical reactions.

Origins:

- Originally chemical reactions were seen as a simple move towards a stable equilibrium which was not very promising for computation.
- This was changed by a discovery made by Boris Belousov, a Soviet scientist, in the 1950s.

Belousov:

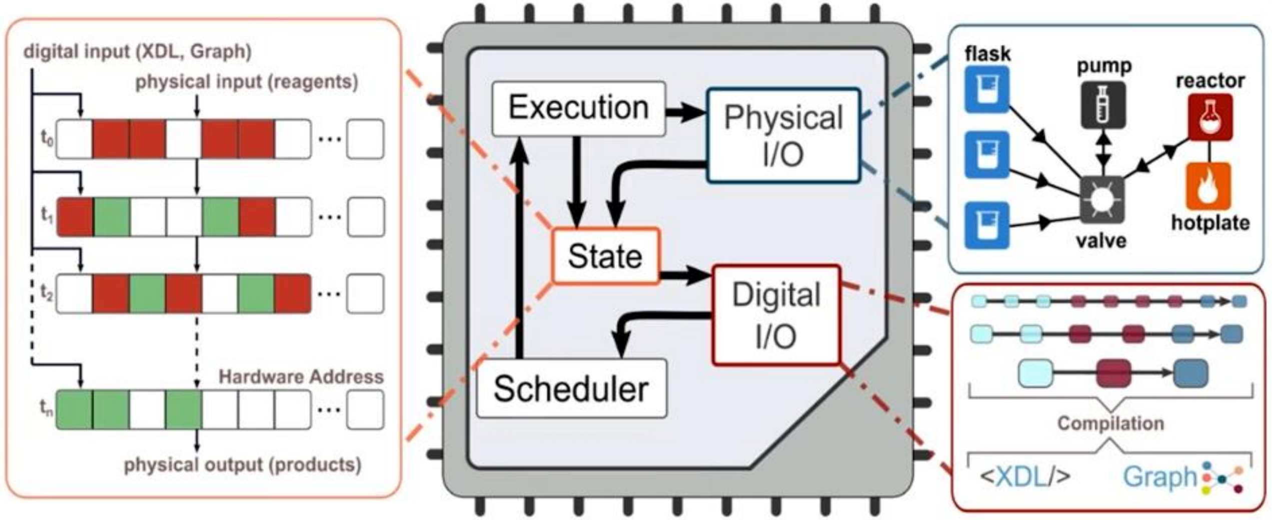
- Belousov created a chemical reaction between different salts and acids that swing back and forth between being yellow and clear.
- This is because the concentration of the different components changes up and down in a cyclic way.

Adamatzky:

- Andrew Adamatzky
  at the University of the West of England
  has demonstrated simple logic gates
  using reaction–diffusion processes.
- An important step towards programmability.

An Implementation:

Why is it Universal? The Chemical Synthesis State Machine



- Abstraction of chemical assembly → state machine that can make any molecule / material
- Inputs are digital and physical → Outputs are physical

Cheмputation is the process of running XDL code reliably on **any** compatible hardware
c.f. Computation - running programs on a digital computer

European Projects

ESF:
• RoMoL: Riding on Moore's Law (0 weeks)
• HiPeac: Parallel Programming Models (0 weeks)

FP7/H20:
• FP7: ProSense (0 weeks)
• FP7: BalCon (0 weeks)

USA Projects

- Nature Based Construction and Computing:
  - Purdue (X weeks)
  - Indiana University (X weeks)
- Nature Based Media and Computing:
  - MIT (X weeks)
  - Harvard (X weeks)

Example Algorithms for Practical Implementations

Engineering:
- Computer Engineering (M weeks)
- Financial Engineering (M weeks)

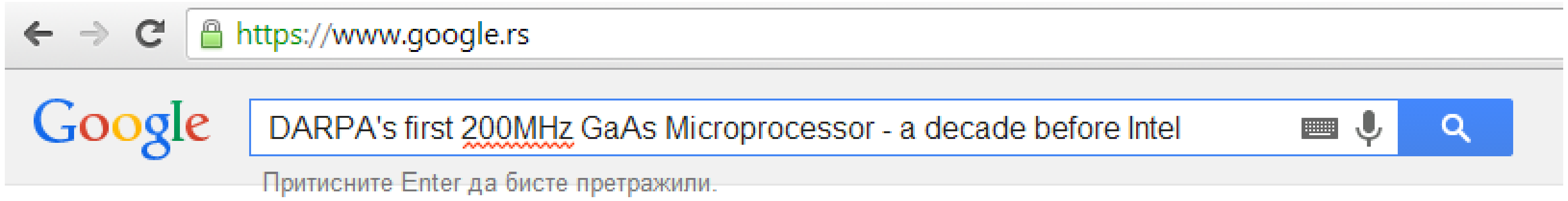
Science:
- Physical Chemistry (M weeks)
- Computer Science (M weeks)

SOME PREVIOUS OFFERINGS OF THIS TECH COURSE

- Purdue, Indiana;
- MIT, Harvard;
- Imperial, Kings;
- ETH, EPFL;
- UNIWIE, TUWIEN;
- Siena, Salerno, Barcelona, Madrid;
- Ljubljana, Koper, Zagreb, Rijeka, Podgorica, Sarajevo;
- Technion, Jerusalem;
- Bogazici, Koc;
- Tsinghua, Shandong.

Previous offerings of the related MGMT course: Amherst, Dartmouth, Harvard, MIT Media, Pisa, Genova.

BOTTOM LINE:
BRINGING ADVANCED INDUSTRIAL EXPERIENCE INTO THE CLASSROOM



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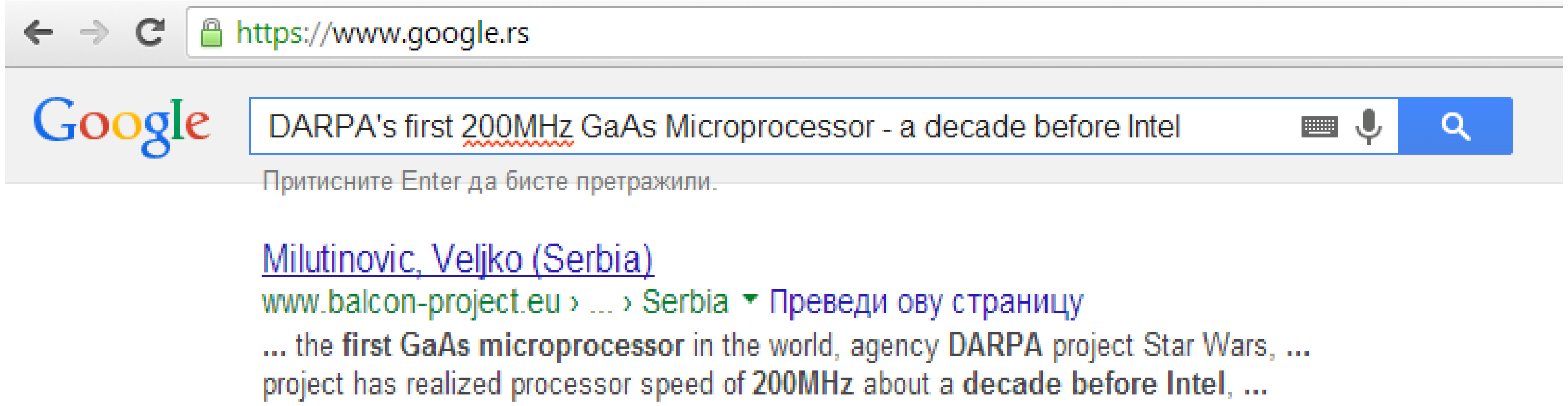

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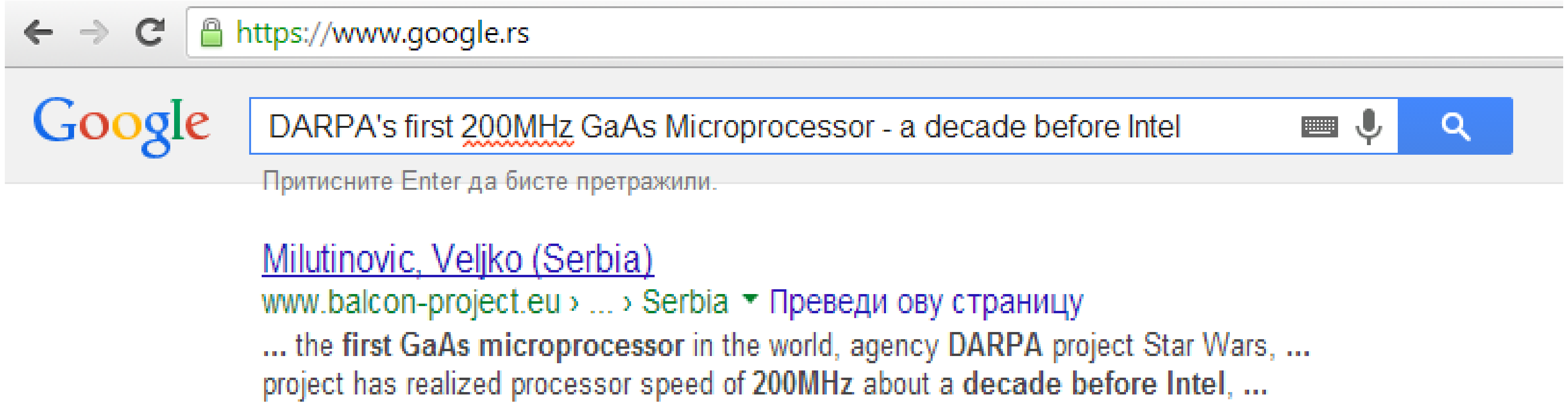

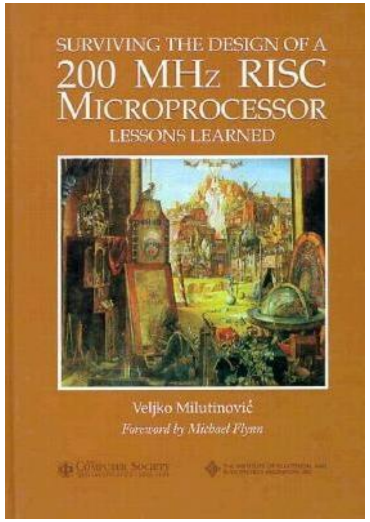

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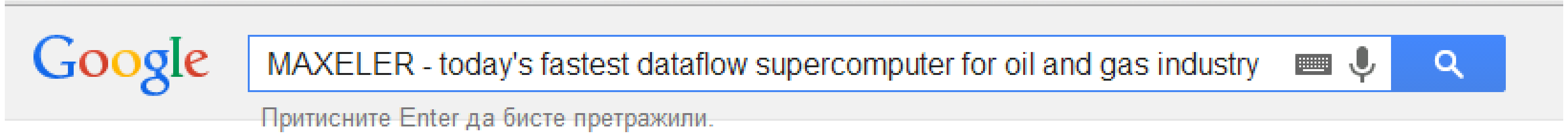

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MAXELER - today's fastest dataflow supercomputer for oil and gas industry

Притисните Enter да бисте претражили.

[HPCwire: **Maxeler** Launches MPC-X Series **Dataflow** Engines]

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Veljko Milutinovic

Dr. Veljko Milutinovic is a professor at the School of Electrical Engineering, University of Belgrade, Serbia. During the 80's, for about a decade, he was on the faculty of Purdue University in the U.S.A, where he co-authored the architecture and design of the world's first DARPA GaAs microprocessor. During the 90's, after returning to Serbia, he took part in teaching and research at a number of major EU schools. He also delivered lectures at Stanford and MIT, and has about 20 books published by leading publishers in the U.S.A. Dr. Milutinovic is a Fellow of the IEEE and a Member of Academia Europaea.

BOTTOM LINE:
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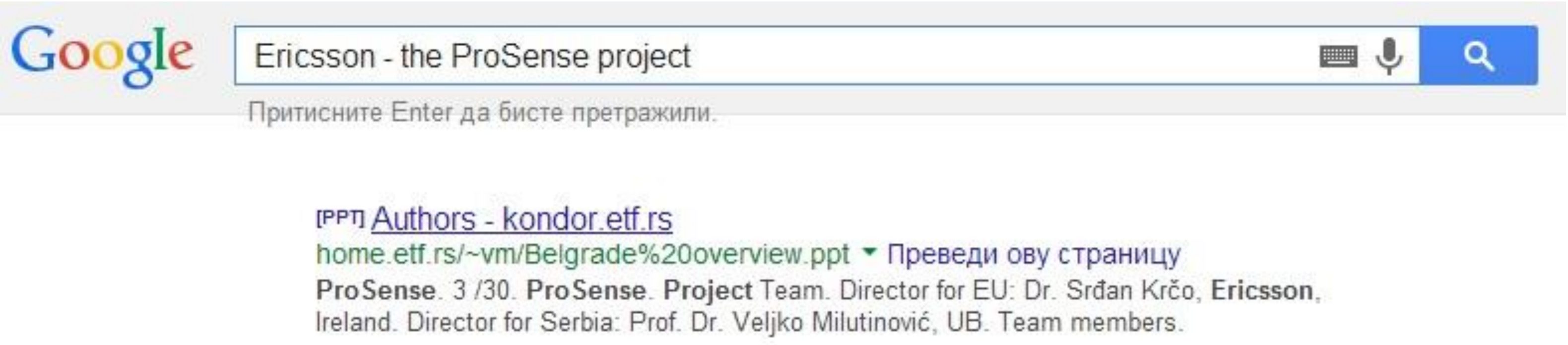

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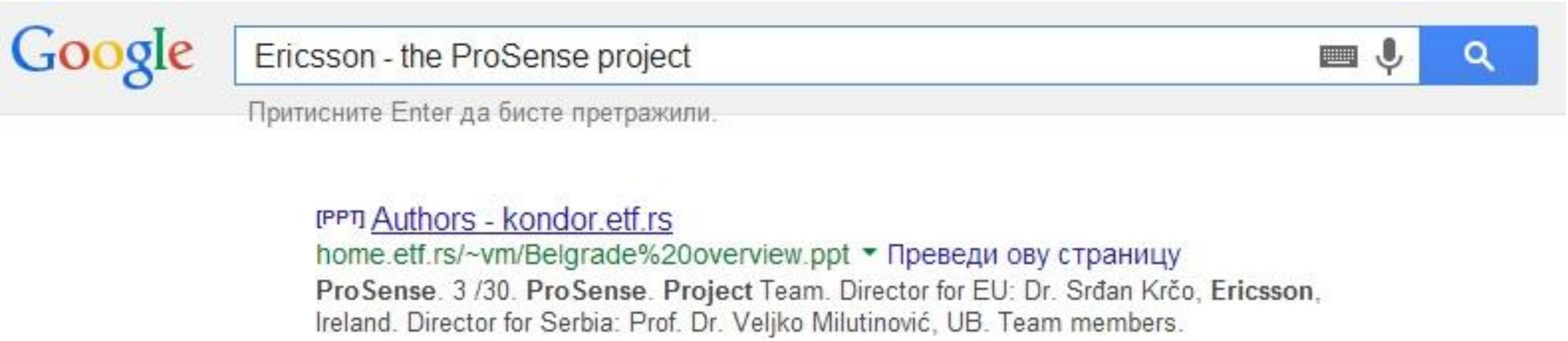

Wireless Sensor Networks: ApplicationDesign and DataMining

SUGGESTED READINGS:

- Z. Babovic, V. Milutinovic, “Novel System Architectures for Semantic-Based Integration of Sensor Networks,” Elsevier, Advances in Computers, 2013.
- Z. Babovic, “DataFlow systems: From their origins to future applications in data analytics, deep learning, and the Internet of Things,”
  In V. Milutinovic et al. “DataFlow Supercomputing Essentials,” Springer, 2017.
- IPSI TIR (Transactions on Internet Research), 2024.
- IPSI TAR (Transactions on Advanced Research), 2024.
- Special Issues of Elsevier – Advances in Computers 2024.
- Special Issues of Springer – Journal of Big Data, 2024, 2025.

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DataFlow SuperComputing

Maxeler Veljko Anegdotic 060 PPT
A Paper by Jakob Salom
A Paper from Tsinghua on Shallow Water Weather Forecast
Tutorial Slides by Sasha and Veljko: Practice (Current Update)
Maxeler Oskar Introduction, 2012
Maxeler Oskar Prediction, 2018
Tutorial Slides by Oskar: Theory (6 parts)
Slides by Jacob, New York
Slides by Jacob, Alabama
Homework Solutions (6 parts)
Maxeler in Meteorology
Maxeler in Mathematics
Tutorial Slides by Sasha: Practice (Current Update)
Maxeler Essence by Nenad Korolija, August 2010
Maxeler Essence by Veljko Milutinovic, August 2012
Discusion of Two Possible Research Avenues
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Maxeler Related Work of PHD Students
Catalogue 2013
Brochure about Risk Analytics
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Oskar@Linz Keynote Speech About GeoPhysics (Weather, etc...)
Oskar@London Course Logistics (if You like to travel to London)
Maxeler Tutorial: The Full Blown Text
Paper, unconditionally accepted for Advances in Computers by Elsevier (March 2015)
Paper, unconditionally accepted for Communications of the ACM (May 2013)
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The IET Best Paper 2014 Premium Award for Computing and Digital Techniques by Jovanovic and Milutinovic
The IEEE/ACM HICSS Best Paper 1986 Track Award for The Architecture Track by Fortes and Milutinovic
AppGallery.maxeler.com How to Prepare an Application for AppGallery? The HowToPackageAnApp link with low level details HowTos for GitHub
The Newest March 2015 Book by Elsevier, on DataFlow Concepts and Applications: Dataflow Processing (Advances in Computers, Volume 96)
The Newest May 2015 Book by Springer, on DataFlow Concepts and Applications: Guide to DataFlow SuperComputing (Computer Communications and Networks)
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Intel Trying to Acquire Altera Intel Aquires Altera for $16.7 Billion Intel After Aquiring Altera (PKGupta) Qualcomm + Xilinx IBM + Xilinx Micron + Convey
The Best SI student HW of January 2016
The Best RI student HW of January 2016
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Oskar's Interview for ODBMS
PPTX: DataFlow Computing by Milos Kotlar
PPTX: Matrix Multiplication by Milos Kotlar
Maxeler@Amazon: A Video
Maxeler@Amazon: Press Release
Training Examples: Math
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Training Examples: ML - Perceptron on Maxeler by Milos Kotlar
Training Examples: Tensor Calculus by Milos Kotlar
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THE ISC-2018 IN FRANKFURT, GERMANY: Maxeler ExaComm
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